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MASTER'S DEGREE IN

BIOINFORMATICS FOR COMPUTATIONAL GENOMICS

GENOTYPING BY SEQUENCING (GBS) FOR POPULATION GENETICS AND PHYLOGENETIC STUDIES OF NON-MODEL PLANTS: THE CASE OF *CAMPANULA THYRSOIDES*, AND OTHERS.

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"May you keep going forward."

Abstract

This thesis aim is to showcase the use of Genotyping by Sequencing (a Rad-Seq method) for conservation-focused genomic analyses of non-model plants, through three different projects.

The first one was a populations genetics study with the aim of supporting the planning of a strengthening action for a population of the endemic alpine species *Campanula thyrsoides* subspecies *carniolica*. The study was carried over 137 individuals spanning 9 populations of interest. For this goal SNPs were called using `Stacks`: `denovo_map.pl` pipeline, which provided a matrix of 3896 polymorphic, non rare ($MAF > 0.005$) SNPs. The results showed a good gene flow between populations with no inbreeding within population (all $F_{IS} < 0$), despite a considerable extent of isolation among all populations (pairwise $F_{ST} > 0.10$, $p - value < 10e - 16$). Only three neighbouring population in Veneto and Friuli-Venezia-Giulia (Italy), had pairwise $F_{ST} \leq 0.10$ ($p - value < 10e - 16$). PCA and Maximum-Likelihood tree further separated the populations in Italian, Slovenian and Austrian groups, highlighting a longitudinal gradient in their genetic variation. Furthermore, a population collected in the Italian region of Lombardy, historically range of *Campanula thyrsoides* subspecies *thyrsoides*, actually clustered with subsp. *carniolica*, showing an expansion in this subspecies' distribution range. The second project was a phylogenetic study of landraces of *Solanum tuberosum* (modern potato), performed using both GBS data from 12 putative Italian old landraces and 2 commercial cultivars of *S. tuberosum* and WGS data from 4 wild and 4 cultivated Andean *Solanum* species. Using SolTub_3.0 reference genome and classic tools for genome-based variant calling, we built PCA plots and maximum likelihood trees over 14945 polymorphic SNPs: the results show that while some of the putative Italian landraces are indeed modern commercial cultivars (Desiree and Kennebec), others are probably the results of different, older domestication events. The

third, and last, project, aimed to use PstI digested GBS data to explore populations structure of urban and wild populations of orchids *Anacamptis pyramidalis* and *Anacamptis berica*. Using a matrix of 13791 polymorphic, non rare SNPs, we explored the structure of 19 natural (61 samples) and 5 urban populations (29 samples), divided in two species (4 populations of *A. berica* and 20 of *A. pyramidalis*). There is no differentiation either between populations or between urban and wild samples: the pipeline is able, though, to capture the distance between the two species.

We discuss each study limitations, focusing specifically on the bioinformatics approach taken to address the three projects goals: we compared software performance and performed parameters used, gave insights on how to correctly execute genomics analyses of GBS data and how to interpret the results, combining bioinformatics and biodiversity points of view.

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1. Introduction

1.1 Biodiversity conservation

Many plant species populations are declining due to human activity (Supple and Shapiro, 2018; Yin *et al.*, 2024): the over-exploitation and habitat destruction threaten diversity (Pence *et al.*, 2022) and lead to fragmentation and isolation of populations, making them susceptible to decline and ultimately extinction (Aguilar *et al.*, 2008; IUCN/SSC, 2014). Diversity can be assessed on different levels depending on species (Habel & Schmitt, 2012), but it is possible to recognize three main types: between species, between populations of the same species and between individuals of the same populations (Salgotra & Chauhan, 2023). All these types of diversity concur together to form the biodiversity of an ecosystem. The main causes of genetic diversity changes within a population of the same species are selection, mutation, gene flow and genetic drift (Salgotra & Chauhan, 2023), but the human influence on both wild and domesticated species can change this delicate equilibrium: while wild species suffer from habitat fragmentation (Aguilar *et al.*, 2008; Habel and Schmitt, 2012), crop plants are instead artificially selected for high yielding, which lowers their genetic diversity (Salgotra & Chauhan, 2023). Under environmental changes, different populations might survive depending on the presence of favorable genetic characteristics; if, however, populations possess little or no genetic diversity, they could become more susceptible to abiotic pressure, their individuals might experience loss of fitness (Habel & Schmitt, 2012) and, in the worst cases, whole populations might fail to adapt (Supple & Shapiro, 2018) and could not survive the stressful event (Salgotra & Chauhan, 2023). Anthropic activity has a big impact on species' ability to adapt so, especially in the case of fragmented populations, it becomes essential to limit human influence and apply strategies of both *in situ* preservation and *ex situ* conservation (IUCN/SSC, 2014). The first term refers to the conservation of genetic resources in a natural habitat (Vijayan *et al.*, 2011), while *ex situ* conservation is carried on in research institutes, botanical gardens or, in general, away from the original habitat of the species, where individuals are maintained under artificial conditions (IUCN/SSC, 2014). *Ex-situ* conservation techniques often have a final *in-situ* output (Volis & Blecher, 2010). For example, seed banking (drying at c.15% relative humidity and then storing at 20°C), arguably the most used, low-cost and effective long-term *ex situ* conservation method (Volis and Blecher, 2010; Pence *et al.*, 2022, Salgotra and Chauhan, 2023) is often used to build collections of seeds as material for *in-situ* recovery actions (Volis & Blecher, 2010). Seed banking, though, is not feasible for a lot of species, for example for those species whose seeds cannot survive freezing (Pence *et al.*, 2022) or have a long germination time (Volis & Blecher, 2010). To bypass these limitations living collections can be used, which are able to maintain a viable population as a source of individuals for population restoration (Volis and Blecher, 2010; IUCN/SSC, 2014; Pence *et al.*, 2022) but they are especially sensitive to genetic erosion, artificial selection and infestation by pathogens (Volis & Blecher, 2010). *In situ* conservation consists in applying conservation strategies while maintaining the species in their original habitat: it is the best way to conserve the entire genetic diversity of a plant species (Whitlock *et al.*, 2016), thanks to random mating and recombination among the individuals of the reinforced population. *In situ* conservation offers healthy competition among members of the population and ensures a dynamic turnover which favors evolution (Vijayan *et al.*, 2011), with no artificial selection (IUCN/SSC, 2014; Salgotra and Chauhan, 2023). The importance of *in situ* preservation is stated by the International Union for Conservation of Nature (IUCN) itself, which stresses the need for a proper maintenance of diversity in the wild (Whitlock *et al.*, 2016) and suggests *ex situ* conservation as part of a broader strategy and as a support for *in situ* conservation (IUCN/SSC, 2014). When *in situ* preservation is impracticable due to invasive species and habitat degradation (Hoban & Schlarbaum, 2014) a complementary *ex-situ* approach (Salgotra & Chauhan, 2023) or proper management in parks and protected areas can help overcome these problems. Unfortunately, many institutions face constraints due to political factors which obstruct conservation measures (Cosma *et al.*, 2023). Additionally, differently from *ex situ* (IUCN/SSC, 2014), IUCN does not provide specific guidance for *in-situ* preservation, especially for wild species with low potential utility value (e.g. species not related to crop plants; Whitlock *et al.*, 2016), which further complicates conservation planning and efforts. Nevertheless, this kind of conservation requires areas to be protected from human activity (Supple & Shapiro, 2018), based on research and constant monitoring activities. Since genetic diversity is essential for survival of both crop (Reddy *et al.*, 2018) and wild plants, its assessment should be taken as the first step towards conservation programs.

1.2 The genomic era of conservation

When traditional *ex-situ* methods for genetic resources monitoring and conservation are not feasible or sufficient, plant biotechnology provides tools to mitigate negative effects, especially on crop and model plants (Yin *et al.*, 2024) but also on rare, ornamental and medicinal species (Salgotra & Chauhan, 2023). These applications include, but are not limited to, genome editing (Supple & Shapiro, 2018) and recombinant technologies (Salgotra & Chauhan, 2023), proven advantageous in comparison to conventional mutagenesis methods for traits improvement, but are also controversial due to their generation of transgenic plants, recognized as GMOs (Yin *et al.*, 2024). Planning any improvement or conservation program on plants needs prior understanding of genetic diversity (Kanaka *et al.*, 2023): it requires consultation from an interdisciplinary team, including of biotechnologists, ecologists and bioinformaticians (Yin *et al.*, 2024). The application of bioinformatics for biodiversity conservation has enhanced researches ability to intervene on the earlier steps of conservation plans. Genomics, transcriptomics and epigenomics techniques are often used in population genetics studies (Yin *et al.*, 2024), and their findings have been more and more often used for conservation research (Theissinger *et al.*, 2023) and interventions planning (Habel & Schmitt, 2012) since they are able to detect wild population dynamics and minimize negative ecological effects (Yin *et al.*, 2024). Genomics technologies can, in fact, estimate genetic variation (Whitlock *et al.*, 2016), infer molecular phylogeny (Unamba *et al.*, 2015) and model species' demographic history (Yin *et al.*, 2024) and, through next-generation sequencing (NGS), it is now possible to both identify at risk-populations and predict, with high resolution, consequences of any conservation action (Supple & Shapiro, 2018): by identifying their weaknesses, genomics approaches can greatly optimize conservation strategies (Theissinger *et al.*, 2023). Any type of genomics study, if properly organized to address limitations, can be applied to model as well as non-model systems, guiding both researchers and institutions choices. Genetic and genomics approaches' applications, though, heavily depend on quantity and quality of materials, laboratory and bioinformatics skills and, lastly, on feasibility and costs (Theissinger *et al.*, 2023).

1.2.1 Challenges with non-model plants

All plant genetics knowledge we possess was built on studies of model plant species, which include *A. thaliana*, *Oryza sativa* and *Zea mays*, for which a reference assembly has been available as early as the first years of the 2000s (Unamba *et al.*, 2015): given the wide range of information that a reference genome provides (He *et al.*, 2014), many other similar plants and species important for the food economy have been thoroughly studied at the genomic level. Any genetic study, ranging from breeding interventions preliminary investigations to phylogenetic analyses, has then relied on molecular markers (He *et al.*, 2014). Although DNA-based techniques have been recognized for their high resolution also in both wild rare plants and crop improvement programs (Salgotra & Chauhan, 2023), non-model organisms often do not have an available reference genome for variant calling (Supple & Shapiro, 2018). The situation is improving for crop plants (Unamba *et al.*, 2015), and not for less abundant and threatened plants, which should instead be prioritized in conservation efforts due to their lower adaptability potential (Yin *et al.*, 2024). Although in 2023 plants accounted for 63% of the vulnerable and endangered species in the IUCN Red List (Kanaka *et al.*, 2023), reference genome-based analyses are being carried mostly on endangered animals like the Atlantic cod (*Gadus morhua*) and the Iberian lynx (*Lynx pardinus*) (Theissinger *et al.*, 2023); Yin *et al.*, in 2024, estimated that there are assemblies for only 0.37% of the total number of plant species included in the list. A reference genome is often desirable in biodiversity projects (Theissinger *et al.*, 2023) as it can help the analyses by simplifying them but, fortunately, it is not strictly essential for many types of studies (He *et al.*, 2014); furthermore, while GWAS and other association studies require 100s to millions of markers to generate sufficient information and coverage (He *et al.*, 2014), phylogenetic and population structure studies often used in conservation plans instead only require several hundred or few thousand of SNP-containing loci (K. R. Andrews *et al.*, 2016), which lowers complexity of applications. Even if some degree of trade-off is possible, both crop and rare plants still face difficulties: non model plants often have poly-ploid genomes (even hexaploid like oats; He *et al.*, 2014) which raise the complexity of genetic studies because they require higher coverage to appropriately asses genotypes (K. R. Andrews *et al.*, 2016), or their genomes have high duplication levels and are full of repetitive sequences, that have to be dealt with accordingly (Unamba *et al.*, 2015).

1.2.2 Reduced representation techniques

Many population genetics studies on non-model organisms employ different molecular markers techniques: in both animals and plants, researches have used techniques like amplified fragment length polymorphisms (AFLPs) (Whitlock *et al.*, 2016; Ilves *et al.*, 2016), sequence tags (ESTs) (Supple & Shapiro, 2018), restriction fragment length polymorphisms (RFLPs) (He *et al.*, 2014) and microsatellities (Ægisdóttir *et al.*, 2009; Frei *et al.*, 2012; Mandolino *et al.*, 2015). The use of microsatellites drastically decreased, between 2010 and 2020, from 81% to 18% of genotyping methods used (Kanaka *et al.*, 2023), in favor of SNP arrays: they cover a much larger portion of the genome compared to microsatellites and are, for example, currently largely preferred in crop improvements programs (Salgotra & Chauhan, 2023). SNP-based maker techniques allow for improved marker density (He *et al.*, 2014) and, using them on a whole-genome scale, lead to more accurate estimates of genetic diversity and population structure (Supple & Shapiro, 2018), necessary for conservation planning, specifically when selecting populations that are genetically distinct and diverse at the same time (Yin *et al.*, 2024). To study a suitable amount of SNPs for such studies, it is necessary to employ Next-generation or high-throughput sequencing (NGS) (Sonah *et al.*, 2013): all NGS strategies comprehend the same steps, where universal adapters are ligated to DNA fragments, which are in turn immobilized on a synthetic surface, iteratively cloned (amplified) and sequenced using signal emission after the incorporation of each base (He *et al.*, 2014). The quality of the data generated depends on the techniques (i.e. Illumina or Oxford Nanopore) and costs keeps decreasing (He *et al.*, 2014; Yin *et al.*, 2024). Unfortunately, most studies and projects focused on biodiversity conservation have limited resources (Supple & Shapiro, 2018), due to shortcomings of governments which hardly distribute biodiversity conservation funding (Cosma *et al.*, 2023). This causes limitations on the number of samples that can be sequenced in their entirety. To tackle the economic difficulty of whole-genome sequencing, RAD-seq (Restriction-Site Associated DNA Sequencing) is often used for non-model plants (Theissinger *et al.*, 2023): it is an easy and quick approach that reduces genome complexity by restriction enzymes-based fragmentation (Elshire *et al.*, 2011) and avoids repetitive regions of genomes (Elshire *et al.*, 2011). It is particularly useful for genotyping when a reference genome is not available (Elshire *et al.*, 2011; He *et al.*, 2014), thanks to its focus on informative portions of the genome with no need to know their exact sequence: for this reason it is commonly used for studies in ecological, evolutionary and conservation genomics of non-model organisms (K. R. Andrews *et al.*, 2016). Sequencing only some portions of the DNA, which may vary from a cut every 4,096 bp to every 65,536 bp, depending on the restriction enzyme used (K. R. Andrews *et al.*, 2016), limits the cost of a whole genome approach, while reaching the high resolution which microsatellites analysis often do not possess (Supple & Shapiro, 2018). GBS (Genotype by Sequencing) is an even lower cost RAD-Seq technique (Elshire *et al.*, 2011;He *et al.*, 2014; Islam *et al.*, 2021) which possesses enhanced resolution in comparison to the use of microsatellites (Martina *et al.*, 2025) and it is one of the RADseq methods able to provide the biggest amount of identified loci (K. R. Andrews *et al.*, 2016). It has been used in makers-based studies in crop plants like maize (Elshire *et al.*, 2011), soybean (Sonah *et al.*, 2013), potato (Martina *et al.*, 2025), lettuce (He *et al.*, 2014), olives (Islam *et al.*, 2021), and can be used in a *de novo* setting too, given proper parameters tuning (K. R. Andrews *et al.*, 2016). It differs from the other protocols of RAD-seq because it is less complicated: it requires a single-well digestion and ligation which reduces sample handling and errors associated with it (He *et al.*, 2014) and, additionally, performs indirect size selection (Elshire *et al.*, 2011), lowering the amount of DNA needed (Sonah *et al.*, 2013). The low cost of this method is not its only advantage: in this study, GBS was performed (i) using enzyme ApeKI which has up to millions of restriction sites in the genome and (ii) with paired-end sequencing, which improves genotyping accuracy towards the ends of the fragments (K. R. Andrews *et al.*, 2016).

1.3 Objective of the study

In our study, we implemented genomics analyses on GBS data in three different projects, spanning some of the critical issues in plant conservation we've introduced: we performed a population genetics study on a rare plant living in a fragmented habitat (*Campanula thyrsoides* subspecies *carniolica*), a phylogenetic reconstruction on the domestication history of Italian landraces of a largely studied crop plant (*S. tuberosum*) and investigated population structure of a widespread plant subject to human activity pressure (*Anacamptis pyramidalis*). Even if all projects aim to answer different research questions, they show how genomics, more specifically based on a low-cost reduced representation technique, can be exploited for biodiversity conservation purposes, both in the presence or absence of a reference genome.

1.3.1 Main project: *Campanula thyrsoides* subspecies *carniolica* conservation

This study aims to support the National Park of Dolomiti Bellunesi (Veneto, Italy) in its goal of preserving a subspecies of an alpine species present in its protected area, *Campanula thyrsoides* subspecies *carniolica*, for which a reference genome is currently not available. Studies of *C. thyrsoides* identify the species' distribution (Ægisdóttir *et al.*, 2009) on the Alpine mountains, with subspecies *thyrsoides* found on the entirety of the Alps (Italy), the adjacent Jura Mountains (France and Switzerland) and north-eastern Austrian Alps (Scheepens *et al.*, 2011), while subspecies *carniolica*'s habitat in Italy is limited to the north-eastern regions of Veneto and Friuli Venezia Giulia (Conti *et al.*, 2005) and, outside of Italy, on the Dinaric Arc of the Balkan Peninsula (Kuss *et al.*, 2007). Small and isolated populations suffering from genetic drift and/or inbreeding are usually a sign of an endangered species (Volis & Blecher, 2010): following this definition, *C. thyrsoides* is currently not part of the Italian IUCN red list of endangered species (Graziano Rossi *et al.*, 2020) since historically speaking its populations had been rare but locally abundant (Ægisdóttir *et al.*, 2009). The subspecies *carniolica* is instead in the IUCN red list with a "Least Concern" label, but we would argue that not only literature information about this species is very limited, but also its populations are very small and isolated (< 20 individuals per locality, personal communication by S. Orsenigo, researcher in the project) which should raise a red flag. The necessity of maintaining a high population genetic diversity can be a strong burden for species with small and isolated populations (Habel & Schmitt, 2012): very rare plant species should rightfully have priority in conservation plans, but *C. thyrsoides* may be more vulnerable to population loss due to local inbreeding caused by lack of connectivity. The goal of the collaboration between the National Park of Dolomiti Bellunesi, the University of Milan and the University of Pavia is to assess the status of the small *Campanula thyrsoides* subspecies carniolica population of Serva mountain, located in the area of the National Park. Our questions were: (i) whether this population, consisting of very few individuals, risked loss of genetic diversity because of isolation and inbreeding and (ii) which other population was closely-related enough to be used for genetic reinforcing (i.e. the translocation of compatible individuals from nearby populations that could reactivate gene flow without changing genetic identity).

1.3.2 Secondary project 1: Reconstruction of *Solanum tuberosum* landraces farming history

In this project we used GBS to reconstruct the domestication history of *Solanum tuberosum* L. (common potato) landraces of the region of Lombardy in Italy, comparing them with commercial cultivars and ancient/wild species of the same genus. Recent studies have analyzed Italian local landraces, but either Lombardy landraces were not the focus of these studies (Martina *et al.*, 2025; Parri *et al.*, 2025) or the research question was not in their evolutionary history (Parri *et al.*, 2025). This analysis is part of the project La Rava e la Fava, financed by the Rural Development Program (Progetto Sviluppo Rurale) financed by Lombardy region and which focuses on the conservation and promotion of the cultivation of landraces. The aim of the whole project La Rava e la Fava is to morphologically and genetically assess at least 17 local landraces, enhance *ex situ* conservation and to reintroduce these landraces to local farms. Preserving genetic diversity in the form of landraces and obsolete cultivars[1], serves as a source of alleles for developing new varieties (Reddy *et al.*, 2018): in fact, the program is particularly interested in minor

[1] Cultivars that have undergone selection in the XX century and which has been largely replaced by modern, comercial cultivars (Vegini *et al.*, 2022).

local cultivars, defined as such because they are no longer cultivated and are marginal for crop production. This projects counteracts the expansion of monocolture cropping, that replaced local landraces of many crop varieties since the 20th century (Salgotra & Chauhan, 2023). In this thesis we study obsolete cultivars, for example the obsolete cultivar (Vegini *et al.*, 2022) of Bossico (BG), which is better adapted to the climatic area of the North of Italy, although less cultivated than commercial varieties[2]. This study isn't focused on populations genetics; instead, its aim is to determine if landraces of *Solanum tuberosum* of hypothesized local origin are more closely related to ancient (here tested with Andean species and wild *Solanum* spp.) or are genetically more similar to commercial cultivars. To answer this research question we analyzed samples from the seed bank of University of Pavia, collected in collaboration with public institutions (Botanical garden of Bergamo) and several farms.

1.3.3 Secondary project 2: Urban and natural populations of *Anacamptis pyramidalis* and *Anacamptis berica*, Orchidaceae

This analysis is part of a more detailed project carried out by PhD candidate Lisa Scramoncin of the University of Ferrara, financed by PNRR. We refer to her work for more information about the project's main goals, which can be summarized as investigating how plant populations of *Anacamptis pyramidalis* have colonized urban habitats and evolved in the new conditions, i.e. whether urban populations are genetically different from those in their natural habitat. *Anacamptis pyramidalis* is a non-endangered orchid currently listed as "Least Concern" in Europe due to its wide distribution (Fay *et al.*, 2024), with no reference genome available. Several populations of *A. pyramidalis* and outgroup *Anacamptis berica* were included in the analysis: these populations were sampled from a big variety of habitats with different disturbance intensity, from natural environments like chalk grasslands (Alec M. Pridgeon *et al.*, 2001) to even road verges (Fay *et al.*, 2024). This study investigates the structure of urban and wild populations of *A. pyramidalis* in Central Italy: the question asked was whether there is evidence of genetic differentiation between the two types or not and and whether conservation actions shall be taken.

[2]More information about the specifics of the project and each variety description, can be found in BSc graduate Giulia Ballarotti's bachelor thesis in Natural Sciences (Ballarotti, 2025).

2. Materials and Methods

2.1 *Campanula thyrsoides*

Campanula thyrsoides is a monocarpic, diploid ($2n = 34$; Ægisdóttir *et al.*, 2009), yellow-flowering bell flower occurring in the European Alps (Scheepens *et al.*, 2011) found on limestone (Ægisdóttir *et al.*, 2009). This species is long-lived with a mean flowering age of 10 years (minimum of 3 to a maximum of 16 years; Frei *et al.*, 2012). It possesses a gametophytic self-incompatible system, common in the genus (Ægisdóttir *et al.*, 2007): this means that flowers can be pollinated from plants of the same seed family (half-siblings) but the plant's own anthers wither before its stigma becomes receptive, so the temporal separation of maturation makes self-breeding close to impossible, even with hand pollination of flowers on the same plant (Ægisdóttir *et al.*, 2007). It includes two subspecies: *Campanula thyrsoides* subspecies *thyrsoides* (the most common (Kuss *et al.*, 2011) and strongly adapted to the higher altitudes of the West Alps; Scheepens *et al.*, 2011) and *Campanula thyrsoides* subspecies *carniolica* which are both morphologically (Fig. 2.1) and geographically distinct (Scheepens *et al.*, 2011). The two subspecies differ mainly with respect to inflorescence height and spike density (Kuss *et al.*, 2011) but they also shave different phenology: subspecies *thyrsoides* flowers from top to bottom of the stem, in July and August while *carniolica* from bottom to top, in the first half of August, so they partly overlap in flowering time (Scheepens *et al.*, 2011). The mismatch is probably caused by the different altitude at which they can be found: subspecies *thyrsoides* is predominantly present between 1,600 to 2,200 m a.s.l., while *carniolica* occurs at submontane to montane elevations (400–1,800 m a.s.l.; Scheepens *et al.*, 2011): the different altitudinal requirements, combined with reproductive isolation due to population fragmentation (Coyne &

Orr, 2004), is a key driver of subpopulations differentiation and, ultimately, of speciation. The species' pollinators are predominantly bumblebees and other Hymenoptera, and the seeds are dispersed by wind (Ægisdóttir *et al.*, 2009), with subspecies *carniolica* dispersing over a larger range due to its taller inflorescence (Scheepens *et al.*, 2011).

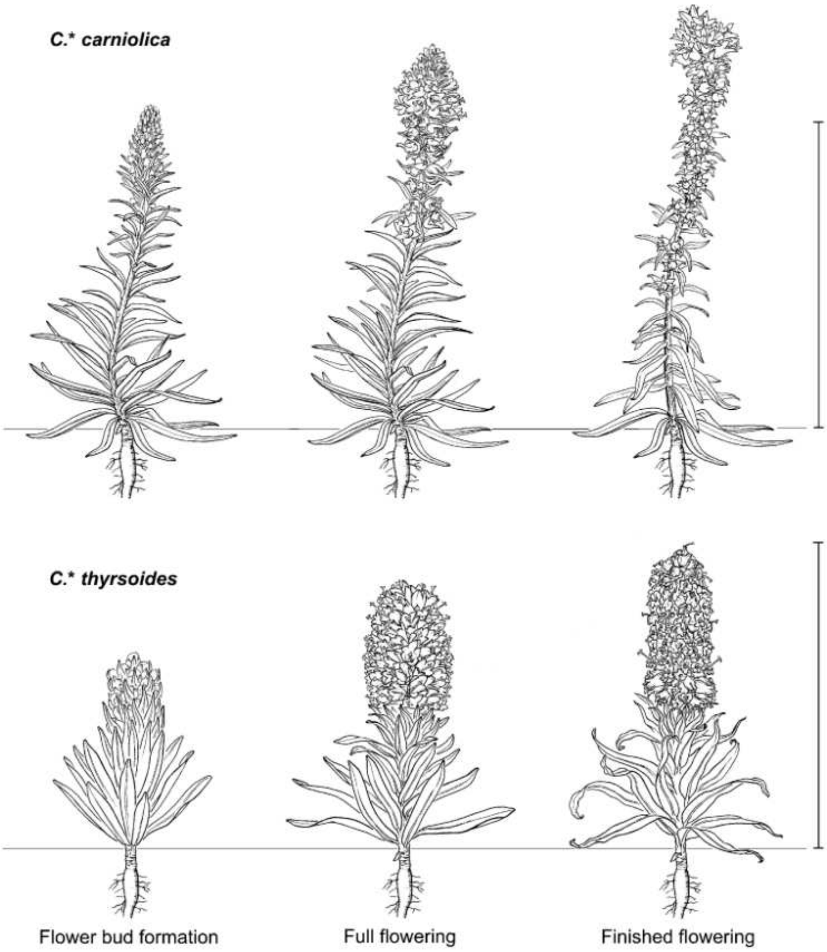


Figure 2.1: ***Morphological difference between carniolica and thyrsoides.*** *The difference between the two subspecies can be mainly attributed to inflorescence height: carniolica is usually taller (top). Other differences, as lower Specific leaf area and higher leaf thickness can be attributed to carniolica adaptation to drought in the submediterranean climate. Drawing by Atelier Guido Köhler, Basel (Scheepens* et al.*, 2011).*

Suitable habitats in the Alps are shrinking as a consequence of pasture abandonment or overgrazing (Frei *et al.*, 2012): moreover, *C. thyrsoides*' self incompatibility system, which is rather uncommon in Alpine plants with fragmented populations (Ægisdóttir *et al.*, 2007), makes the species even more sensitive to the increasing isolation within population. In Germany, *C. thyrsoides* is protected at the national level, in Austria and Switzerland at regional level (Kuss *et al.*, 2011): in Italy it is protected at the regional level in Lombardy (Regione Lombardia, 2008) and in Valle d'Aosta (Regione Autonoma Valle d'Aosta, 2009) where collection is strictly forbidden. Despite subspecies *carniolica* belonging to the "Least Concern" category in the Italian IUCN red list of endangered species (Graziano Rossi *et al.*, 2020), it is categorized as "Near Threatened" in the region of Veneto (Buffa, 2016). While prior research that had been conducted on the genome size of other European diploid species of *Campanula* with the same number of chromosomes (*C. garganica, C. latifolia, C. medium, C. portenschlagiana, C. poscharskyana* and *C. pyramidalis*; Zonneveld, 2019) reported an average DNA weight of $3pg$, *C. thyrsoides* subsp. *carniolica* shows a bigger genome (2n = 2x with x = 13, 2C = 4.944 ± 0.021 pg; Fainelli, 2025). DNA weight can be roughly converted to the number of base pairs, where $1pg = 978Mbp$ (Doležel *et al.*, 2003) so we could assume *C. thyrsoides* subsp. *carniolica* having diploid genome roughly $4835.2Mb = 4.83Gb$ long, so as much as 76.4% of the total human diploid genome (averaging 6.32 Gb; Piovesan *et al.*, 2019).

2.1.1 Sampling and conservation

137 fully grown individuals of *Campanula thyrsoides* subsp. *carniolica* were sampled, collecting 3-4 leaves from each plant between July and August 2022 from 9 locations spanning the North of Italy, South of Austria and West of Slovenia, while the outgrup formed by individuals (n = 16) of *Campanula thyrsoides* subspecies *thyrsoides* were sampled from 2 populations in Italy and France (Figure 2.2).

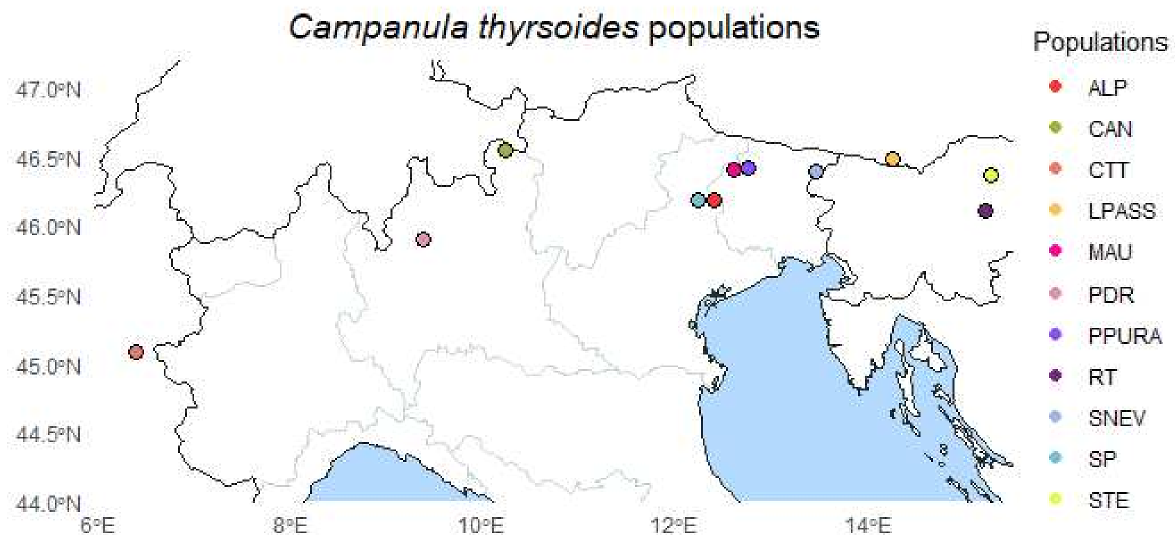


Figure 2.2: ***Map of sampled populations.*** *Populations location over the political map of Italy, France, Slovenia and Austria.*

In table 2.1 are reported more information about the location, coordinates and number of samples collected for each population. A minimum of 15 different individuals sampled per population was our target sample size, but some populations (i.e. CAN and MAU) weren't big enough. The leaves were stored in bags full of silica to dry, and stored at room temperature (i.e. between 20°C and 25°C) until DNA extraction was performed in 2024.

Table 2.1: *Populations' sampling information.*

Pop	Locality	Taxon	Region	State	Coordinates	Altitude	Date	N
CAN	Cancano-Valdidendro Lake	*C. thyrsoides* subsp. thyrsoides	Lombardy	Italy	N 46.5444991 E 10.2547278	1960 m	30/08/22	13
CTT	Col du Télégraph	*C. thyrsoides* subsp. thyrsoides	Auvergne-Rhône-Alpes	France	N 45.0879 E 6.4327	2044 m	30/07/24	3
ALP	Alpago	*C. thyrsoides* subsp. carniolica	Veneto	Italy	N 46.192300 E 12.4156899	1140 m	28/08/22	15
MAU	Mauria pass	*C. thyrsoides* subsp. carniolica	Veneto	Italy	N 46.4035951 E 12.6109512	860 m	28/08/22	11
PPURA	Pura pass	*C. thyrsoides* subsp. carniolica	Friuli-Venezia Giulia	Italy	N 46.422130 E 12.752440	1320 m	02/08/22	15
SNEV	Sella Nevea-Chiusaforte	*C. thyrsoides* subsp. carniolica	Friuli-Venezia Giulia	Italy	N 46.3909564 E 13.469300	1115 m	02/08/22	15
RT	Gračnica-Rimske Toplice	*C. thyrsoides* subsp. carniolica	Savinja Statistical Region	Slovenia	N 46.1111551 E 15.225581	245 m	01/08/22	16
LPASS	Loiblpass	*C. thyrsoides* subsp. carniolica	Carinzia	Austria	N 46.483864 E 14.261965	705 m	02/08/22	16
SP	Serva mountain	*C. thyrsoides* subsp. carniolica	Veneto	Italy	N 46.191773 E 12.243491	1590 m	03/08/22	19
STE	Stenica-Vitanje	*C. thyrsoides* subsp. carniolica	Slovenske Konjice	Slovenia	N 46.366841 E 15.281769	420 m	01/08/22	15
PDR	Pian dei Resinelli	*C. thyrsoides* subsp. carniolica	Lombardy	Italy	N 45.906065 E 9.394220	1290 m	28/07/22	15

Information about populations' sampling, carried out in 2022.

2.1.2 Wet lab protocol

DNA extraction

A total of 153 samples' DNA was extracted following the SILEX method (CTAB-Silica) illustrated in chapter 8. After the extraction, the presence of DNA in each sample was checked on agarose gel 0.8%. The electrophoresis set up was 90V for 45 minutes. Figure 2.3 shows DNA quality assessed through electrophoresis: samples RT12, RT13, SP4AI15, SP4B16 showed lower quality and were excluded from the study. Similarly, we could extract high quality DNA (necessary for NGS sequencing) from 106 samples out of 153.

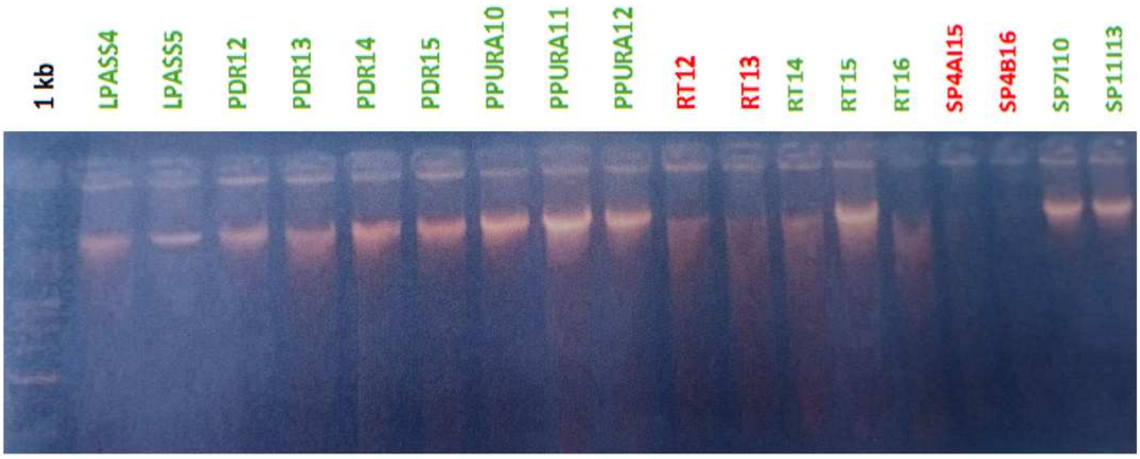


Figure 2.3: ***Electrophoresis gel of 18 samples.*** *In green, the samples that show a strong band indicating DNA presence, while in red those samples that show presence of fragmented DNA. In black, the ladder used as reference.*

Digestion

After the assessment of the presence of DNA, the samples went through digestion which ensures that the DNA is fragmented in pieces of various length, but homogeneous in their distribution. We used a methylation-sensitive endonuclease, ApeKI, that does not cut the sites in the DNA where cytosine is methylated or regions full of repetitive sequences (Sonah *et al.*, 2013). ApeKI also cuts DNA in precise restriction sites, leaving so called 'sticky ends' at the extremes of the fragments (Figure 2.4). The fragments will end with the restriction site sequence (CWG, where W is either an A or a T) that is where the Illumina adapters will be ligated.

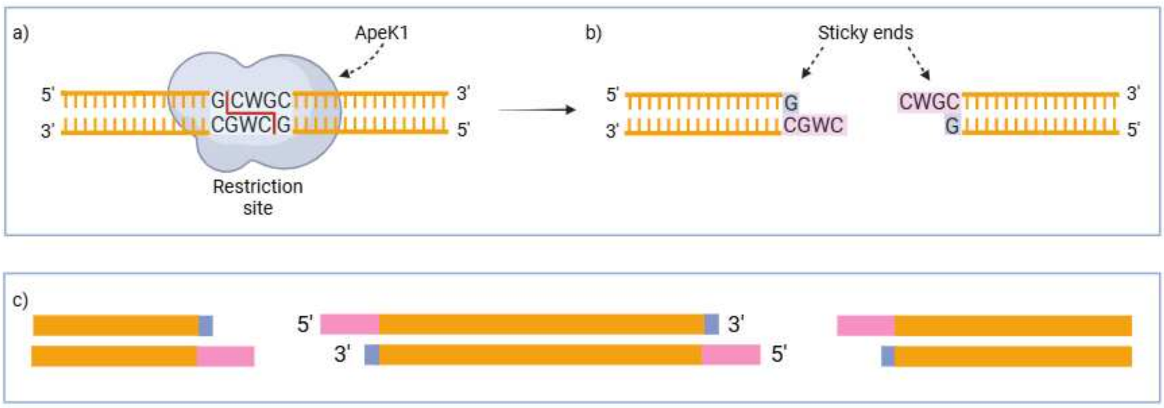


Figure 2.4: ***ApeKI restriction site and action.*** *a) ApeKI restriction site CWG. b) Creation of sticky ends on DNA fragments. c) Several fragments of dsDNA. Created in https://BioRender.com*

The same amount of DNA (50 ng) was digested for each sample to ensure a similar number of fragments. Samples were diluted with ddH_2O accordingly to their concentration, until desired volume ($44\mu L$), to which $2\mu L$ of ApeKI and $4\mu L$ of NEB r3.1 10x buffer are added. More information about the protocols used for digestion and adapters preparation can be found in chapter 8. For *Campanula thyrsoides* the digestion time was increased because the DNA was hard to digest, probably because of the presence of latex in the leaves of this species (Kuss *et al.*, 2007). The digestion protocol used is illustrated in the Additional Materials section. At this step the whole population of Cancano (CAN) and 34 other individuals (refer to Table 8.1) had to be scrapped because of DNA absence from the samples.

Ligation

During ligation, the DNA fragments are ligated with adapters that can either be indexed or not, depending on the sequencing method employed. We used double indexed adapters, necessary for Illumina TruSeq combinatorial-dual DNA sequencing: in Figure 2.5 the barcodes (4 to 8 bp long) present only on the forward read are highlighted in blue, the indexes in sky blue and the Truseq adapters are shown in red/orange: index sequences (i5 and i7) are composed of specific 6–8 nucleotides to distinguish one sample from another and enable multiple sequencing on a common flow cell. Both the barcodes and common adapters are present in the same solution (called stock solution) prepared before ligation. More information about the preparation of the stock solution and the ligation protocol can be found on Additional Materials, chapter 8.

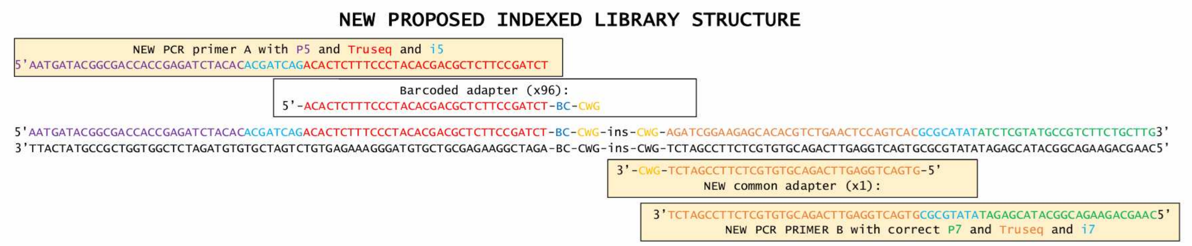
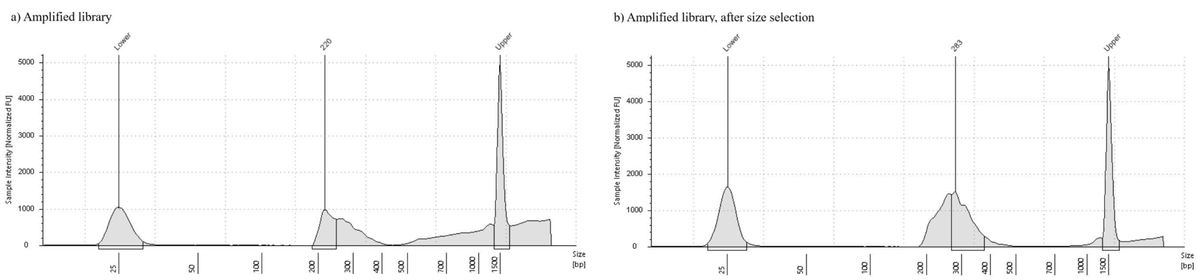


Figure 2.5: ***Library structure.*** *In blue the barcode (labelled as "BC") which univocally identifies each sample: the barcode is only ligated to the forward read and the complete list can be found in Table 8.2. In red and orange Truseq adapters (sequencing primers), and in light blue the i5 and i7 Illumina library indexes. In purple and green the P5 and P7 adapters, that facilitate clonal amplification and sequencing reactions. Lastly, in yellow, the restriction sites where ApeKI cuts the insert, here labelled as "ins".*

Pooling and amplification

After ligation fragments were amplified to produce the sequencing library (the full protocol can be found in the Additional Materials chapter). PCR amplification requires that forward P5 primer and reverse P7 primer (in Fig. 2.5 in purple) are also attached, respectively on the 5' and 3' end of the reads: these sequences allow the fragments to be attached to the flow cell. After this step the library was quantified with Qubit station (we obtained a concentration of 60 $ng/\mu L$) and the fragments size was checked with Tapestation. GBS protocols usually do not involve a direct size selection prior to PCR (Elshire *et al.*, 2011; Sonah *et al.*, 2013), libraries are only checked to ensure that the majority of the fragments are in the correct size range. To improve our library that still had several long fragments, we used Sera-Mag Select Size Selection kit from cytiva, employing Dual-Side size selection between 150 to 500 bp, with an average size of 250 bp: this paid off, and Tapestation showed only one peak between 200 and 500 bp of length (Figure 2.6). Size selection's protocol is reported in the appropriate section in Additional Materials.



Size [bp]	Calibrated Conc. [ng/µl]	Assigned Conc. [ng/µl]	Peak Molarity [nmol/l]
25	4.98	-	307
220	3.01	-	21.0
1500	6.50	6.50	6.67

Size [bp]	Calibrated Conc. [ng/µl]	Assigned Conc. [ng/µl]	Peak Molarity [nmol/l]
25	6.57	-	404
283	6.22	-	33.8
1500	6.50	6.50	6.67

Figure 2.6: ***Library before and after size selection.*** *Reads length distribution in the library before (a) and after (b) size selection: on the y axis the sample intensity, and on the x axis the fragments length in base pairs. A table with peaks' concentration and molarity information is also added on the bottom of each plot.*

2.1.3 DNA sequencing

Initially, a subset of 10 test-samples (ALP1-ALP11) was sent to GENEWIZ from Azenta Life Sciences (Leipzig, Germany) for sequencing in January 2025, then the remaining 96 were sent in April 2025: standard Illumina TruSeq combinatorial-dual DNA sequencing was used for both sequencing batches and sequences' data were received in `.fastq` format shortly after submission.

2.1.4 The bioinformatics pipeline

All the projects mentioned in this thesis followed the same pipeline (Figure 2.7), which was built on the objective of the main project, namely the reconstruction of *Campanula thyrsoides* subspecies *carniolica* populations' structure. The secondary projects differ from it by just the choice of starting and/or ending step, as well as some of the parameters used. These discrepancies are thoroughly explained in each project's section.

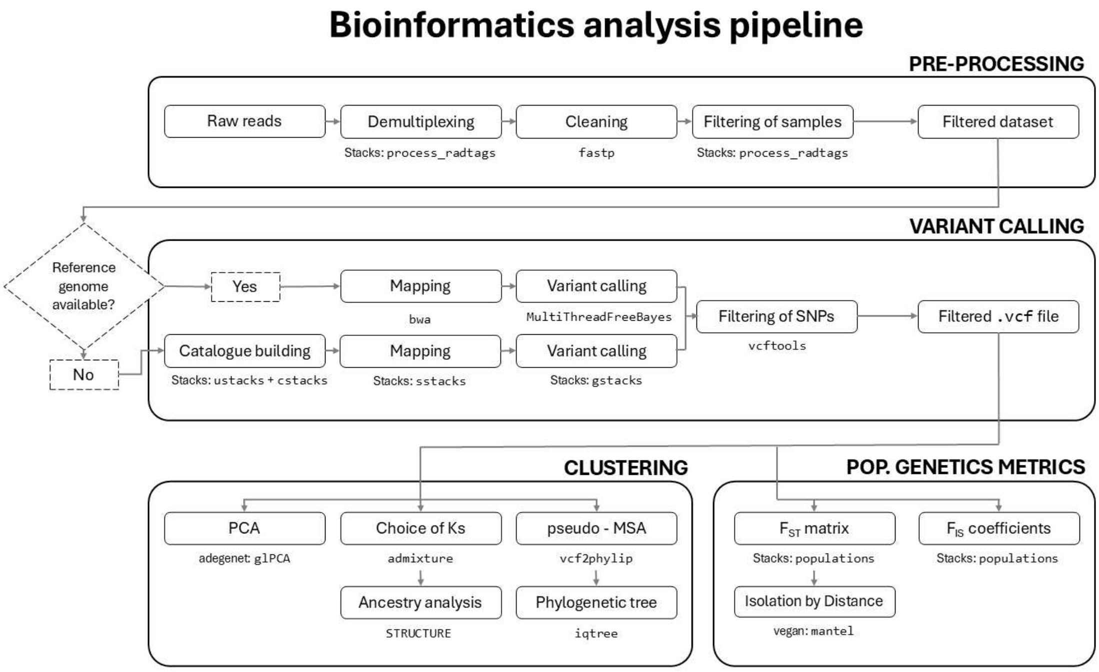


Figure 2.7: ***Bioinformatics analysis' pipeline.*** *Full pipeline followed in all the projects, except for a few adjustments in parameters chosen. The function used at each step is shown undereneath it, as well as the software or R library it belongs to, if the name differs.*

2.1.5 Pre-processing

Individuals from ALP1 to ALP11 were sequenced in a different batch than the other samples. Considering the time that passed before receiving the second batch of data, samples ALP1 to

ALP11 were used as a subset to test some of the tools employed in the pipeline, and underwent pre-processing separately from the second batch (ALP12 onward).

Demultiplexing and quality checks

Demultiplexing, namely the process of separating the DNA sequences depending on the samples they come from, is based on the barcode attached to it during the ligation step (section 2.1.2). For this step, a custom script was written (`01_demultiplex.sh`) which runs the command `process_radtags` from software `Stacks` (version: 2.68; Catchen *et al.*, 2011) and collects all the unpaired sequences in one folder. It takes as input a compressed file in `fastq` format which contains all the sequences, the name of the restriction enzyme used (ApeKI) and finally a text file containing sample and barcode couples. This step not only separates the sequences but also removes the barcode from the forward read; the full list of samples accompanied by their barcodes (table 8.2) can be found in the Additional Materials chapter.

Trimming using fastq-mcf and fastp

After separating them, all the sequences were cleaned, a loose term which encompasses several operations:

- Eliminate the Illumina adapters contamination;
- Trim low quality scores sequences sections. Quality is denoted by Phred-score (q), calculated as (Ewing & Green, 1998):

$$q = -10 \cdot log_{10}(p) \tag{2.1}$$

  where p is the estimated error probability for that base-call. Phred quality scores that are less than 30 are considered unreliable (O'Rawe *et al.*, 2015);
- Trim poly-G tails. These artifacts of repeated, high-confidence Guanine bases appear on two-channel sequencing systems when the signal is too weak to be detected, and is erroneously recognized as a G in the base calling stage;

- Finally, after trimming, discard the sequences that are shorter than 50 base-pairs.

Two different softwares were tested on the ALP1-ALP11 subset to determine the most efficient tool: `fastq-mcf` (v1.05; Aronesty, 2015, August 3/2011) and `fastp` (v0.24.0; Chen *et al.*, 2018). The performance was evaluated in terms of computational time, quality of cleaning and number of reads retained on average. Ultimately, the latter was chosen as the cleaning tool for the second batch of samples too and used in a custom script (`02_trim_fastp.sh`), which automatizes the process by iterating the command through all the samples. `fastp` also requires a `fasta` file which contains the Illumina adapters sequences to remove from within the reads: the list also included sequences of adapters containing the barcodes used during the demultiplexing step, as it was noticed that some reverse reads still contained them after cleaning.

2.1.6 Variant calling

Variant calling, which aims to identify single-nucleotide polymorphisms (SNPs) in the sample's DNA, is usually performed by aligning to a reference genome. To account for the fact that *Campanula thyrsoides* lacks a reference genome for its nuclear DNA, the pipeline `denovo_map.pl` from `Stacks` was employed: such tool is able to identify loci *de novo* and calls genotypes using a maximum likelihood statistical model (Catchen *et al.*, 2011).

`Stacks`: `denovo_map.pl` pipeline

`denovo_map.pl` is a modular pipeline that runs six different components of `Stacks`: in order, it calls `ustacks`, `cstacks`, `sstacks`, `tsv2bam`, `gstacks` and lastly `population` (Figure 2.8). The pipeline takes as input a dataset of demultiplexed and cleaned reads, a population map and the following parameters:

m: in `ustacks`, the minimum number of reads to create a stack (default 3, figure 2.8, A to B);

M: in `ustacks`, the maximum distance, in nucleotides, allowed between stacks (default 2, figure 2.8, C to D);

n: in `cstacks`, the number of mismatches allowed between stacks of different individuals to be merged into one (default is set equal to **M**, figure 2.8, F).

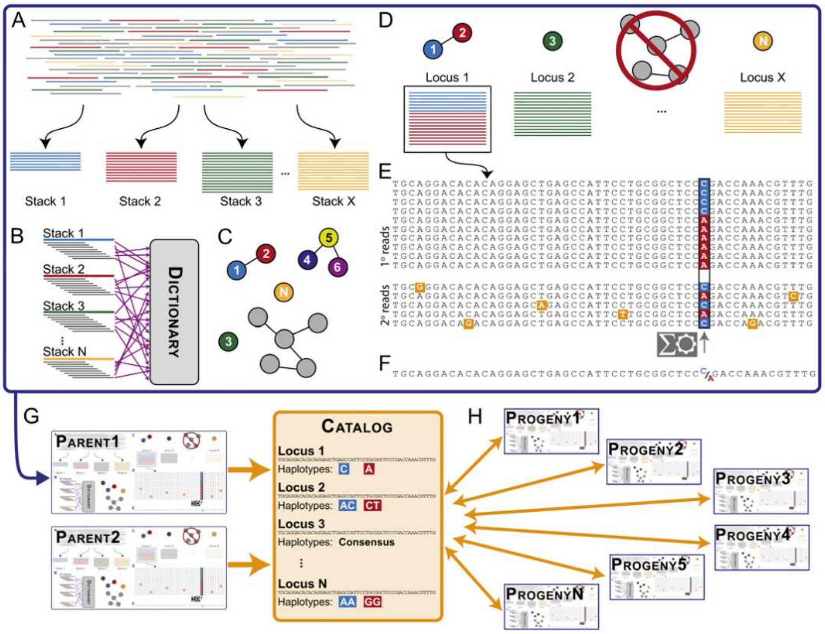


Figure 2.8: ***Stacks:*** `denovo_map.pl` ***pipeline.*** *Full view of the Stacks:* `denovo_map.pl` *pipeline from Catchen* et al. *(2011). It starts with (A)* `ustacks`*, which, for each individual, forms stacks of minimum m reads that match exactly, (B) breaks down each stack into k-mers and stores them in a dictionary (C). Then matches secondary reads (D) based on a certain number M of mismatches to increases stack depth. Each locus at each nucleotide position is checked for polymorphisms and a consensus sequence is called (E). (F)* `cstacks` *then loads all stacks into a Catalog to create a set of all possible loci, merging those that have at maximum n mismatches between each other. (G)* `sstacks` *matches map cross progeny against the Catalog to determine the haplotypes at each locus in every individual (H).* `populations`*, at the end, computes genetic populations metrics based on the SNPs called and a population map provided.*

The population map is a text file that associates each sample to its population of origin: it is used solely to optimize the number of polymorphic loci found in a certain percentage of the individuals of each population. This allows to discard uninformative loci from the final matrix. Other filtering methods that consider all samples dataset in its entirety are employed in the next steps.

Monte Carlo and Taboo Search for hyperparameters' tuning

To optimize the parameters for `denovo_map.pl`, a variation of the r80 method (Paris *et al.*, 2017) was employed. In the classical r80 method, the parameters are selected depending on which ones maximize the number of polymorphic loci in at least 80% of the individuals of the study, hence the name. The percentage could be changed depending on how strict the user of the pipeline needs to be: for example, small populations usually require more loci to quantify levels of variation (K. R. Andrews *et al.*, 2016). An increase in percentage equals a stricter selection, but Paris *et al.* (2017) found that 80% was the optimal percentage for a good trade-off between quality and quantity of loci found. The change we introduced was that the evaluation of parameters performance was based not only on the number of polymorphic loci, but also on the mean coverage obtained over all the samples, and on the number of unique, independent SNPs (counting only one SNP for each locus to reduce the effect of linkage disequilibrium).

To perform the search of the optimal parameters, performed on the ALP1-ALP11 subset, we first defined a search space for the parameters we wanted to test. The ranges of tested parameters were set, after an initial quick assessment of the proposed ones in Paris *et al.* (2017), as 3 to 5 for m and 0 to 4 for both M and n. Then two custom scripts were prepared: `gen_random_pop.sh` which randomly generates a set of parameters from the ranges defined in the search space, checks if it was already explored (i.e. if they are not present in a `log_combinations.txt` file), calls the `Stacks`: `denovo_map.pl` pipeline and then executes the second custom script `add_to_log.sh`. This last script extracts some information from the log file of the pipeline such as mean and standard deviation of read coverage, the number of total loci, the number of SNPs per locus and adds a FAIL flag to determine if that combination fails some minimum requirements. Only loci that were genotyped in at least 80% of the samples were selected for further analysis ($r = 0.80$), minimizing missing data. Crucially, for this parameters search, the $r = 0.80$ flag was used as to apply the filter to the whole dataset, although in `Stacks`: `populations` SNPs are selected based on the percentage of missing data in each population. In our case, our subset ALP1-ALP11 is part of the same population, so the two filters overlap, but when tuning on a bigger dataset, we recommend using a temporary population map which places all samples in the same population, even if they actually belong to different populations.

This parameters search can be seen as a Taboo restricted Monte Carlo search. A Monte Carlo method is essentially a technique that gathers information about an object (in our case, the parameters' performance on the pipeline) by observing random realizations of it (Kroese *et al.*, 2014): we exploit this method by randomly generating parameters combination and testing them. During each iteration of a simple Taboo search, on the other hand, a subset of *taboo moves* is created out of all the possible ones, and membership to this group is determined by a set of taboo conditions (i.e. a logical relationship; Glover, 1989). In this case a combination enters the subset of taboo moves simply if it was already explored. Any algorithm usually stops after either a chosen number of iteration has elapsed in total or since the solution as explored: we decided to stop after $n = 45$ combinations were explored. Each combination i is the evaluated with a custom-prepared scoring function, that assigns the following score (s):

$$s_i = \frac{\frac{\text{tot SNPs}_i}{\text{tot unique loci}_i}}{\max\limits_n \left(\frac{\text{tot SNPs}_n}{\text{tot unique loci}_n} \right)} + \frac{CV_i}{\max\limits_n (CV_n)} + FAIL_i - \frac{\text{tot unique loci}_i}{\max\limits_n (\text{tot unique loci}_i)} \tag{2.2}$$

Where:

$$FAIL_i = f(x) = \begin{cases} 5, & \text{if unique loci}_i < 2400 \text{ or } \mu_{\text{coverage}_i} < 10 \\ 0, & \text{otherwise} \end{cases} \tag{2.3}$$

The best set of parameters has the minimum score s out of all the explored ones. This equation ensures:

- a **penalty** for each locus that contains more than one SNP (first term), which would mean that they are in linkage. This quantity is normalized over all the explored combinations to not skew the scores;
- a **penalty** for average coverage with a large standard deviation across samples (second term), normalized over all the explored combinations as well. To ensure that an higher average coverage would not be penalized by a predictably big, in absolute values, standard deviation, we did not penalize the standard deviation itself, but instead the coefficient of variation (CV). CV for a combination i is computed as:

$$CV_i = \frac{\sigma_{\text{coverage}_i}}{\mu_{\text{coverage}_i}} \tag{2.4}$$

- a **penalty**, which also meant an instant **fail** thanks to the sum of 5 points to the score, to those combinations which did not reach a minimum of average 10x coverage and/or did not find at least 2400 loci. This third term of the equation ensures that no values lower than these are accepted, because they would not be enough for a strong statistical analysis, since SNPs are filtered on the next steps and are only bound to decrease. These values can be changed on a case-by-case basis: the average coverage could be set higher for a stricter selection (we would not recommend a lower value) while the minimum number of loci found should be raised with the number of samples provided;

- finally, a **reward** to combinations that resulted in an higher number of loci (fourth term), normalized over all the combinations.

This edit in the r80 method ensures a stricter evaluation of parameters aimed at the optimization of different outputs of the pipeline: this aids the statistical strength of the next analyses, while retaining an overall simplicity in its application.

Filtering out SNPs

After performing hyperparameters' tuning on the ALP1-ALP11 subset, we used `denovo_map.pl` with parameters $m = 4$, $M = 1$ and $n = 1$ on the whole dataset. The output of this step was a standard `.vcf` file containing called SNPs and their positions, except for the fact that, since they were called on a catalog of stacks and not on a reference genome, the "chromosome" section showed sequentially increasing numbers according to the numerical ID of the stack. The file was later filtered using `vcftools` (v0.1.16; Danecek *et al.*, 2011) with the following parameters:

- **max-missing 0.9**: While the r80 filter in the `Stacks: denovo_map.pl` ensures that SNPs are retained if present in at least 80% of each populations, `vcftools` instead filters out those with more than 10% of missing data through the whole dataset. This ensures we retain only real polymorphic loci and not those whose information is only the presence or absence of the called SNP;
- **min-meanDP 5**: Includes only sites with average depth values (over the whole dataset) greater than 5, based on the "DP" tag in the `.vcf` file. Such value is enough for a *de novo* assembly combining reads from multiple samples (K. R. Andrews *et al.*, 2016);
- **thin 1000**: retains only one SNP every 1000 base pairs, to ensure independence of SNPs, assumption on which we operate. The more genetic markers in a study, the more of them are required to be independent in order to assign an individual to a certain sample pool with known allele frequencies (a population genetics principle used already in forensic identification; Visscher and Hill, 2009). SNPs that assort independently are said to be unlinked, and have a recombination frequency of 50%, and the higher the frequency of recombination between two markers, the further away they are situated on a chromosome

(Semagn *et al.*, 2006). There are two formulas that can be used to find the map distance d in cM between two loci, depending on their recombination rate R: Haldane's (2.5) and Kosambi's (2.6) (Huehn, 2010).

$$d_H = \begin{cases} -0.50 \cdot \ln\left(1 - 2R\right), & \text{for } R < 0.50 \\ \infty, & \text{for } R = 0.50 \end{cases} \tag{2.5}$$

$$d_K = \begin{cases} 0.25 \cdot \ln\left(\frac{1+2R}{1-2R}\right), & \text{for } R < 0.50 \\ \infty, & \text{for } R = 0.50 \end{cases} \tag{2.6}$$

In theory, these formulas could be used to assess a minimum distance between SNPs to have no linkage or a partial one. When two loci are located on the same chromosome they can never be truly unlinked, since these functions possess a vertical asymptote at $R = 0.5$ (Figure 2.9). Only two loci located on different chromosomes are truly independent since they cannot physically recombine: their recombination rate is arbitrarily set to 50% and their distance cannot be computed. We could consider independent those loci with a suppressed recombination rate ($R_i < average(R)$) (Semagn *et al.*, 2006), but in order to compute that, a genome-wide genetic mapping of *Campanula thyrsoides* would be needed. In other studies on crop plants (i.e. maize) it was observed that LD structure decays quickly after 100s to 2000 bases (Buckler & Thornsberry, 2002), but this relationship has not been thoroughly studied in other types of plants. Since there is no reference genome available for*C. thyrsoides*, we could not compute the average recombination rate, nor we could assess the SNPs location on a chromosome: our only information is about the locus position on a certain stack. By setting a minimum distance of 1000 bp between SNPs, we filter out SNPs on the same stack: they are built using reads with a median length of 100 bp, so a distance of 1000 bp is enough to ensure picking one SNP per stack.

- **maf 0.005**: Include only sites with a Minor Allele Frequency greater than 0.005. It is defined as the number of times an allele appears over all individuals at that site, divided by the total number of non-missing alleles. While it is often set to 0.05 (Kanaka *et al.*,

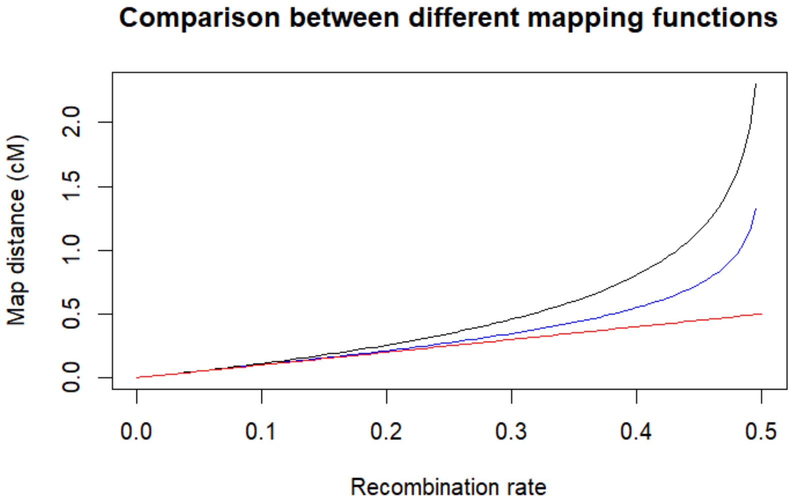


Figure 2.9: ***Comparison of Haldane's and Kosambi's mapping functions.*** *The red line shows a linear relationship between map distance (in cM) and recombination rate (R), while the black and the blue one show how the map distance grows exponentially with the recombination rate following, respectively, Haldane's (2.5) and Kosambi's (2.6) functions. Haldane's assumes there no interference, namely the effect of a crossover in one region on the likelihood of a crossover in another one (Huehn, 2010). there is no direct linear relationship between units of genetic distances in centimorgan (cM) and physical distances in kilobase pairs (kb), and depends solely on the organism in the study (Semagn* et al.*, 2006).*

2023) we instead computed the minimum maf threshold as:

$$maf_{threshold} = \frac{1}{\text{number of alleles} \cdot \text{number of samples}} = \frac{1}{2 \cdot 103} \tag{2.7}$$

This metric helps to distinguish between common and uncommon SNPs in a population: the loci with low MAF have limited heterozygosity and are thus less informative. MAFs lower than 0.5% usually refers to SNPs that describe rare variants behind, for example, uncommon diseases (Kanaka *et al.*, 2023) in a specific individual. Our study objective is a population-wide analysis, so SNPs that refer only to a variant in a single, or a small group of individuals can be discarded. Furthermore, most novel (initially rare) alleles arising through mutation are deleterious and could have negative impacts: this means that they should not be targets for conservation (Whitlock *et al.*, 2016).

2.1.7 Population genetics analyses

After associating the SNPs to each genotype, this information can be used for a series of analyses: metrics for a quantitative analysis of the genetical structure of the populations and different clustering methods to individuate their belonging to bigger groups or a subdivision in smaller subpopulations. We are going to talk about each method to perform these analyses in detail, reserving the results and discussion to the appropriate chapters.

Metrics

To determine the genetical structure of the populations in a quantitative way, the most common used metrics are those defined by Wright (Wright, 1949) as different fixation indexes. Assuming that the total putative original population is called T, each subpopulation deriving from its subdivision is referred as S and is formed by individuals I, we can define:

F_{IS} - The inbreeding index that is most used in literature, called by Wright (1949), "current inbreeding coefficient". It can be computed as:

$$F_{IS} = 1 - \frac{H_{ob}}{H_{exp}} = 1 - P_{IS} \tag{2.8}$$

where H_{exp} is the expected heterozigosity within the same subpopulation assumed generated by random mating, H_{ob} is the observed heterozigosity and P_{IS} is the panmictic index associated to F_{IS}. This is the only index that can be estimated from frequency of data of a single population (Weir & Cockerman, 1984) and ranges from -1 to 1, were -1 describes a population of individuals different from each other and 1 is associated to a population composed by fixed lines. We computed this value using R package `hierfstat`;

F_{IT} - This index is defined as the version of F_{IS} which is computed for individuals relative to the total, and it is what Wright defines as, simply, "inbreeding coefficient". A clearer definition of this is given by Weir and Cockerman in 1984 (Weir & Cockerman, 1984), as the correlation of genes within individuals of the full population. Differently from F_{IS}, F_{IT} depends on the population's (T) size and history so it needs the information given by all the individuals (I) in the dataset (Weir & Cockerman, 1984). It is unaffected, though, by the number of alleles sampled, the number of individuals within subpopulations (S) and the number of subpopulations themselves. Similarly to F_{IS}, F_{IT} ranges from -1 to 1 and its interpretation reflects the same as the former and it can also be associated with a panmictic index $P_{IT} = 1 - F_{IT}$ (Wright, 1949).;

F_{ST} - Defined by Wright (1949) as the "correlation of randomly drawn gametes from the same population, relative to the total population", it is calculated as:

$$F_{ST}^{W} = \frac{F_{IT} - F_{IS}}{1 - F_{IS}} = \frac{P_{IT}}{P_{IS}} \tag{2.9}$$

F_{ST} is strictly positive, with values ranging from 0 to 1, where 0 quantifies a little difference between subpopulations and 1 a strong difference between them. However, Wright did not specify what is the total population, so different authors interpreted it in different manners. While for Nei (1973) it is the combination of the two population samples, for Cockerman (1969) and Weir and Hill (Weir & Hill, 2002), the total population is the most recent common ancestral population to the two subpopulations being considered. In synthesis, the former viewed F_{ST} as a statistic from observed samples, which varies according to the number of sub populations observed (Weir & Cockerman, 1984), while the others as a parameter of the evolutionary process (Bhatia *et al.*, 2013). For this reason Weir and Cockerman (1984) define F_{ST} also as "coancestry", and compute it as:

$$F_{ST}^{WC} = 1 - \frac{2 \cdot \frac{n_1 n_2}{n_1 + n_2} \cdot \frac{1}{n_1 + n_2 - 2} \cdot [n_1 p_1 (1 - p_1) + n_2 p_2 (1 - p_2)]}{\frac{n_1 n_2}{n_1 + n_2} (p_1 - p_2)^2 + \left(2 \cdot \frac{n_1 n_2}{n_1 + n_2} - 1\right) \frac{1}{n_1 + n_2 - 2} [n_1 p_1 (1 - p_1) + n_2 p_2 (1 - p_2)]} \tag{2.10}$$

where n_i is the population i sample size and p_i is its allele frequency. If the sample sizes are unequal, the estimate will depend upon the ratio of sample sizes: this makes F_{ST}^{WC} not comparable across studies with different sample sizes. It is possible to correct this estimate considering the ratio of sample sizes, but another, more direct method, was proposed in 2002 by Weir and Hill (Weir & Hill, 2002) which introduced population-specific estimates of F_{ST} that is not influenced by the aforementioned limitation. Assuming p_i^s allele frequency of the derived allele in population i at SNP s and p_{anc}^s the allele frequency of the derived allele in the ancestral population at that SNP position, Weir and Hill define the following relationship between them:

$$\mathtt{E}[p_i^s | p_{anc}^s] = p_{anc}^s \tag{2.11}$$

for those SNPs that were already present in the ancestral population, and are not derived from a recent mutation. The variance present in the studied sample of the population i is then used to calculate the population-specific F_{ST}^{i} for a specific SNP s as:

$$F_{ST}^{WH,i} = \frac{Var(p_i^s|p_{anc}^s)}{p_{anc}^s(1-p_{anc}^s)} \tag{2.12}$$

Estimates of F_{ST} across multiple SNPs can be combined, but this can be done in two different ways: either first averaging the two components of equation 2.12 and then calculating the ratio (”ratio of averages” (Bhatia *et al.*, 2013)), or averaging genome-wide the single SNP estimates (”average of ratios”). Different softwares use distinct approaches, although the first method will converge to the correct underlying value as the number of independent SNPs increases, so it is the most recommended (Bhatia *et al.*, 2013). For two populations, the quantity is simply an average of the two population-specific F_{ST} values:

$$F_{ST}^{WH} = \frac{F_{ST}^{1} + F_{ST}^{2}}{2} \tag{2.13}$$

We reported here different equations because we analyzed this quantity pairwise over all the populations using different softwares: F_{ST} estimates can vary substantially based on the choice of estimator, but different results can be used to better describe the populations history and relationship between them. The softwares that were used to analyze the data were:

- `Stacks: populations` which computes SNPs F_{ST}^{WC} (2.10), corrects it for a Fisher exact test p-value and computes an ”average of ratios”;
- `vcftools` which calculates both a normal and weighted F_{ST}^{WC} [1];
- R package `BEDASSLE` (v1.6.1; Bradburd, 2024) that measures F_{ST}^{WH} (2.12);
- R package `hierfstat` (v0.5.11; Goudet and Jombart, 2015, March 18/2024) which computes SNPs F_{ST}^{WC} , unweighted, and then performs an average of ratios exactly as `Stacks: populations`. Differently from `populations`, `hierfstat` does not correct the values with a statistical test, but supports bootstrapping even in the absence of a reference genome, so it is able to compute confidence intervals;
- R package `StAMPP` (v1.6.3; Pembleton *et al.*, 2013) which can perform bootstrapping over weighted F_{ST}^{WC} and additionally computes its confidence intervals and p-value associated to these values.

[1] A weighted F_{ST}^{WC} is akin to F_{ST}^{WH}.

Isolation by Distance (IBD)

Isolation by distance refers to the phenomenon of restricted gene flow due to geographic distance: the goal is to test the relationship between population differentiation metrics (F_{ST}) and geographic distance between the samples. Under IBD, random mating is more likely to occur between nearby sites than those more distantly located (Loaiza *et al.*, 2012). To test the relationship between genetic and geographical distance, the most common approach is the use of Mantel test, first proposed in 1967 (Mantel, 1967): it was originally created to test the temporal and spatial separations for $n(n-1)/2$ possible pairs that can be formed from n observed cases of disease, but it can be applied to the pairwise F_{ST} between populations, and their distance in meters.

Mantel test's strength in comparison to the procedures that statisticians had developed until that time, is that its statistical power is improved by applying a reciprocal transform to the pairwise separations ($d'_{ij} = 1/d_{ij}$) to increase the strength of the test on shorter distances. For the sake of simplicity, from now on, when using d_{ij} we will refer to the already transformed distance. The observed association Z_m is tested relative to its permutational variance (Mantel, 1967), which is the variance obtained by permutation and avoids strong assumptions. The test is given by:

$$Z_m = \sum_{i=1}^{n} \sum_{j=1}^{n} g_{ij} \times d_{ij} \tag{2.14}$$

where g_{ij} and d_{ij} are, respectively, the genetic and geographic distances between populations i and j, considering n total populations. Then the Z_m value is compared with a null distribution, which was originally proposed by Mantel as a standard normal deviate (SND):

$$SND = \frac{Z_m}{var(Z_m)^{1/2}} \tag{2.15}$$

The hypothesis under Mantel's test is simple: if there is a relationship between two matrices, the sum of the products of their elements will be high and randomizing rows and column will destroy this relationship. Values of Z_m can be randomized and the p-value of the test can be computed as the probability of observing a Z_m value higher than the one obtained by formula 2.14 (Diniz-Filho *et al.*, 2013).

Mantel's test is usually applied using R package `vegan` (v2.7.1; Oksanen *et al.*, 2025): this package also allows the user to choose the most appropriate correlation metric between Pearson's, Spearman's and Kendall's. When applying the Pearson correlation r, the standardized version of Mantel's test (Z_N) is actually computed as (Diniz-Filho *et al.*, 2013):

$$Z_N = \frac{\sum_{i=1}^{n}\sum_{j=1}^{n}(g_{ij}-\bar{G})\times(d_{ij}-\bar{D})}{var(G)^{1/2}\times var(D)^{1/2}} \tag{2.16}$$

where $\bar{G}$ and $\bar{D}$ are the means of the genetic and distance matrices. Similarly to standardizing Mantel's test using Pearson's correlation coefficient, one could employ Spearman's coefficient:

$$\rho = 1 - \frac{6\sum_{i=1}^{n}(R_i - S_i)^2}{(n(n^2-1))} \tag{2.17}$$

where R_i represents the order of the values of G, while S_i the order of the values of D, from smallest to the largest. A final metric that could be used is Kendall's τ, which examines the relationship between all $n(n-1)/2$ possible comparisons between any pair of rank (g_i, d_i) and (g_j, d_j):

$$\tau = \frac{(num_C - num_D)}{n(n-1)/2} \tag{2.18}$$

where num_C is the number of concordant pairs (how many larger ranks are below a certain rank) and num_D is the number of discordand pairs (how many smaller ranks are below a certain rank). Each metric is recommended when its assumptions are valid (Table 2.2): when the distribution of data is normal, a parametric coefficient like Pearson's is the most used, while Spearman's and Kendall's, being non-parametric, are more robust in many other cases (El-Hashash & Shiekh, 2022). Before applying any of these methods for IBD testing, we checked for normality, linearity and presence of outliers in our samples' data.

Table 2.2: ***Comparison between correlation metrics.***

Assumptions	**Correlation metrics**		
	Pearson	Spearman	Kendall
Bivariate normality	Yes	No	No
Linearity	Yes	Yes	No
Monotony	Yes	Yes	Yes
Sample size $N > 30$	Yes	No	No
Absence of outliers	Yes	No	No

Phylogenetic tree

A phylogenetic tree, also known as a cladogram or evolutionary tree, is a graphical representation that illustrates the evolutionary and phylogenetic relationships between taxa based on their genetic features (Zou *et al.*, 2024). The branching structure might resemble the output of a hierarchical clustering, but phylogenetic trees aim to model evolutionary history (not just similarity) between ancestral lineages and their descendants by grouping them into clades depending on several levels of hierarchy. Trees can be unrooted or rooted: the latter have a root node from which the rest of the tree diverges, indicating an evolutionary direction.

We used two different methods when inferring a phylogenetic tree for the *Campanula thyrsoides* samples in the study, and the same methodology was applied to the other projects:

Neighbor joining, a distance-based method : for this tree we used R package `poppr` (v2.9.7; Kamvar *et al.*, 2013). The function implemented by this package automates the process of bootstrapping genetic data to create a dendrogram: bootstrapping is any test or metric that uses random sampling with replacement and falls under the broader class of resampling methods (Ojha *et al.*, 2022). It first builds a distance matrix between pairs of samples j and z based on the fraction of dissimilar SNPs: this is comparable to the computation of Hamming distance ($D_H(j,z)$) where SNPs information is treated as a string where the presence of a different SNP at locus i increments the dissimilarity value by 1. The Hamming distance is thus calculated as:

$$D_H(j,z) = \sum_{i=1}^{n} 1(j_i \neq z_i) \tag{2.19}$$

Missing data is set it to the average allele frequency in the dataset. At each iteration, the matrix is updated by merging the two nodes with the smallest distance and a new node connecting these two clusters is created (Zou *et al.*, 2024). Iterations are repeated until only one cluster remains, so this method produces only one tree;

Maximum likelihood, a character-based method : for this method we used software `iqtree` (v2.4.0; Minh *et al.*, 2020), which generates a large number of hypothetical trees using as input a multiple sequence alignment (MSA) of the samples, and reconstructs the tree that is best explained by the data. To respect the requirement of MSA input, we converted the data from the `.vcf` file in a `.phylip` file using python script `vcf2phylip` (v2.0; Ortiz, 2019), prior to running this command. A suitable evolutionary model (or sequence evolution model, SE; Kalyaanamoorthy *et al.*, 2017) is selected based on the sequence data being studied: it includes both a substitution model and a RHAS (rate heterogenerity across site) model, which might assume that either all sites evolved at the same rate or that it follows a probability distribution. The software we used (`iqtree`) accepts a flag (`-MFP`) which runs ModelFinder (Kalyaanamoorthy *et al.*, 2017) and implements a greedy strategy that searches tree space for every model of SE considered and selects the one with the lowest error metric: we decided to use BIC value (Bayesian Information Criterior) as it successfully penalizes evolutionary models with a large amount of parameters that do not substantially improve performance. Lastly, we made sure to run ModelFinder including an ascertainment bias correction (+ASC) model which should be applied if the alignment does not contain constant sites (such as SNPs data; Lewis, 2001): without it, the branch lengths could have been overestimated (Wong *et al.*, 2025). This method not only produces numerous trees by performing ultrafast bootstrap (UFBoot) approximation (-bb 3000; Minh *et al.*, 2013, Hoang *et al.*, 2018), but based on them it can build a network of evolutionary relationships between samples: for the sake of simplicity of visualization and comparison, a consensus tree is usually chosen as the final output of such method, which represents the most common features or relationships among a set of trees. On this tree branches, bootstrap values indicate how many times, in percentages, the same branch is observed when repeating the generation of a phylogenetic tree on a resampled set of data. (Ojha *et al.*, 2022).

Population structure

To study the populations structure and to delve more in depth in the genetic ancestry shared by the samples, we used two different softwares for the estimation of individual ancestries from unlinked markers: `Admixture` (v1.3.0; Alexander *et al.*, 2009) and `STRUCTURE` (v2.3.4; Falush *et al.*, 2003). They use the same statistical model-based (Alexander *et al.*, 2020) clustering method: the idea is that there exists a model of K populations, each characterized by its set of allele frequencies at every locus of the dataset provide (Pritchard *et al.*, 2010) and each individual is assigned to one population, or more than one if the genotype shows to be admixed. The base assumptions of both software is that, within populations, the loci are at Hardy-Weinberg equilibrium and linkage equilibrium: so the individuals are assigned to populations with a probability that increases the more these conditions are met. Additionally, `Admixture` assumes that samples are admixed, while, as will be discussed later, `STRUCTURE` is more customizable.

One of the strengths of `Admixture` is its computational speed: it uses a fast optimization method which alternatively updates allele frequencies and ancestry fraction parameters. Each update is solved as a set of convex optimization problems, each with its own solution, and convergence is sped up by a quasi-Newton acceleration method, which stops when the log-likelihood increases by less than $\varepsilon = 10^{-4}$ between iterations and outperforms, computationally speaking, EM and MCMC algorithms. The final parameters are then evaluated calculating a 5-fold cross validation error. It is calculated by estimating the genotype g_{ij} at each locus using a Binomial Distribution, fitting a model using the first 4 folds and then averaging the squares of the deviance residuals $d(n_{ij}, \hat{\mu}_{ij})$ in the last fold:

$$d(n_{ij}, \hat{\mu}_{ij}) = n_{ij} \log\left(\frac{n_{ij}}{\hat{\mu}_{ij}}\right) + (2 - n_{ij}) \log\left(\frac{2 - n_{ij}}{2 - \hat{\mu}_{ij}}\right) \tag{2.20}$$

where $\hat{\mu}_{ij}$ is the estimate of the observed genotype n_{ij}. The best K is chosen as the one that produces the lowest cross validation error. `STRUCTURE`, instead, uses the Bayes Theorem to obtain the posterior probability of the data, combining the prior probability of each sample of belonging to cluster K, arbitrarily initialized as equal for each, with a *model evidence*. The latter is the compatibility of the observed data to a certain ancestry model, out of four available:

- one can assume **no admixture**, so individuals can only be sorted to one population;
- one can allow instead **admixture**;
- a **linkage** model can be used to account for linked SNPs. This is not our case, since we applied filters in the upstream analysis to ensure picking SNPs as independent from each other as possible;
- lastly, the software can use information about **sampling locations**.

For our analysis, we used an admixture model which is often recommended as a starting point since, in real data, admixture is a common feature (Pritchard *et al.*, 2010; Supple and Shapiro, 2018). The program starts from a random configuration, takes a series of steps through the parameter space (where each step depends only on the previous one) and stops when convergence is reached. This is a common Markov Chain Monte Carlo algorithm: the states of the chain at different points are correlated to each other, but by running the simulation for a long period of time the correlations becomes negligible. To account for this, two parameters need to be adjusted: the burnin length, which is the number of starting iterations skipped to minimize the effect of the random initialization point, and then the number of repetitions needed to get accurate parameter estimates.

`Admixture` focuses on maximizing the likelihood of the data, and for this reason is usually much faster than the high-dimensional MCMC used by `STRUCTURE` (Alexander *et al.*, 2009; Alexander *et al.*, 2020), but the latter can reach more accurate ancestry estimates as parameters of the statistical model (Stift *et al.*, 2019). Considering the two softwares weaknesses and strengths, we decided to run `Admixture` first on a larger range of parameter K, find the optimal K value that minimized the cross validation error, and then repeat the analysis on `STRUCTURE` over a smaller set of K values around the optimal one. `Admixture` was run over $K \in [2,30]$ and `STRUCTURE` over $K \in [1,20]$, repeating the analysis 5 times for each K, with $burnin = 20000$ and $numreps = 80000$: the same workflow for `STRUCTURE` was replicated both including and excluding the population of the outgroup CTT (subspecies *thyrsoides*). The final optimal value of K for both datasets as estimated by `STRUCTURE` was chosen using both Evanno (Evanno *et al.*, 2005) and Puechmaille (Puechmaille, 2016) methods:

- Evanno uses ΔK, an *ad hoc* measure related to the log-likelihood of the data. It is obtained first by calculating the mean difference between successive likelihood values ($L'(K) = L(K) - L(K-1)$), then computing the absolute value of the difference between successive values ($|L''(K)| = |L'(K+1) - L'(K)|$) and finally estimating ΔK as the mean (m) of the absolute values of $L''(K)$ averaged over 20 runs, divided by the standard deviation (s) of $L(K)$. To summarise the last step, it is calculated as such:

$$\Delta K = m(|L''(K)|)/s[L(K)] \tag{2.21}$$

  The strength of this method is that it minimizes the numbers of clusters found while maintaining a good amount of information, by identifying the number of K groups corresponding to the uppermost hierarchical level in which the dataset can be divided into;

- Since it was noted that subpopulations with reduced sampling tended to be merged together (Puechmaille, 2016), the more recent study from Puechmaille proposed four new supervised methods to detect the number of clusters. Although it is still less used (791 citations; ISI Web of Science database; 06/09/2025) than Evanno (which went from 4715 in 22/03/2015 to 18400 citations in 06/09/2025), it introduces the concept of spurious clusters to account for uneven sampling, like in the case of our data. This means that this method maximizes the number of ”meaningful” clusters, by ignoring those represented just by a single or few individuals: a limitation of this method is that it utilizes prior sampling information, so its use is not always feasible. The four supervised methods function in slightly different ways. For a range of K values $K = 1, 2, ..., k$, and for each run r_k:

MedMeaK (median of means): the software calculates the normalized membership coefficient m_k to cluster k, so that for each sample i, we have $\sum_{k=1}^{K} m_k = 1$. A spurious cluster is defined as a cluster that never achieves a **mean** membership coefficient greater than a threshold value (which in our analysis we set to the default 0.5) in any sub-population of the dataset. This means that, using the prior information that sample i belongs to subpopulation p, if:

$$\frac{1}{N_p} \sum_{i=1, i \in p}^{N_p} m_{k_i} > 0.5 \tag{2.22}$$

then the subpopulation p is clearly assigned to cluster k. If a cluster k has no subpopulation clearly assigned to it, then it is considered spurious. For each run (r_K) of that value of parameter K, the number of real, non spurious clusters is counted, and the **median** of these values across all runs r_K is computed as the **corrected** number of clusters for K;

MaxMeaK (maximum of means): similarly as before, assigns subpopulation p to a cluster depending on the **mean** coefficient of membership, but the corrected K is calculates as the **maximum** number of non-spurious clusters over all the runs;

MedMedK (median of medians): this method assigns a subpopulation if the **median** membership coefficient for cluster k is over 0.5, and, as MedMeaK, the corrected K is given by the **median** of real clusters across all the runs;

MaxMedK (maximum of medians): an hybrid between MedMedK and MaxMeaK, it assigns subpopulations depending on the **median** membership coefficient, and returns the corrected K as the **maximum** number of clusters across runs.

After exploring the whole range of K values and having obtained the set of corrected number of clusters for each value of K, each method takes the **maximum** of these values as the final estimate of clusters. Since no estimator is universally the best, we computed all four estimators as suggested (Puechmaille, 2016) and compared their results, to exploit the differences between them and to better explain biological properties of the dataset.

2.2 *Solanum tuberosum* landraces

The modern potato (Solanum tuberosum L.) is an annual species, belonging to family *Solanaceae* which includes other important crop plants like tomato (*S. lycopersicum*), eggplant (*S. melongena*) and pepper (*Capsicum annuum*). It is herbaceous with roots rich in fibers which produce, upon favorable climatic conditions (semi-humidity, 20°C - 25°C; Wang *et al.*, 2022), enlarged portions called tubers that are used as food source due to their high starch content (Xu *et al.*, 2011): its fruits, instead, are usually inedible due to poisonous alkaloids (Reddy *et al.*, 2018). It is mostly self-pollinated but when cross pollination occurs, it is mainly entomophily by genus *Bombus* (Ackerman, 2000). It was originally domesticated in the Andes between 8000 to 10000 years ago (Kui *et al.*, 2022) probably in the vicinity of Lake Titicaca on the border of Peru and Bolivia (De Jong, 2016), introduced in Europe in the late 1500s (Wang *et al.*, 2022) and since then cultivated for food, alcoholic beverages (Reddy *et al.*, 2018) and animal feed (Wang *et al.*, 2022). Cultivars can be distinguished using morphological markers based on height, stem and flower color, grow habits etc., but these characteristics are susceptible to phenotypic plasticity and environmental fluctuations (Reddy *et al.*, 2018), so molecular markers are more accurate.

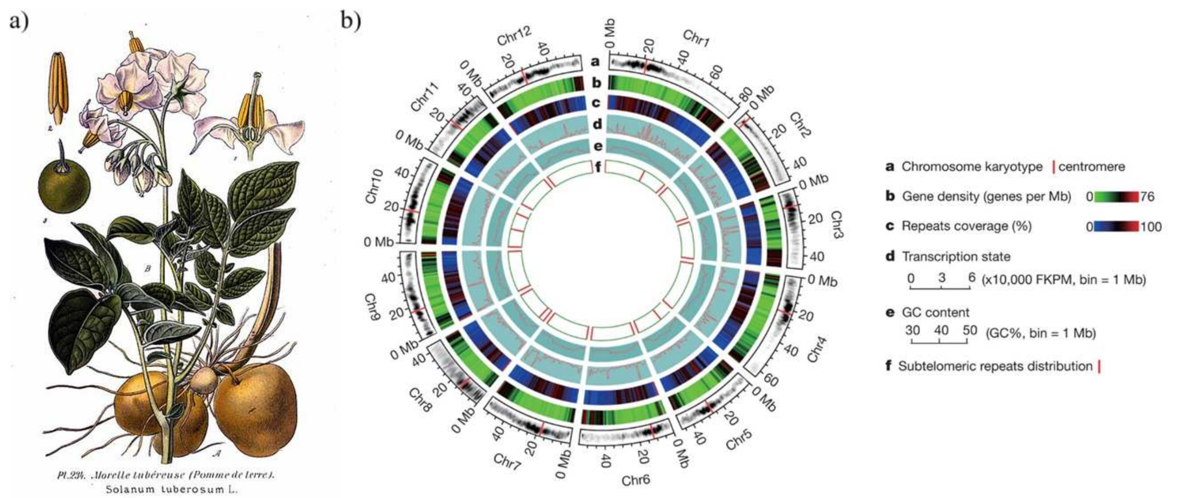


Figure 2.10: ***S. tuberosum* morphological and genetic characteristics.** *Morphology of S. tuberosum (a) illustrated by Amédée Masclef (Masclef, 1891), and ideograms (b) of the 12 pseudo- chromosomes (Xu* et al.*, 2011) that compose the assembly we used as reference.*

The potato genome consists of 12 chromosomes (Figure 2.10, b) with an haploid length of approximately 840 Mb, but the majority of potato cultivars are autotetraploid (2n = 4x , with x

= 12) and, additionally, suffer from acute inbreeding depression (Xu *et al.*, 2011), due to their pedigree containing the same cultivars more than once in their ancestry (Sun *et al.*, 2022). In fact, a recent study on the cultivar Otava showed that more or less 50% of its genome is identical by descent (IBD), which results in 1.9 diverse allelic copies per gene (Kui *et al.*, 2022): the maximal allelic diversity in the potatoes is 4, so a value as low as 1.9 confirms a reduced genetic diversity shared by many of the modern cultivars of potatoes. This problem is exacerbated by the production method used almost globally: potatoes are usually propagated through the tubers - whole, a portion (Reddy *et al.*, 2018) or through the buds (Wang *et al.*, 2022) - which means that individuals are cloned and not generated by crossing. This reduces costs and ensures that optimal characteristics (larger production per plant, less bitterness, bigger tubers...) are selected (Wang *et al.*, 2022). Artificial selection can cause enrichment of specific genes: for example, in another study again on cultivar ”Otava”, it was shown that there was a positive correlation between selection-induced reduction of allelic diversity and enrichment of genes involved in photosynthesis, chlorophyll binding and translation (Sun *et al.*, 2022). The inbreeding depression of *S. tuberosum* raises concern because it decreases both breeding success (Sun *et al.*, 2022) and, more importantly, genetic diversity, which is essential for developing resistance to pathogens. Resistance to diseases is extremely important for crop plants: an example of this can be found in history, when in the 1840s blight disease attacked potatoes crops and caused the infamous Irish Potato Famine (Wang *et al.*, 2022), which ultimately was the starting point of modern plant pathology.

2.2.1 Sampling

The samples used in this study can be divided into two categories: local landraces and modern cultivars. The aim was to reconstruct the domestication history to understand the variation in local landraces and their conservation value. The study involved a total of 12 local landraces of *S. tuberosum*, 2 commercial cultivars (Desiree and Kennebec) and WGS data of 4 wild and 4 ancient cultivated species of the same genus: the WGS data was collected from the NCBI database and included in the bioinformatic analysis as reference of ancestral varieties (Table 2.3). Classification of cultivated potatoes can change depending on their identification as species (as stated by the International Code of Botanical Nomenclature) or as cultivar groups (International Code of Nomenclature of Cultivated Plants; Huamán and Spooner, 2002): in this analysis, we followed ICBN classification as separated species (Turland *et al.*, 2018).

Table 2.3: *List of commercial cultivars and local landraces of S. tuberosum and wild/ancient species of genus Solanum.*

***S. tuberosum* landraces**		
Species	**Cultivar**	**Origin**
S. tuberosum	Kennebec	Commercial, USA
S. tuberosum	Desiree	Commercial, Netherlands
S. tuberosum	Bossico (BG) 1	Teglio (SO)
S. tuberosum	Bossico (BG) 2	Bossico (BG)
S. tuberosum	Starleggia Rossa	Campodolcino (SO)
S. tuberosum	Starleggia Bianca	Campodolcino (SO)
S. tuberosum	Quarantina Genovese Bianca	Oltrepò (PV)
S. tuberosum	Quarantina Genovese	Genova (GE)
S. tuberosum	Comasca Biancona	Como (CO)
S. tuberosum	Rossa	"La Campagnola" farm, Gordona (SO)
S. tuberosum	Bianca	"La Campagnola" farm, Gordona (SO)
S. tuberosum	Rossa Oltrepò, Romagnese	Ca' Novelli nel Romagnese (PV)
S. tuberosum	Rossa Oltrepò di Pietragavina	Pietragravina (PV)
S. tuberosum	Schilpario	Botanical garden of Bergamo (BG)
Ancient cultivated species of genus *Solanum*		
Species	**Accession Number**	**Origin**
S. chaucha	SRR10248511	Andes
S. phureja	SRR10244439	Bolivia, Peru, Venezuela, Colombia, Ecuador
S. stenotomum	SRR10244438	Peru
S. tuberosum subsp. *andigenum*	SRR10248514	Andes
Wild species of genus *Solanum*		
Species	**Accession Number**	**Origin**
S. microdontum	SRR24976252	Bolivia, Argentina
S. vernei	SRR24976381	Bolivia, Argentina
S. brevicaule	SRR24976310	Bolivia, Peru
S. chacoense	SRR24976359	Bolivia, Peru, Argentina, Paraguay, Uruguay, Brazil

List of commercial cultivars and local landraces of S. tuberosum and wild/ancient species of genus Solanum. The accession number associated with the wild/ancient species refers to specific runs on NCBI's SRA database.

The choice of which wild/ancient species to use in this analysis was based on: (i) their phylogenetic position in other studies with respect to *S. tuberosum* (i.e. *S. phureja*, Xu *et al.*, 2011; *S. stenotomum*, Wang *et al.*, 2022; *S. chauca*, Huamán and Spooner, 2002), (ii) being wild closely related species (i.e. *S. microdontum*, Huamán and Spooner, 2002; *S. vernei*, Spooner *et al.*, 2014), prioritizing those (i.e. *S. brevicaule* and *S. chacoense*; Huamán and Spooner, 2002) currently present in the wild were domestication started (Bolivia and Peru; De Jong, 2016), (iii) reads quality and sequences availability. Instead, data for landraces was obtained from living plants: seeds were provided by the seed bank of University of Pavia, after collection from its research institutes, public institutions and farms in the Lombardy region. Samples were then grown in a growth chamber, and, after germination, seedlings' leaves were collected and stored at -80°C until genetic analysis in 2024.

2.2.2 Wet lab protocol

DNA extraction was carried out in University of Milan following the same CTAB protocol employed for *C. thyrsoides*, and then purified using Genomic DNA Clean & Concentrator-10 kit by Zymo Research. Library preparation for GBS data was also performed in the same manner as described in the previous sections, using again restriction enzyme ApeKI to fragment the DNA. Ligation, Pooling and PCR amplification protocols were the same as described for *C. thyrsoides*, minus the size-selection step which was skipped since the library was of Illumina compatible size. The 14 landraces' DNA libraries were sent to GENEWIZ from Azenta Life Sciences (Leipzig, Germany) for standard Illumina TruSeq combinatorial-dual DNA sequencing in the second half of 2024.

2.2.3 Data analysis

Pre-processing of the data was performed by PostDoc Chiara Paleni for BSc graduate Giulia Ballarotti's thesis (Ballarotti, 2025) prior to our analysis, so the sequences' files were already de-multiplexed, trimmed and filtered at the beginning of this study. Nevertheless, the softwares and parameters used widely overlapped, with the only exception that cleaning was performed using software `fastq-mcf` (v1.05) with trimming of base-pairs based on the quality Phred score threshold of 30 ($-q$ 30), minimum length after trimming of 50 bp ($-l$ 50), disabling skew filtering ($-k$ 0) and adding a window size set to 2 to further remove low quality bases ($-w$ 2). The full list of commands used for these and the next steps can be found in the supporting file `Final_report_potatoes_cultivars.html`, freely available in the `Github` repository associated with this thesis `flavialeotta\master-thesis`.

Variant calling with a reference genome

The advantage of working with crop plants is that, more often than not, a reference genome is available: this is the case of *S. tuberosum* for which an official reference genome can be found on NCBI, and which we used for variant calling. The chromosome-level genome assembly we used is file "Solanum_tuberosum.SolTub_3.0.dna.toplevel.fa" corresponding to GenBank assembly GCA_000226075.1 (available in release-62 of Ensemble genomes at https://ftp.ensemblgenomes.ebi.ac.uk/pub/plants/release-62/fasta/solanum_tuberosum/dna/), which, in turn, was built on scaffold-level assembly NCBI RefSeq GCF_000226075.1 submitted by Beijing Genomics Institute in Shenzhen, China (Xu *et al.*, 2011). This sequence accession number corresponds to an homozygous diploid cultivar *S. tuberosum* group Phureja strain DM1-3 516 R44[2]. Sequences files associated with other wild species of genus *Solanum* were downloaded through their accession number (Tab. 2.3) using SRA Toolkit. All samples, both landraces and wild/ancient species, were mapped on the same reference genome using a custom script `map_bwa.sh` which recursively maps (`bwa mem`, v0.7.19; H. Li, 2013) on the reference genome and then converts the generated `.bam` files in a `.sam` file, using `samtools` (v1.21; Danecek *et al.*, 2021). After doing this for every sample, all `.sam` files are sorted (`samtools`) and merged (`bamaddrg`[3]). The resulting `.bam` file is indexed (`samtools`) and then converted in a `.bed` file using `bedtools` (v2.29.0; Garrison, 2011, April 28/2025). To speed up the analysis, we employed `MultiThreadFreeBayes`(v0.1[4]), a script that supports multithreading of the golden standard `freebayes`, and performed the final step of variant calling: we only kept SNPs called with a minimum base quality of 30 ($-m = 30$), minimum mapping quality of 30 ($-q = 30$), and lower and upper coverage limits of 10 and 1000 (–limit-coverage = 100, –min-coverage = 10). After some intermediate steps to prepare the input file, we filtered the `.vcf` using `vcftools` with the same parameters used for *C. thyrsoides*, with the exception of minor allele frequency threshold being set to 0.023 (calculated using formula 2.7 and $n = 22$ biallelic samples) and maximum missing data to 5% (–max-missing 0.95). The latter is a slightly more stringent threshold than what was used with *C. thyrsoides* (–max-missing 0.90)

[2]DM1-3 516 R44 is the result of chromosome doubling of a monoploid (1n = 1x, with x = 12) derived by a heterozygous diploid (2n = 2x, with x = 12) *S. tuberosum* group Phureja clone (PI 225669; Xu *et al.*, 2011).

[3]Version 9baba65f88228e55639689a3cea38dd150e6284f (Garrison, 2011, April 28/2025).

[4]From GenoToolBox (Bombarely, 2020). It depends on `freebayes` (v1.38; Garrison and Marth, 2012).

since working with a reference genome allows us to be stricter with the quality of called SNPs, without losing information. Additionally, for each sample, we obtained `.bed` files with the intervals of the regions covered by samples' reads, along with their coverage (`bedtools`). These files are intersected to find the shared sites between all of the samples and we stored the information found in a text file. Lastly, the generation of the phylogenetic tree followed the same workflow as *C. thyrsoides*: the evolutionary model chosen by program `ModelFinder` was TVM+F+ASC+G4.

2.3 *Anacamptis* spp.

Orchids are one of the largest family of flowering plants, which accounts for about a tenth of them (Yin *et al.*, 2024): Orchideae comprise about 62 genera and more or less 1800 species (Alec M. Pridgeon *et al.*, 2001), and a large percentage of them are regarded as threatened or endangered (Yin *et al.*, 2024). *Anacamptis* genus itself encompasses 16 taxa (Alec M. Pridgeon *et al.*, 2001): one of the most common is *A. pyramidalis*, target of our study.

2.3.1 *Anacamptis pyramidalis*

Anacamptis pyramidalis is a widespread perennial (Ilves *et al.*, 2016) species, originating in England (Doro, 2020) and then spanning through all the temperate European countries to south-western Asia and North Africa (Fay *et al.*, 2024), where it grows on well-drained, open, calcareous soils (Alec M. Pridgeon *et al.*, 2001). In the Northern border of its range (namely Estonia and Öland, a Swedish island), it is found to be less widespread and abundant (Ilves *et al.*, 2016). Its broad presence on the Italian peninsula (San Marino included; Alessandrini *et al.*, 2022) is well documented: it can be found from the very northern Valle d'Aosta (Bovio, 2014) at altitudes over 2000 m (Alec M. Pridgeon *et al.*, 2001), through all the other regions, islands included (Conti *et al.*, 2005). Its conservation status has not been assessed globally (Fay *et al.*, 2024) nor nationally in Italy (Graziano Rossi *et al.*, 2020), but it belongs to the Least Concern category at European level (Fay *et al.*, 2024) and regional level in Veneto (Buffa, 2016): in region Valle d'Aosta is categorized as Vulnerable (Regione Autonoma Valle d'Aosta, 2009). Despite its vast distribution, achieved thanks to its high levels of seed dispersal (Alec M. Pridgeon *et al.*, 2001), it is regarded as a specialist species since it requires mycorrhyza with species of Family Tulasnella for germination (Fay *et al.*, 2024). Its name derives exactly from its flowers (Fay *et al.*, 2024), characterized by a pyramidal shape (Fig. 2.11), which can be appreciated from mid-June until the end of July in UK (Alec M. Pridgeon *et al.*, 2001) and from late April to mid-June in Mediterranean countries (Fay *et al.*, 2024). Many *Anacamptis* species attract pollinators but do not produce nectar (Alec M. Pridgeon *et al.*, 2001); similarly, *A. pyramidalis* is a nectar-less mainly outcrossing species (Ilves *et al.*, 2016), primarily pollinated by Lepidoptera

(Fay *et al.*, 2024), but it can also perform vegetative propagation through daughter tubers, although it is less common since it may take even 3 years before the first mother tuber appears (Alec M. Pridgeon *et al.*, 2001). Genetic studies on this species are rather limited and focused on karyomorphology (Doro, 2020): from cytogenetic studies we know that *A. pyramidalis* is mainly a diploid species (2n = 2x, with x = 16; Alec M. Pridgeon *et al.*, 2001) but some triploids (2n = 54; Doro, 2020) and tetraploid (2n = 72; Fay *et al.*, 2024) were also reported. Triploids are usually due to hybridization between diploid and tetraploid individual (Doro, 2020), while tetraploid individuals are of auto-polyploid origin (Alec M. Pridgeon *et al.*, 2001). In other cytogenetic experiments, researchers used fluorescent in situ hybridization to identify the physical location of sites of highly repeated genes (Alec M. Pridgeon *et al.*, 2001), exploited as markers in some groups of Orchidinae to investigate taxonomic relationships (Turco *et al.*, 2021). Little has been done to explore genetic diversity in this species: to our knowledge, none of the marker-based studies employed SNP arrays so far, but AFLPs (Amplified Fragment Length Polymorphisms; Pegoraro *et al.*, 2016; Ilves *et al.*, 2016).

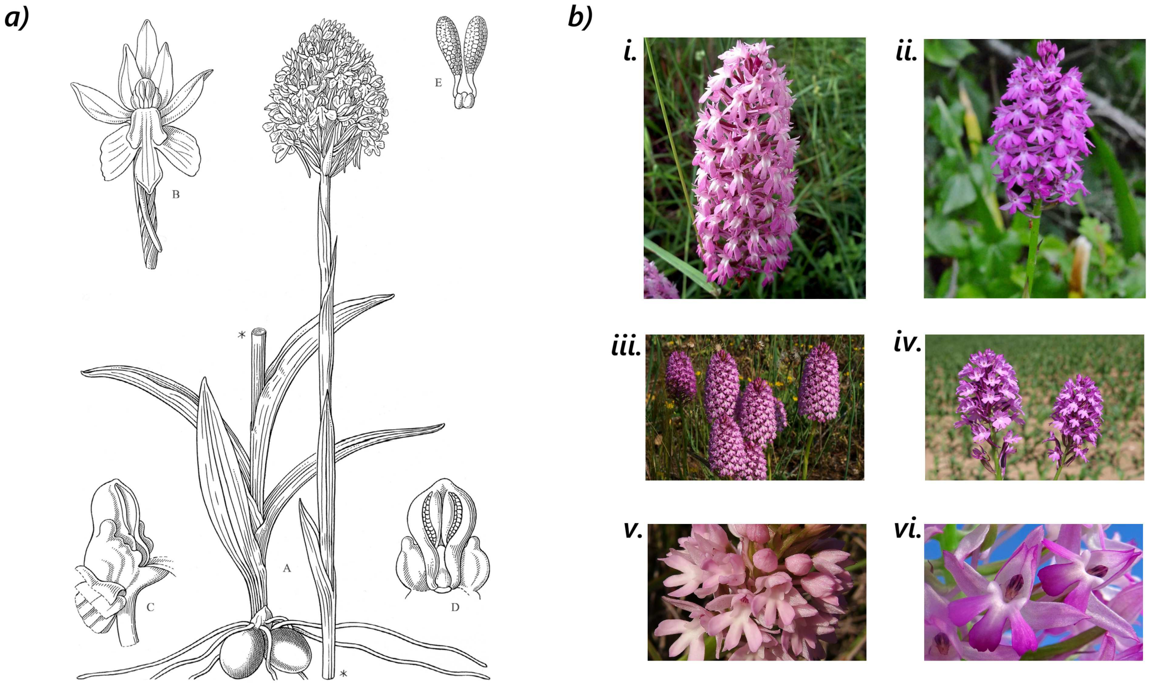


Figure 2.11: ***A. pyramidalis* and *A. berica* morphology.** *a) Illustration of A. pyramidalis by Stella Ross-Craig (Ross-Craig, 1971); b) Comparison between A. pyramidalis (left) and A. berica (right), focusing on observed morphological differences (Doro, 2020) in the shape of the flowers (i vs ii), number of flowers (iii vs iv) and color (v vs vi). Photos by: Brunello Pierini 2006 (i), Piera Pellizzer 2021 (ii), Vito Buono 2015 (iii), Silvio Colombo 2023 (iv), Quintino Giovanni Manni 2011 (v), Michele Aleo 2025 (vi). Acta plantarum: www.actaplantarum.org.*

2.3.2 *Anacamptis berica*

We identified *A. berica* as the outgroup species in our analysis, another Italian endemic orchid occurring in regions of Liguria, Trentino Alto Adige, Emilia-Romagna, Molise (Bartolucci *et al.*, 2024), Veneto, Abruzzo (Pica *et al.*, 2023) and Puglia (Griebl & Presser, 2021). It was recorded in Lombardy and Sicily too, although its presence is yet to be confirmed. It is a newly identified species of genus *Anacamptis*, with ovoid rather than the characteristic pyramidal-cylindrical shaped inflorescence (Figure 2.11, b, `i` vs `ii`), less but bigger flowers in comparison to *A. pyramidalis* (`iii` vs `iv`) and with a vivid pink colour (`v` vs `vi`) (Doro, 2020). *A. berica* was first observed in the Berici Hills (VI) by Doro in 2007 (2020) and was identified by Pegoraro *et al.* (2016) as simply an early-blooming, tetra-ploid (2n = 72) *A. pyramidalis*. After AFLPs analysis by Doro (2020), *A. berica* was identified as separate species evolved by auto-polyploidy in sympatry with *A. pyramidalis*. Despite *A. pyramidalis* and *A, berica* are inter-fertile (Pegoraro *et al.*, 2016), the hybrids were found to be sterile (Doro, 2020). This supports two distinct species distinction. The similarities between them, combined with the genetic and morphological distinction, guided our choice of *A. berica* as the outgroup species in this study.

2.3.3 Sampling

A total of 90 individuals were sampled spanning 23 sites in three regions of the Center-North of Italy (Emilia-Romagna, Lombardy and Veneto): leaves from 79 individuals of *A. pyramidalis* (20 populations) and 11 of *A.berica* (4 populations) were collected. Coordinates are shown in Figure 2.12.

Urban population are highlighted by a yellow circle, but more information about specific localities and population IDs used in the map can be found in Table 2.4. Samples from population P_NI were indicated as of dubious taxonomy due to the morphological similarity to both species (Fig. 2.11, b): in all the next analyses they are treated as species *A. pyramidalis*.

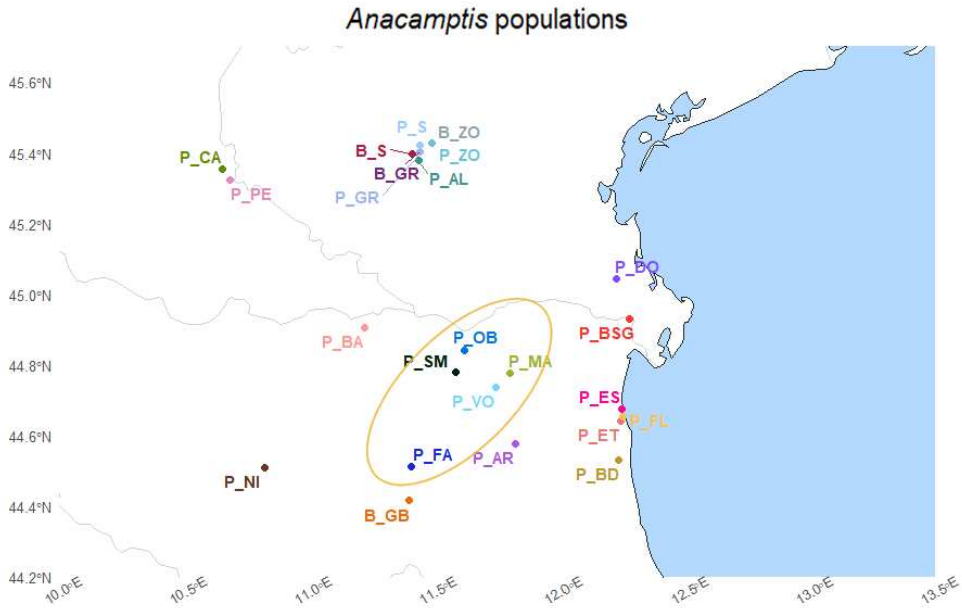


Figure 2.12: ***Anacamptis pyramidalis and Anacamptis berica coordinates.*** *Populations of A. pyramidalis ID start with 'P', while those of A. berica start with 'B'. Moreover, a yellow circle highlights urban populations, while the rest are classified as natural.*

2.3.4 Wet lab protocol

DNA was extracted using the CTAB protocol with sorbitol while digestion followed the RAD-seq protocol modified by Paun *et al.* (2016), employing the PstI restriction enzyme. DNA extraction and library preparation was done by the PhD student Lisa Scramoncin at the University of Vienna. Sequencing was performed using Illumina NovaSeqX 10B XP flowcell PairedEnd 150- single lane.

2.3.5 Data analysis

The data for *Anacamptis pyramidalis* and *Anacamptis berica* was provided by University of Vienna already filtered and cleaned, so the same pipeline built on dataset *C. thyrsoides* (Fig. 2.7) was applied starting from the variant calling step (subsection 2.1.6). Hyperparameters optimization for `STACKS`: `denovo_map.pl` pipeline was not needed, as the chosen parameters were previously assessed by PhD candidate Scramoncin Lisa, and selected as $m = 3,\ M = 4,\ n = 4$. All of the subsequent analyses were carried in the same way as previously discussed, twice when

Table 2.4: *Sampling information for Anacamptis populations.*

		A. pyramidalis			
Pop ID	**Locality**	**Region**	**Coordinates**	**Type**	**N**
P_AL	Alonte (VI)	Veneto	N 45.380005 E 11.43727597	Natural	5
P_AR	Natural Reserve of Argenta (FE)	Emilia-Romagna	N 44.577870 E 11.82361401	Natural	3
P_BA	Mirandola (MO)	Emilia-Romagna	N 44.906427 E 11.22175816	Natural	3
P_BD	Bardello Reserve (RA)	Emilia-Romagna	N 44.530852 E 12.23498823	Natural	5
P_BSG	Bosco di Santa Giustina (FE)	Emilia-Romagna	N 44.929103 E 12.27648002	Natural	4
P_CA	Mantova (MN)	Lombardy	N 45.355515 E 10.65290536	Natural	3
P_DO	Dune Fossili di Donado (RO)	Veneto	N 45.042720 E 12.22362915	Natural	4
P_ES	Lido degli Estensi (FE)	Emilia-Romagna	N 44.675212 E 12.24747726	Natural	3
P_ET	Lido di Spina (FE)	Emilia-Romagna	N 44.640628 E 12.24127214	Natural	3
P_FA	Fanin Street (BO)	Emilia-Romagna	N 44.513445 E 11.40683206	Urban	5
P_FL	Bagno Florida (FE)	Emilia-Romagna	N 44.652368 E 12.24951796	Natural	3
P_GR	Grancona (VI)	Veneto	N 45.401857 E 11.44525321	Natural	1
P_MA	Masi San Giacomo (FE)	Emilia-Romagna	N 44.777000 E 11.80178303	Urban	7
P_NI*	Salse di Nirano (MO)	Emilia-Romagna	N 44.510188 E 10.82085703	Natural	3
P_OB	Botanical garden (FE)	Emilia-Romagna	N 44.841896 E 11.62238398	Urban	6
P_PE	Petacchi (MN)	Lombardy	N 45.324625 E 10.68484643	Natural	4
P_S	Sarego (VI)	Veneto	N 45.422654 E 11.44386531	Natural	3
P_SM	San Martino (FE)	Emilia-Romagna	N 44.780659 E 11.58659894	Urban	6
P_VO	Voghiera (FE)	Emilia-Romagna	N 44.735430 E 11.74827479	Urban	5
P_ZO	Zovencedo (VI)	Veneto	N 45.426675 E 11.49167732	Natural	3
		A. berica			
Pop ID	**Locality**	**Region**	**Coordinates**	**Type**	**N**
B_GB	Parco dei Gessi (BO)	Emilia-Romagna	N 44.415677 E 11.39921705	Natural	3
B_GR	Grancona (VI)	Veneto	N 44.929103 E 12.27648002	Natural	3
B_S	Sarego (VI)	Veneto	N 45.396867 E 11.41201601	Natural	2
B_ZO	Zovencedo (VI)	Veneto	N 45.426689 E 11.49116853	Natural	3

*List of sampling information for all Anacamptis populations. 'N' indicates the number of samples obtained for each population. *There were some uncertainties about population P_NI being A. pyramidalis: it was reported that it could've been composed by individuals of A. berica.*

necessary: once including and another excluding samples from the species *A. berica*. When excluding them, the only parameter that diverged was Minor Allele frequency (MAF) threshold in the filtering stage, set as 0.006 reflecting the lower number of samples in the analysis.

3. Results

3.1 *Campanula thyrsoides*

Unfortunately, the whole CAN population consisting in the majority of the outgroup of *Campanula thyrsoides* subsp. *thyrsoides* had to be excluded from the library preparation step because the samples showed little to no DNA material after digestion (chapter 2). Nevertheless, the individuals from population CTT could still be used as the outgroup of subspecies *carniolica*. From the original 153, only 106 individuals across 10 populations were sequenced, with a mean of 10.6 individuals per population. Of this dataset, another 3 samples were excluded during the dry lab analysis, precisely on the filtering step: ALP11, LPASS5 and LPASS6 were excluded because they comprised less than 1 million pairs of reads (ALP11 had 576,927 pairs, LPASS5 934,941 and LPASS6 only 25,314). Such low value of reads would have reduced the mean coverage of their populations (ALP and LPASS), and moreover, considering the filters applied to the SNPs later in the analysis, would have led to the exclusion of informative loci because of an higher percentage of missing data across samples.

3.1.1 Pipeline refinement

Improvement of trimming

Using `fastp` instead of `fastq-mcf` resulted in a substantial improvement in performance. The wall-clock execution time decreased from around 30 minutes to more or less 4.5 minutes, corresponding to a speedup of approximately 7x and a reduction in runtime of 85%. The user CPU time reported by `fastp` was higher (62 minutes vs. 40 minutes for `fastq-mcf`), indicating that `fastp` is more CPU-intensive, due to parallelized operations. However, because this additional computation is efficiently distributed across cores, the overall elapsed time is cut to half (from slightly more than 1 minute to just 29 seconds), suggesting less overhead in I/O operations. The quality of the cleaning remained the same if not better with `fastp`, since this software is able to efficiently cut off the majority of the poly-G tails (Figure 3.1) by sacrificing slightly the average length of reverse reads (from 83bp to 80bp) but still retaining the same amount of reads (i.e. 2.6 millions for ALP1) as `fastq-mcf` does. Length of reads might seem short, but it is common in GBS protocols (Elshire *et al.*, 2011).

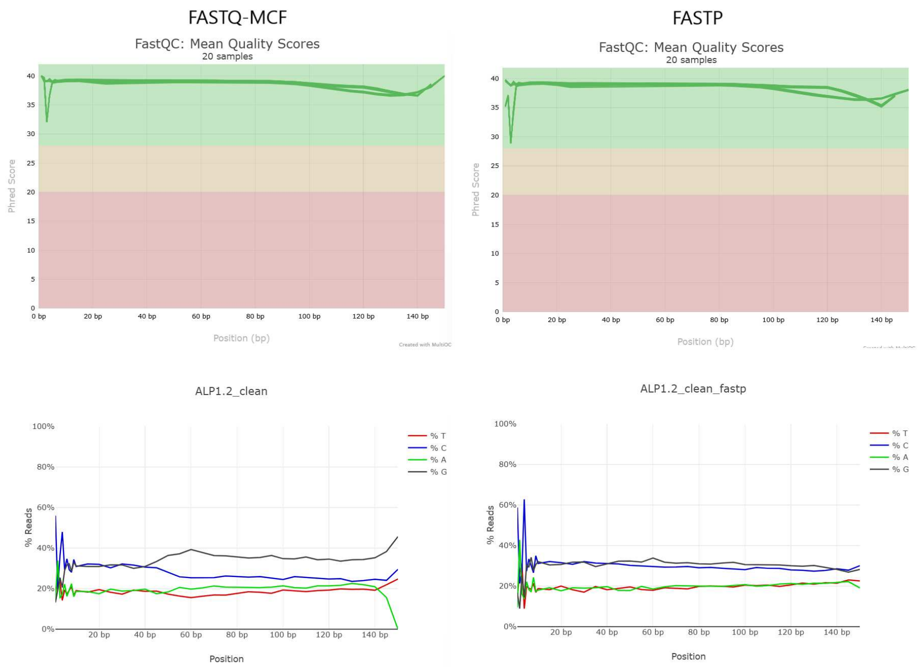


Figure 3.1: ***Comparison between cleaning tools*** *When using the correct parameters,* `fastp` *maintains a good overall reads' quality (top) and trims poly-G tails better (bottom plot). Plots built using software* `FastQC` *(v0.11.5; S. Andrews, 2010) and* `multiqc` *(v1.27.1; Ewels* et al.*, 2016).*

Hyperparameters' tuning outcome

When exploring the percentage of lost reads during the process of variant calling, we've noticed that the majority of the information is lost at the last step, meaning the `Stacks: denovo_map.pl` pipeline implementation (Figure 3.2). This highlights the need to accurately tune this pipeline, as to maximize the amount and the quality of information we are able to gather from sequencing data.

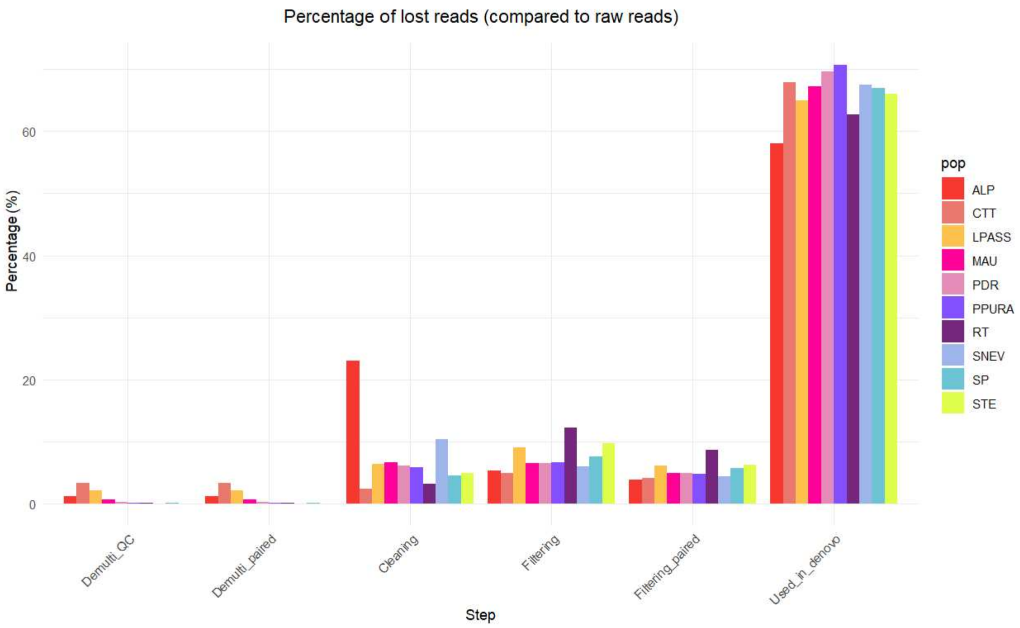


Figure 3.2: ***Lost reads at each step*** *Barplot showing the percentage of discarded reads (in comparison to the amount of raw reads) at each step, from pre-processing to variant calling, grouped by population.*

We tested a total of 46 combinations of parameters for `Stacks: denovo_map.pl` on a subset of 9 samples from population ALP, and whose performance is reported in Table 8.3 in the Additional Materials chapter. The best combination out of them was $m = 4$, $M = 1$ and $n = 1$: interestingly enough, by using the score function 2.2 to assess their performance, the only combinations that passed the required conditions (an average coverage of 10x and at least 2400 called loci) always had $m = [4,5]$ (Fig. 3.3), suggesting that this might be the most important parameter to tune.

To assess how changing the parameters influenced our results, we built regression models to find the mathematical relationship between 'm', 'M' and 'n', and the metrics we were interested in. We used R package `stats` to build the following models:

- For the **mean coverage** (μ_{cov}) we found a significant linear relationship (p-value $<$ 2.2e-16) that is able to well explain the variability of our data (for a 10-fold cross validation repeated 30 times, $R^2 = 0.9980 \pm 0.0017$):

$$\mu_{cov} = -0.93293 + 2.88617 \cdot m + 0.07947 \cdot M \quad (3.1)$$

with 'm' being very significative (p-value $<$ 2.2e-16) and 'M' slightly less (p-value $=$

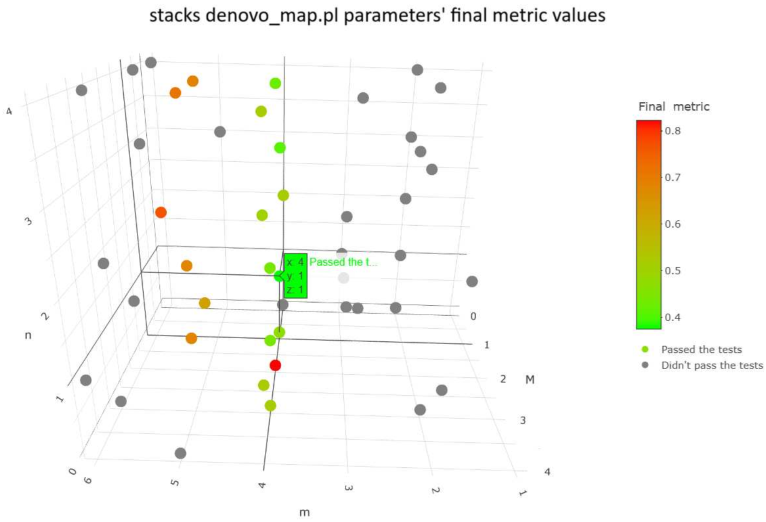


Figure 3.3: ***Score function values plotted for 45 combinations*** *3D plot of the combinations of* `Stacks: denovo_map.pl`*'s parameters tested, colored by score function (Equation 2.2) results. The best combination (m = 4, M = 1, n = 1) is highlighted.*

0.00244). The fact that 'n' is not significative at all is to be expected, since the coverage of samples is computed after the first step of the pipeline (`ustacks`) which uses only parameters 'm' and 'M'.

- The **standard deviation of the coverage** σ_{cov}, instead, depends only on the value of 'm', with p-value $< 2.2e\text{-}16$ and 10-fold cross validation (30 repetitions) $R^2 = 0.9149 \pm 0.1881$:

$$\sigma_{cov} = 0.32135 + 0.25138 \cdot m \quad (3.2)$$

The goal of parameter tuning is to find the correct parameters that result in an high coverage with a small standard deviation: unfortunately, increasing 'm' leads to higher values of both, so $m < 4 \vee m > 5$ both do not pass the requirement of minimum coverage.

- To summarize both quantities, we tried to build a model that describes the **coefficient of variation (CV)** as the ratio between standard deviation and mean of the coverage (p-value= 5.251e-08, 10-fold cross validation $R^2 = 0.8205 \pm 0.2667$):

$$CV = 0.189629 - 0.044843 \cdot log(m) \quad (3.3)$$

We found again that only the value of 'm' was significative, but the model was less robust than the previous ones so we decided to consider the two quantities separately;

- Building a model for the number of called SNPs and unique loci proved to be more complicated. When building a linear model using all parameters, we get a less robust model ($R^2 \sim 0.70$) than what was explored before with mean and standard deviation of coverage. Using only 'm' and increasing the complexity to the fourth polynomial would have raised the training R^2 up to almost 0.99, but it would have drastically reduced the test R^2 to levels as low as 0.59 ± 0.33 for SNPs and 0.52 ± 0.34 for unique loci, a textbook case of overfitting. The best model for the number of **SNPs** included all the parameters and was built to predict SNPs on a logarithmic scale. With p-value = 6.061e-06 and 10-fold cross validation (30 repetitions) $R^2 = 0.7603 \pm 0.2805$, the model is the following:

$$log(SNPs) = 9.08393 - 0.3486 \cdot m + 0.19492 \cdot M + 0.19022 \cdot n \tag{3.4}$$

- Similarly as before, for the number of **unique loci** we obtained:

$$log(unique\ loci) = 8.98525 - 0.36520 \cdot m + 0.13172 \cdot M + 0.13584 \cdot n \tag{3.5}$$

  with p-value = 1.157e-05 and 10-fold cross validation (30 repetitions) $R^2 = 0.7747 \pm 0.2645$.

We used these regression models to test their ability to predict how the metrics could change in any of the combinations we had not tried up to that moment. We used library `scikit` on Python which run these regression models on the whole search space discussed in the Materials and Methods chapters: the optimal solution found was $m = 4$, $M = 0$ and $n = 4$. We tried running these parameters since they had not been explored during training, and we show the comparison between observed and predicted results in Table 3.1.

Table 3.1: **Observed and Predicted metrics for combination $m = 4$, $M = 0$ and $n = 4$.**

Metric	**Observed**	**Predicted**
μ_{cov}	10.8323	10.5964
σ_{cov}	1.42895	1.3248
CV	0.1319	0.1250
SNPs	5264	4618
Unique loci	3274	4055

Comparison between the observed results when running `denovo_map.pl` pipeline, and the ones predicted by the regression models.

These values show that the models underestimate all results: undervaluing the mean coverage, SNPs and unique loci would have led this solution to be considered not ideal (big values of unique loci are a reward in the score function), but it also underestimated the penalty values, namely the standard deviation of the coverage (thus CV too) and the ratio of SNPs per locus. Underestimating the penalties, which are numerically more than the rewards and can easily outweigh them[1] in the model, led to an overall better score for this combination of parameters, which in reality showed a worse combination than combination $m = 4, M = 1, n = 1$. We concluded that those models that include 'M' or 'n' (i.e. the ones for SNPs and unique loci) do not greatly improve the results from those that only included 'm', so while the optimal 'm' is easily identified, the ideal 'M' and 'n' might be more difficult to find. These regression models are of course specific to the subset we built them on and cannot be used in general to predict the parameters in other contexts: we suggest to test datasets case by case, prioritizing the choice of 'm' over both 'M' and 'n'.

De novo assembly

In the pipeline `Stacks: denovo_map.pl`, the first step `ustacks` builds loci *de novo* in each sample: each individual has its own coverage, computed across its own loci. We obtained a mean coverage of 16.31 ± 4.37 ($median = 15.71$) over 103 samples across 10 populations: only one sample had less than 10x coverage (ALP6, having 8.94x) and only one sample had an outlier value (STE8, coverage = 41.09x; Figure 3.4).

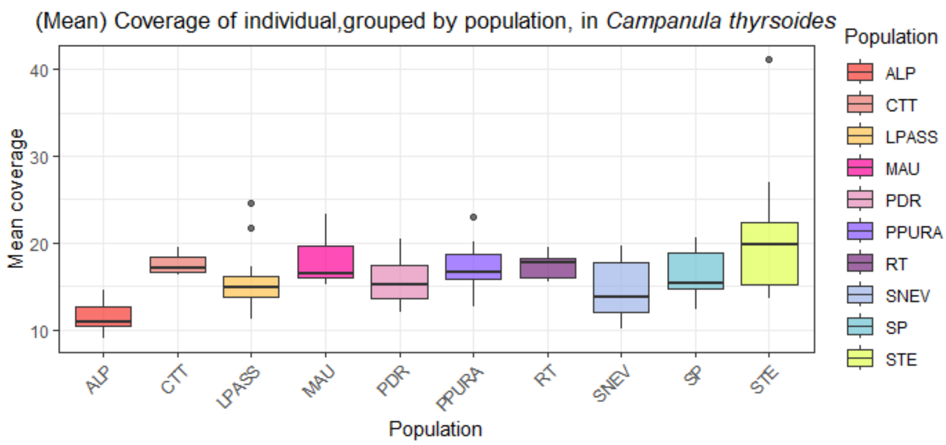


Figure 3.4: ***Coverage across samples*** *Boxplots of coverage of samples of C. thyrsoides, grouped by population.*

[1]In function 2.2 these quantities are all normalised, carrying the same weight $\in [0, 1]$.

After building the loci catalogue using `cstacks` and mapping each samples through `sstacks` as described in the Materials and Methods section, the second to last step is running program `gstacks`. It was able to build consensus sequences for 4,313,775 loci by assembling paired-end contigs for 99.9%[2] of them: of these, 97.7% overlapped (average overlapping region was 30.9 bp long). The average length of contigs was 103.1 bp. Then, out of 398 million paired-end reads, aligned 99.9% on the assembled contigs and performed SNP calling. Since we are in a *de novo* setting SNPs can not be exactly located on chromosomes, so their position was defined over the aforementioned assembled contigs. The estimated mean insert size was 110.1 bp ($\pm$ 49.3 bp). In total, 4,307,333 loci were genotyped with an effective mean coverage of 21.1x $\pm$ 7.2x (ranging from 10.9x to 61.6x). These results prove the high quality of the *de novo* assembly, with a large fraction of the assembled loci having sufficient read coverage and being successfully genotyped. This ensures that the downstream population genomic analyses are going to be as reliable as possible.

Filtering

Filtering out non informative and low-coverage SNPs is an important step of our pipeline for many reason, including but not limited to reducing F_{ST} bias (K. R. Andrews *et al.*, 2016). We tested two software, `population` and `vcftools`; when using the former, we run command:

```
1) populations -r 0.80 -R 0.80 --write-random-snp --min-
   mac 2
```

which kept 9556 out of a possible 126379 Sites. Instead, when using `vcftools` we run:

```
1) populations -r 0.80
2) vcftools--max-missing 0.9 --minDP 5 --min-meanDP 5 --
   thin 1000 --maf 0.005
3) populations --write-random-snp
```

which kept 3,896 out of a possible 126,379 Sites. To keep the comparison between the two methods simple, in the code above we only reported the filtering parameters and not those for input and generation of output files. While flags $-R$ 0.80 in `populations` and *–max-missing* 0.9 in `vcftools` equally define a minimum percentage of individuals to process a locus, the rest

[2]The remaining 0.1% of reads didn't successfully form a paired-end contig.

have a slightly different effect, although they try to apply the same filtering principle. When trying to filter out rare SNPs, we applied –*min*–*mac*2 in `populations` and –*maf*0.005 in `vcftools`. The first is criterion sets a minimum minor allele count process a SNP (**absolute** value), while the second defines a threshold for minor allele frequencies (**relative** value). We've noticed that while filtering using `populations` retains more SNPs (9556 vs 3896), it also increases the amount of missing data (Fig. 3.5, a), with the whole population of ALP having between 50 to 60% of missing SNPs information.

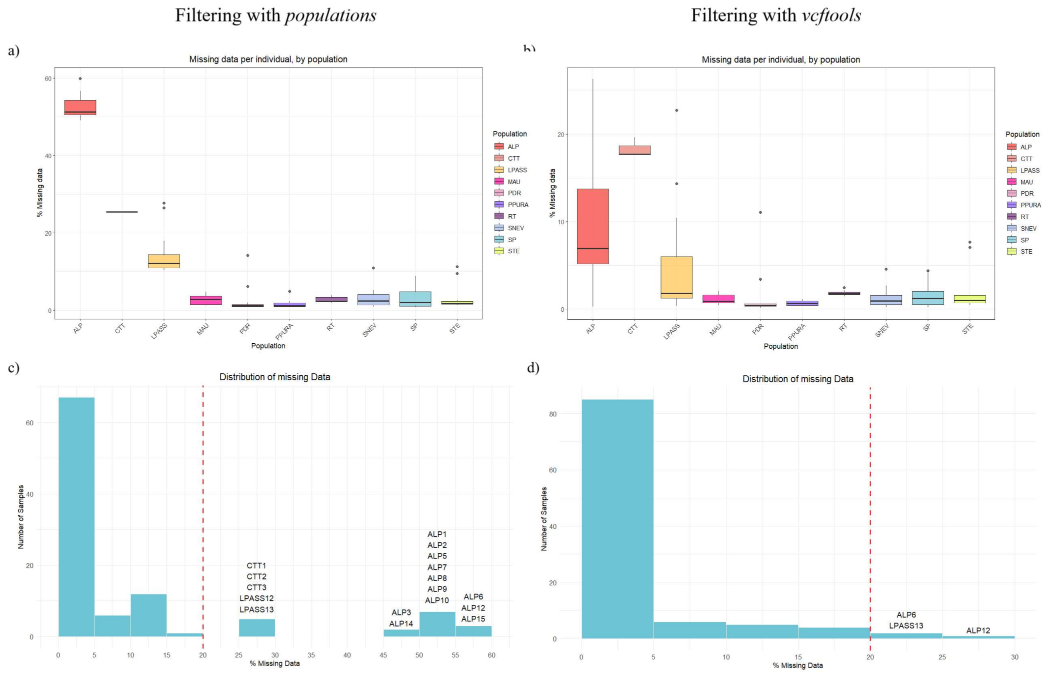


Figure 3.5: ***Comparison between filtering methods*** *Boxplots (a, b) and barplots (c, d) of missing data in each individual. When filtering through* `populations` *(a, c), these plots are based on 9556 SNPs, when filtering through* `vcftools` *(b, d) it is based on 3896 SNPs.*

This is not ideal, since the individuals from that population might cluster together just on the basis of missing the majority of the data, instead of real similarity between them. While populations of Alpago (ALP), Loiblpass (LPASS) and the outgroup *thyrsoides* (CTT) are still missing more data than the others even when using `vcftools` (Fig. 3.5, b), the distribution of the percentage of missing data drastically decreases. Not only that, but when setting a threshold of maximum 20% of missing data to be considered in the next analysis, when filtering through `populations` 17 samples would not pass this condition (Fig. 3.5, c), while when using `vcftools`, only 3 samples would be excluded (Fig. 3.5, d). This is why we ultimately chose to filter SNPs using `vcftools`.

3.1.2 Data analysis

Variant calling

We filtered the file using `vcftools` and run software `populations` again to compute populations' statistics (Table 3.2). Some loci in each population deviate from HWE: in literature they are often filtered out from the analysis (Kanaka *et al.*, 2023), but, because the percentage is small (0% to 5.1%) and their presence is expected when there is overlapping of generations and very small population sizes (Pearman *et al.*, 2022), we did not remove them from our dataset. The final array of SNPs comprehended 3896 polymorphic SNPs when including the outgroup CTT and 3558 when excluding it.

Table 3.2: **Populations' statistics**

Population	π	All sites	Variant	Polymorphic	Private alleles	Sites out HWE	% out HWE
ALP	0.098848	3896	3896	1223	135	128	3.29%
CTT	0.076954	3211	3211	480	297	0	0.00%
LPASS	0.103980	3896	3896	1206	183	106	2.72%
MAU	0.085217	3884	3884	859	49	91	2.34%
PDR	0.066905	3896	3896	750	107	127	3.26%
PPURA	0.112200	3896	3896	1523	157	197	5.06%
RT	0.129100	3855	3855	1411	287	88	2.28%
SNEV	0.093674	3896	3896	1098	115	116	2.98%
SP	0.102140	3896	3896	1369	172	140	3.59%
STE	0.119470	3896	3896	1576	364	141	3.62%

All statistics are calculated by `populations`. "π" represents per-SNP heterozygosity; "All sites" is the total number of loci analyzed in that population; "Variant" is the number of loci with at least one variant allele in the whole dataset; "Polymorphic" are loci variable within the population; "Sites out HWE" are variable sites that significantly deviate from Hardy-Weinberg equilibrium; "% out HWE" is the percentage of sites out of HWE, calculated over variant loci.

PCA analysis

The final set of 3,896 polymorphic SNPs revealed a good separation of the populations: in Figure 3.6 we show plots of the first three principal components for datasets including (a, b, c) and excluding (d, e, f) the outgroup CTT (subspecies *thyrsoides*). Almost all populations' clusters are well defined and separated. Furthermore, in the top plots, CTT separates from the subspecies *carniolica* along the third Principal Component (PC; 6.92% of variance explained), which confirms the divergence between the two subspecies and validates the choice of this population as the outgroup of the study. When excluding the outgroup CTT, the first three principal components are able to catch different characteristics in the dataset:

- PC1 (12.55% of explained variance), captures a longitudinal gradient in the distribution of the genetic diversity of the populations from East to West, starting from the Slovenian-Austrian populations and ending with the population from Lombardy (PDR). Moreover, it is able to further discern two groups of populations from Veneto and Friuli: a cluster that includes SP, MAU, PPURA and ALP and another formed only by SNEV, located on the border of Italy and Slovenia;
- Along PC2 (7.93% of explained variance) the populations of regions Veneto and Friuli (SP, MAU, PPURA, ALP, SNEV) group together towards positive values, while the other populations can be found towards negative ones. The Slovenian-Austrian populations, although separated from the Veneto-Friuli group, cluster together;
- PC3 (6.19% of explained variance) captures the separation of the three populations from outside of Italy (LPASS, RT and STE) that were clustering in the other PCs.

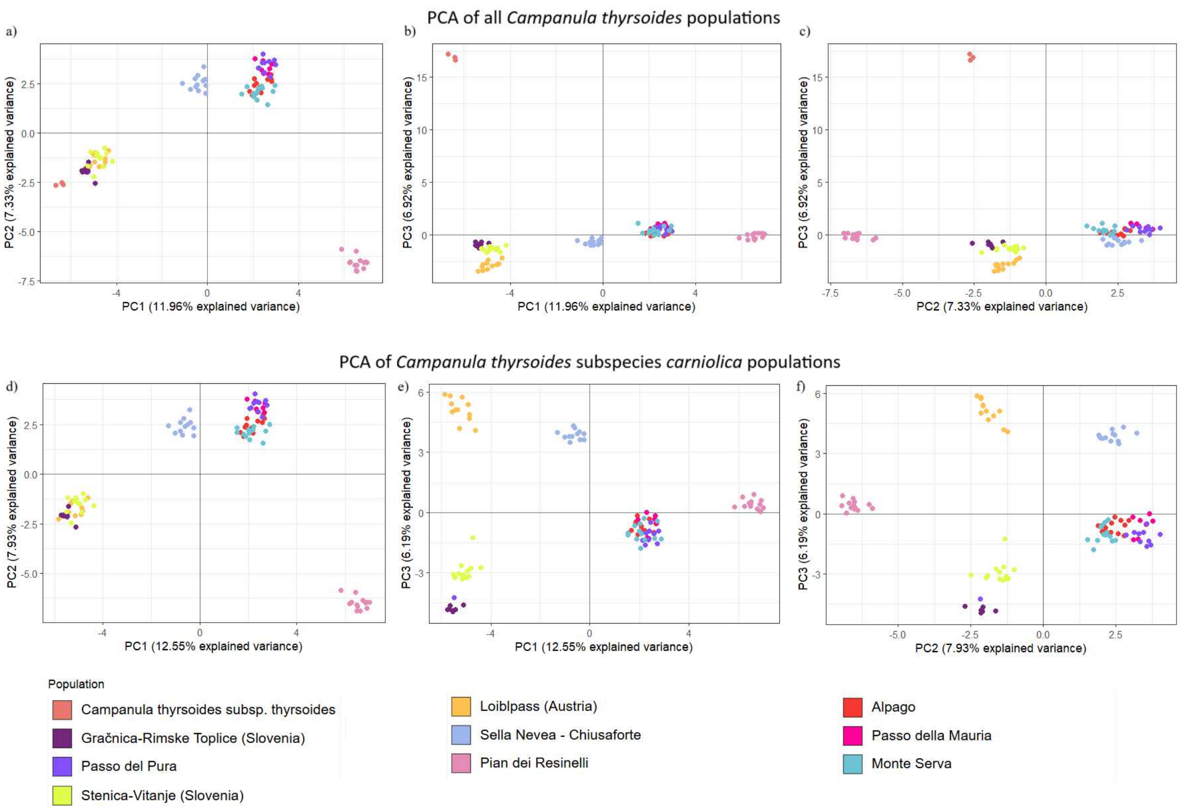


Figure 3.6: ***PCA plots of Campanula thyrsoides populations*** *The PCA plots with all the populations (a, b, c) were built over 3896 polymorphic SNPs, while the plots portraying only populations of subspecies carniolica (d, e and f) were based on 3558 SNPs.*

Phylogenetic analysis

The ancestry shared by populations was assessed using using Maximul Likelihood (ML) phylogentic trees and the programs `Admixture` and `STRUCTURE`. The unrooted ML consensus tree (Fig. 3.7) catches the big genetic distance of the outgroup CTT (subspecies *thyrsoides*) from subspecies *carniolica* with high support ($bootstrap = 100$).

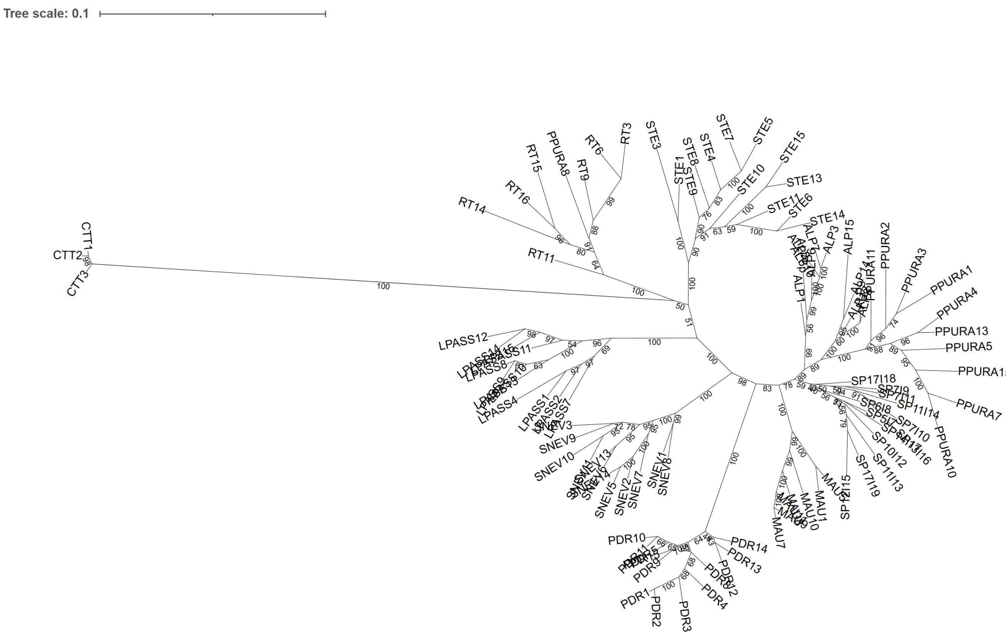


Figure 3.7: ***ML unrooted tree*** *ML consensus tree built with* `iqtree` *and visualized with* `iTOL` *(v7; Letunic and Bork, 2024).*

In the collapsed rooted tree in Figure 3.8, we can see that all population clades have bootstrap support equal to 100, with the exception of Monte Serva (SP) whose clade has a weaker support (59) and Alpago (ALP) whose individuals are split into two branches ($bootstrap = $ 99-100). The separation between Slovenian populations and the others, is supported by weak boostrap values: for example, population RT separates with $boostrap = 50$ and STE with $boostrap = 51$. The Austrian population (LPASS) and the whole Italian clade (SNEV, PDR, MAU, SP, ALP, PPURA) diverge from the Slovenian populations with boostrap support of = 100. The populations of SNEV and PDR are more closely related to the Italian populations than to the Slovenian and Austrian population, but they separate from the Italian populations with high support ($bootstrap = 100$ and $bootstrap = 98$). The population of Monte Serva is closely related to all the populations from the close Veneto-Friuli group (ALP, PPURA and MAU): although they form different clades the branches length is shorter in comparison to all the other populations, showing a close relationship between these samples.

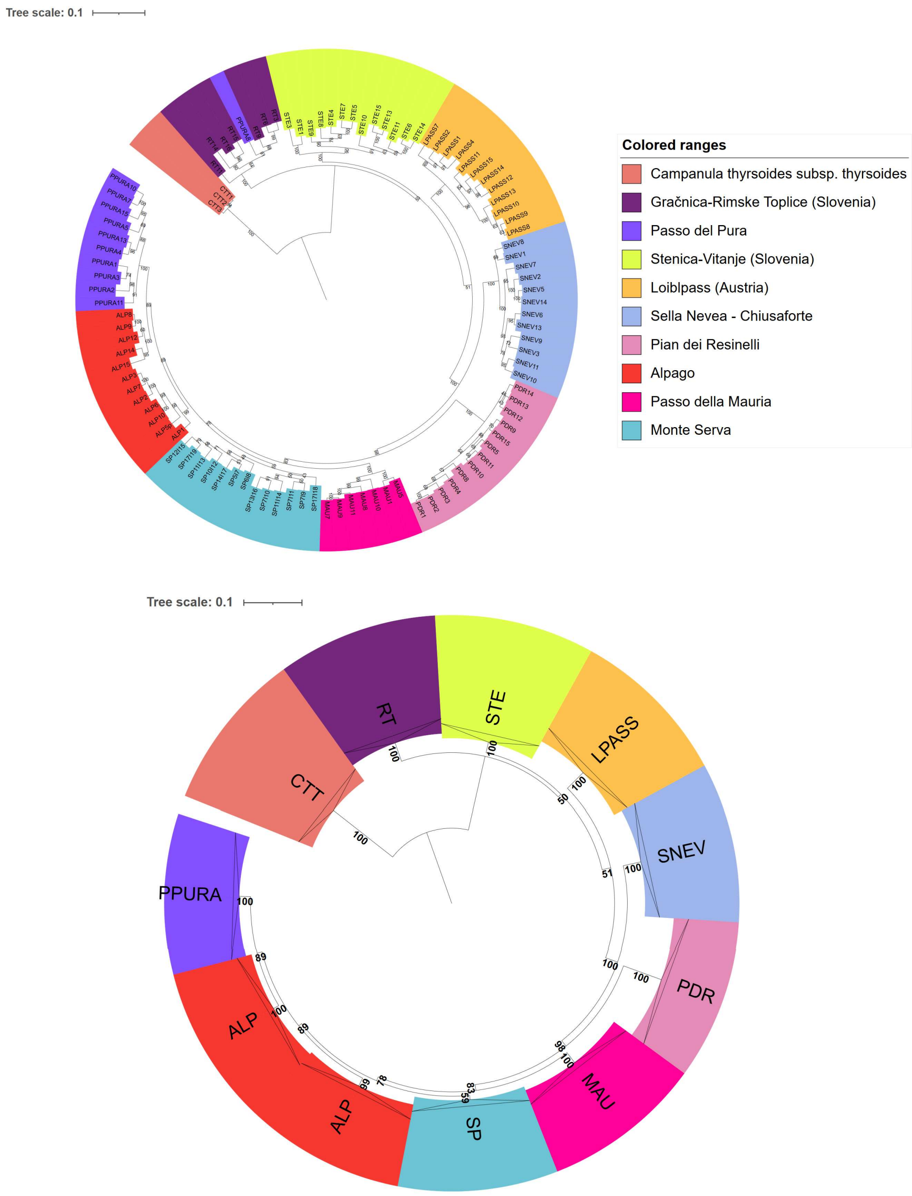


Figure 3.8: ***ML rooted tree*** *On the top, consensus maximum likelihood phylogenetic trees rooted on outgroup CTT clade, constructed with* `iqtree` *and visualized with* `iTOL`*. On the bottom, the same tree but with collapsed populations' clades for visualization purposes.*

With the use of `STRUCTURE` it is possible to study in more detail the shared ancestry between samples, by separating them in ancestral groups. As already mentioned in the methods section, we first used program `Admixture` to quickly assess a large range of K values: by running the program with default parameters in a range of $K \in [2,30]$ we found that $K = 7$ had the minimum value of BIC error (Figure 3.9). We then run program `STRUCTURE` in the neighbourhood of the optimal value $K = 7$ found by `Admixture`, so we tested for $K \in [1, 20]$.

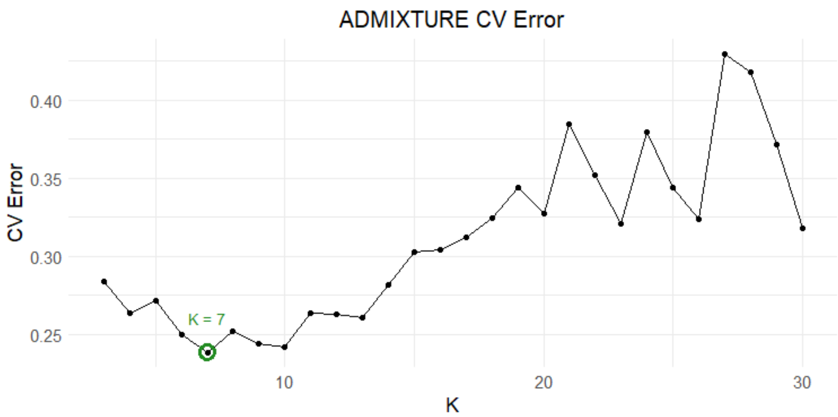


Figure 3.9: ***`Admixture` cross validation error against K values*** *Cross validation BIC error computed by `Admixture` to find the optimal K number of clusters. The minimum is found at $K = 7$.*

We employed both Evanno and Puechmaille methods to assess the best value of K: when including or excluding outgroup CTT. Evanno method found smaller optimal Ks (Figure 3.10) both because of the goal of this method (find the highest hierarchical level) and because uneven sampling leads to a downward-biased estimate of K (Puechmaille, 2016). Evanno detected two ancestral groups for the studied population ($K = 2$): one group included the samples of subspecies *thyrsoides* and the other *C. thyrsoides* subspecies *carniolica* populations.

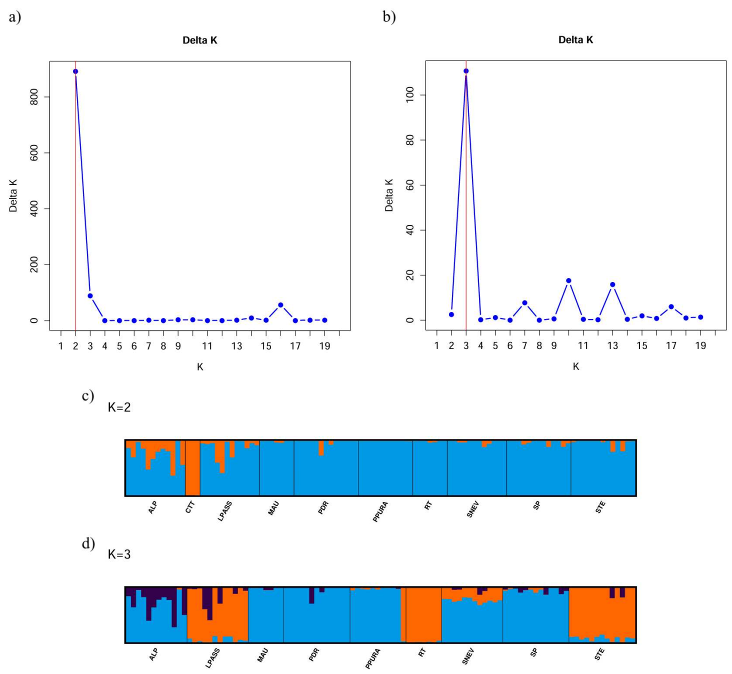


Figure 3.10: ***Structure results (Evanno method).*** *Results for `STRUCTURE` with (a) and without (b) outgroup CTT, using Evanno method to select the best number of K (maximum point) $\in [1, 20]$ clusters. Visual representation offered by StructureSelector (Y.-L. Li & Liu, 2018) and CLUMPAK (Kopelman* et al.*, 2015).*

ALP and LPASS showed a certain degree of introgression. When CTT was excluded from the analysis, K increased to 3 and it was possible to observe a clear separation between Italian (ALP, MAU, PDR, PPURA, SNEV, SP) and non-Italian populations (LPASS, RT, STE). ALP and LPASS showed an introgression pattern from a third ancestral group similar to that from $K = 2$, when CTT was included. Even if Puechmaille method performs better with uneven samples sizes (Puechmaille, 2016), the comparison between the two methods can show interesting dataset characteristics: as shown in Fig. 3.11, when CTT was included the best K found by Puechmaille method was $K = 7$. When CTT was excluded, two optimal values of K were found (6 and 7), depending on the estimator used.

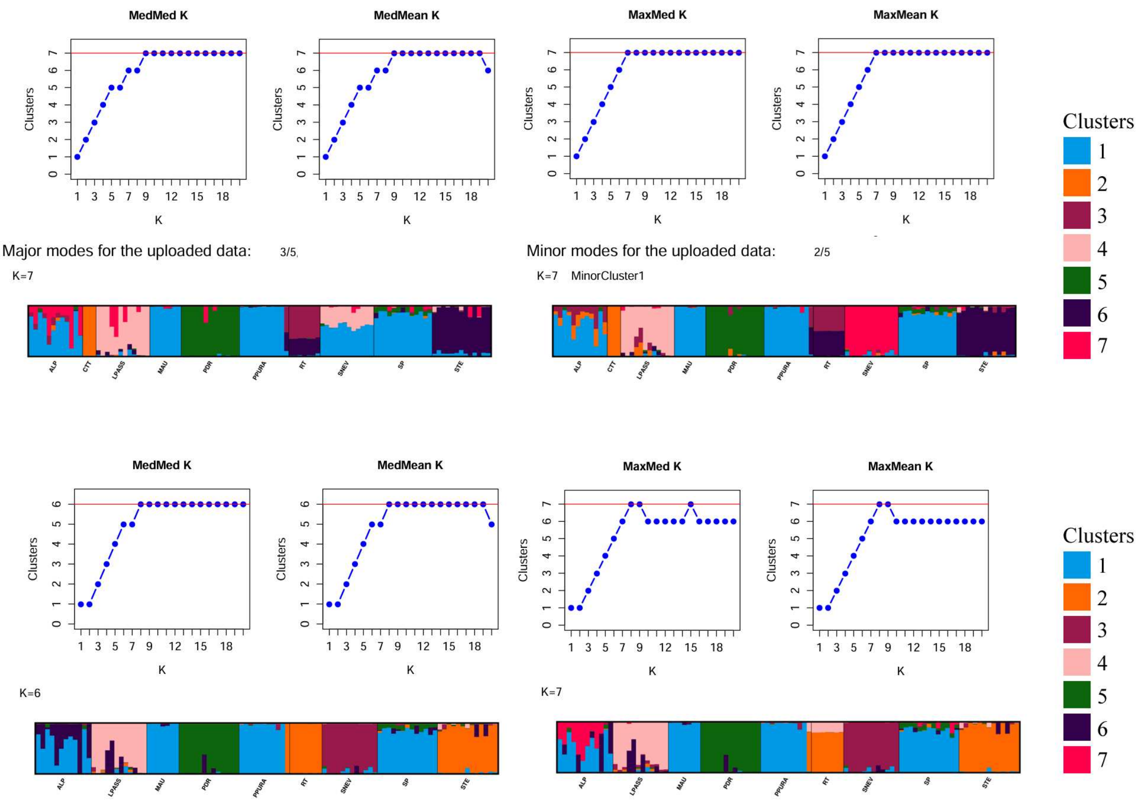


Figure 3.11: ***Structure results (Puechmaille method).*** *Results for* `STRUCTURE` *with (top) and without (bottom) outgroup CTT, using Puechmaille method to select the best number of K clusters. When CTT was involved, two main modes of clustering were equally represented (they're labelled as major and minor modes) so they are both shown in the figure. The maximum point in the plots corresponds to the optimal value of K. Visual representation offered by StructureSelector and CLUMPAK.*

Differently from PCA, when including CTT (top), the Slovenian populations do not cluster together (STE is part of cluster 6, and RT has admixed ancestry between cluster 3 and 6) while the exclusion of the outgroup groups RT and STE in the same cluster 2 (bottom plots). The uncertain relationship of these two populations is caught also by the low bootstrap support in the consensus tree (Fig. 3.8). In both cases, the populations SP, MAU and PPURA cluster together in both analyses (cluster 1), while ALP shows a mixed ancestry: the four populations constitute a Veneto-Friuli group. This group does not include SNEV, which is instead part of a different cluster. Interestingly, when CTT is included (Fig. 3.11, top plots) in 3 runs out of 5 SNEV shares ancestry with both the Veneto-Friuli samples (cluster 1) and the Austrian ones (cluster 4). In the remaining 2 runs, this population shows no admixture (cluster 7) like it does when CTT is excluded (Fig. 3.11, bottom plots, where it belongs solely to cluster 3). This is in agreement with the weak bootstrap of 51 observed in the phylogenetic tree. Finally, as PCA (Fig. 3.6), phylogenetic trees (Fig. 3.8) and `STRUCTURE` (Fig. 3.11) showed that sample PPURA8 was likely a sample of population RT (Gračnica-Rimske Toplice, Slovenia) that got mis-labelled during DNA extraction.

Inbreeding metrics

To assess within population diversity, we computed the inbreeding factor (F_{IS}) using R package `hierfstat`: this metric ranges from -1 (no inbreeding) to 1 (strong inbreeding). In Table 3.3 we reported the 95% confidence intervals we obtained from 1000 bootstrap iterations: none of the values is positive, nor the ranges contain 0, so we can confidently state no population suffers from inbreeding. In fact, all values are strongly negative, going from a minimum of -0.76 for population CTT to a maximum of -0.31 for PPURA. This confirms good gene flow and no risk, at least currently, of inbreeding depression for any of the populations.

Table 3.3: F_{IS} **95% confidence intervals for each population.**

Pop	$C.I._{95\%}$	Pop	$C.I._{95\%}$
ALP	$[-0.3784, -0.3419]$	**PPURA**	$[-0.3501, -0.3152]$
CTT	$[-0.7673, -0.7221]$	**RT**	$[-0.3823, -0.3497]$
LPASS	$[-0.4128, -0.3799]$	**SNEV**	$[-0.4266, -0.3903]$
MAU	$[-0.4663, -0.4236]$	**SP**	$[-0.3645, -0.3286]$
PDR	$[-0.5390, -0.4919]$	**STE**	$[-0.3651, -0.3337]$

F_{IS} 95% confidence intervals estimated per population.

To quantify population differentiation we computed pairwise F_{ST} metric using R package `StAMPP`, which uses the formula by Weir and Cockerman Weir and Cockerman, 1984 and corrected by Weir and Hill Weir and Hill, 2002 for few, uneven and small populations sizes: the results are reported in Table 3.4. As seen before, MAU, ALP, SP and PPURA are close together, since they show low to moderate differentiation ($F_{ST} < 0.20$) with Monte Serva's population (SP) being closest to ALP ($F_{ST} = 0.08$) and PPURA ($F_{ST} = 0.09$), as what was already observed in the consensus tree (Fig. 3.8). As expected, the population that is most genetically distant from all the others is CTT ($0.50 < F_{ST} < 0.68$). We used F_{ST} values to test Isolation by Distance (IBD). For this analysis we excluded the outgroup samples CTT to better capture the pattern of geographical distribution of genetic diversity within subspecies *carniolica*.

Table 3.4: **Pairwise F_{ST} (all p-values $< 10^{-16}$).**

Pop	ALP	CTT	LPASS	MAU	PDR	PPURA	RT	SNEV	SP
CTT	0.56								
LPASS	0.26	0.55							
MAU	0.16	0.61	0.30						
PDR	0.28	0.68	0.39	0.35					
PPURA	0.10	0.52	0.24	0.13	0.28				
RT	0.27	0.50	0.25	0.30	0.40	0.22			
SNEV	0.20	0.57	0.26	0.24	0.35	0.19	0.28		
SP	0.08	0.54	0.24	0.14	0.26	0.09	0.24	0.19	
STE	0.22	0.50	0.20	0.24	0.34	0.20	0.19	0.22	0.20

Pairwise F_{ST} values among populations. The matrix is symmetrical; only the lower half is reported.

IBD can be assessed employing the Mantel test which, as previously stated, can use Pearson, Spearman and Kendall correlation metrics. Following Table 2.2 we tested for Pearson's r assumptions: no outliers, linearity and normality of data (Fig. 3.12). We checked for the presence of outliers by means of a boxplot (Fig. 3.12, a) and found none. To test for linearity we first visually inspected the residuals for homoscedasticity (Fig. 3.12, b) and then used a Studentized Breusch-Pagan test with 1 degree of freedom. The null hypothesis (H_0) is that residuals are distributed with equal variance. Since we obtained p-value = 0.3047, we could not reject the null hypothesis, thus we confirmed presence of homoscedasticity. Lastly, we show QQ-plot of residuals (Fig. 3.12, c) to check for normality, which is confirmed by Shapiro-Wilk normality test (p-value = 0.6462). Since all assumptions were met, we used Pearson metric and we found a strong correlation ($r = 0.817$, p-value = 8e-04) between geographic and genetic distance. This also reflects the longitude gradient we observed in the PCA.

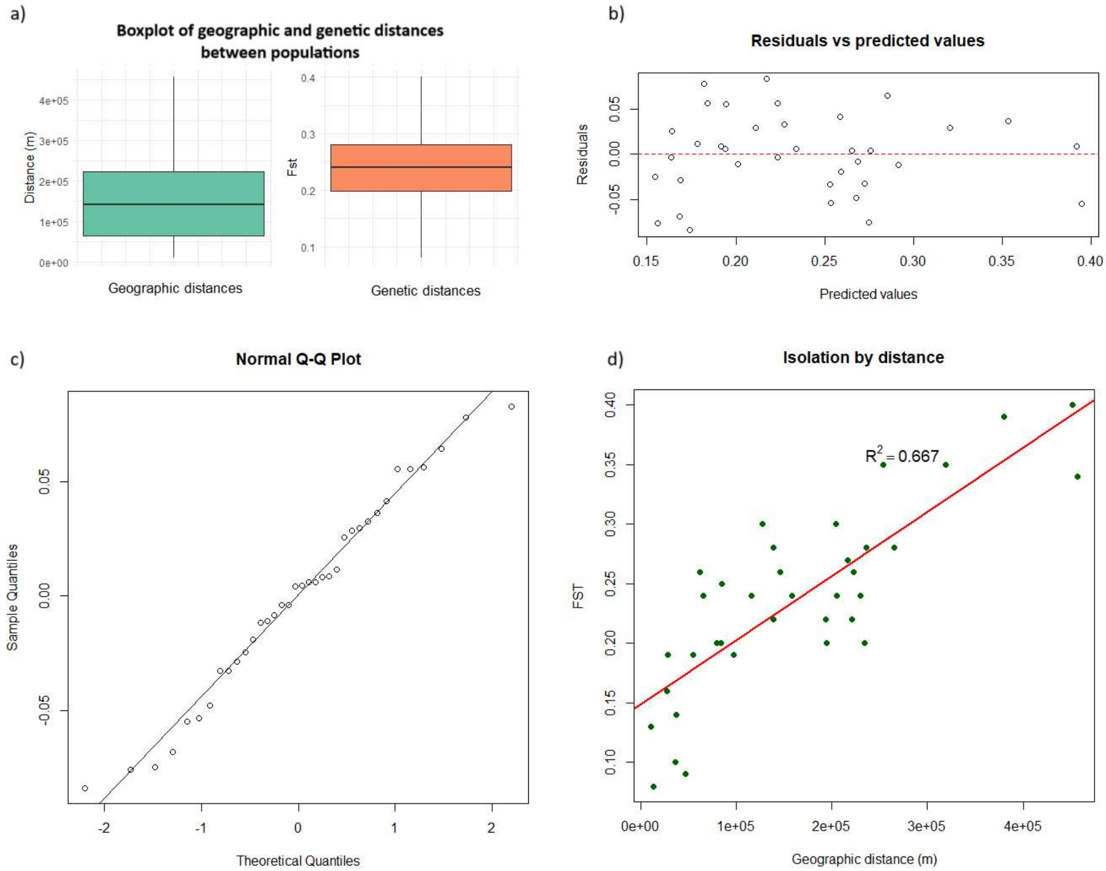


Figure 3.12: ***Mantel test assumptions and results*** *To employ Pearson's correlation in the Mantel test, we checked the presence of outliers (a) and tested for linearity (b) and normality (c) of residuals. In (d) the scatterplot of data (in green) with the regression line (red) corresponding to model* $F_{ST} = 1.488e\text{-}01 + 5.384e\text{-}07 \cdot distance$ *(*$R^2 = 0.667$*, p-value* $= 1.222e\text{-}09$*).*

Comparison of software for F_{ST} estimation

Although we ultimately used the F_{ST} values as computed by package `StAMPP`, we tested 2 software and 2 R packages as well, in order: `stacks`: `populations`, `vcftools`, `BEDASSLE` and `hierfstat`. They can be roughly separated depending if they estimate an unweighted (Group 1: `populations`, `hierfstat` and unweighted `vcftools`) or weighted F_{ST} value (Group 2: weighted `vcftools`, `BEDASSLE`, `StAMPP`): within their group, all software perform similarly, but the difference between the two formulas is noticeable when observing the color scale of the heatmaps of their results (Figure 3.13). We grouped together the software in the aforementioned groups and computed the average $\bar{F}_{ST}^{\text{Group}}$ for each pair of populations. Then we performed a (non parametric) Wilcoxon paired test, with H_0 : median$(\bar{F}_{ST}^{\text{Group 1}} - \bar{F}_{ST}^{\text{Group 2}}) = 0$ and H_A : median$(\bar{F}_{ST}^{\text{Group 1}} - \bar{F}_{ST}^{\text{Group 2}}) < 0$. There is statistical significance showing that the software using unweighted estimates of F_{ST} tend to produce lower median values than those that use the weighted formula (p-value $= 2.842 \cdot 10^{-14}$).

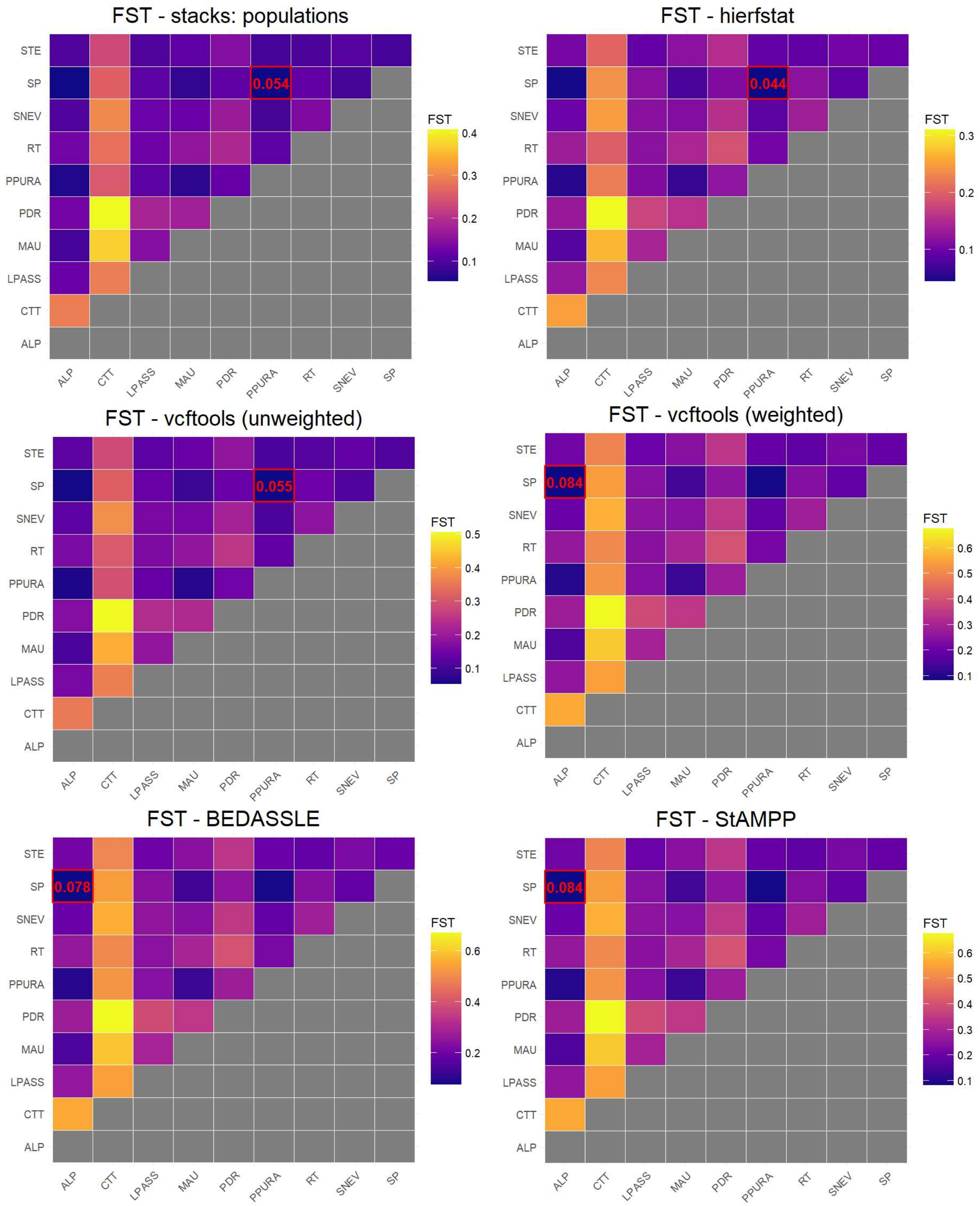


Figure 3.13: ***Heatmaps of software estimates of F_{ST}*** *We compare software performance by plotting heatmaps of F_{ST} estimates for each software tested. In red, the lowest value for Monte Serva population (SP), target of the study.*

Group 1 always underestimated the results in comparison to Group 2: despite the difference in absolute values, this shows that the relationships between populations (computed as the genetic distance between each other) behaved consistently. Populations that are closer in comparison to others, remain the same even if another software is employed to estimate this quantity. It is the absolute value of this distance that changes, but in the case of close populations the difference is minimal: for example, the population closest to the target population SP (Monte Serva) changes when using unweighted (PPURA) or weighted estimates (ALP) but this divergence is not alarming, since PPURA is the second-lowest result for weighted estimates, and ALP for the unweighted ones.

The situation is slight different when considering distant populations: for example, the out-group CTT, when using unweighted estimates, is the furthest from PDR with $F_{ST} = 0.51$ in (unweighted) `vcftools`. As shown by all the software in Group 2, it is still the most distant from PDR, but the value is as large as $F_{ST} = 0.68$ in `StAMPP`: a difference of 0.17 in other analysis could easily lead to two very distinct conclusions. Nevertheless, we believe that `StAMPP` was the best choice for our analysis, both considering the assumptions already discussed in the Material and Methods Chapter, and its ability to compute the associated p-values. The two other weighted estimates software also have their advantages: R package `BEDASSLE`, similarly to `StAMPP`, is able to compute confidence intervals, which can give insight on the strength of its results. And `vcftools`, which is a software that many researchers are already familiar with, is able to operate on the terminal and with faster computations, so `.vcf` files do not need to be exported and converted to the appropriate input format which R packages expect.

3.2 *Solanum tuberosum* landraces

We performed variant calling for the genus *Solanum*, with broadly the same pipeline used for *Campanula* but accounting for the presence of a reference genome. The final `.vcf` file contained 14,940 SNPs, and only one sample had more than 20% missing data (the sample OL_PIET, Oltrepò di Pietragavina, with 23.2% missing data), while the rest had less than 10% of missing data (Fig. 3.14).

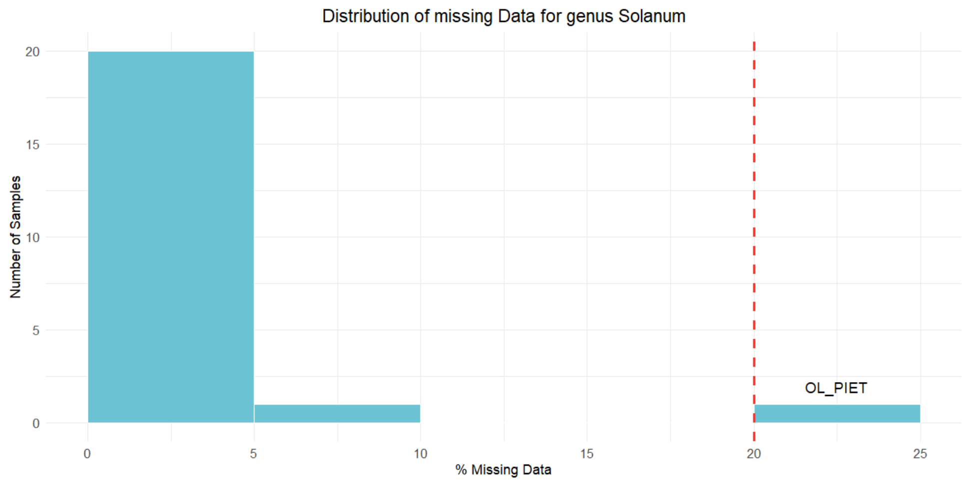


Figure 3.14: ***Missing data distribution of Solanum genus' samples.*** *This barplot shows the distribution of the missing data (in percentage) over 14,940 SNPs: a threshold of 20% is highlighted by a vertical red line.*

When we included the data from the ancient and wild species 14,940 SNPs were retrieved (Fig. 3.15, a). Instead, when we excluded all the ancient/wild species and we used only *S. tuberosum* subsp. *andigenum* as the outgroup, the final matrix contained 10,496 SNPs (Fig. 3.15, b, c and d). In the PCA including all species (Fig. 3.15, a), the first principal component explains 27.6% of the variance and separates cultivated and wild species. In fact, cultivated species are all distributed towards negative values of the x axis, with the cultivated Andean species grouping together in the Cluster A and all commercial cultivars and landraces in Cluster C, and *S. tuberosum* subsp. *andigenum* located between clusters A and C. All the wild species group together in cluster B towards positive values of the x axis, distant from cultivated species. When only *S. tuberosum* subspecies *andigenum* is used as outgroup and plotted, the first PC (20.25% of variance explained) separates white and red peel potatoes. Cluster C is split in Cluster D (white peel potatoes close to commercial cultivar Kennebec) and Cluster E (red peel potatoes close to commercial cultivar Desiree). One landrace, Bianca (in yellow) is placed towards the middle point between white and red peel landraces. Obsolete cultivars (Bossico and Shilpario) and Starleggia Bianca are not close to the other landraces, being placed between *S. tuberosum* subsp *andigenum* and cluster D (along PC2, Fig. 3.15, b).

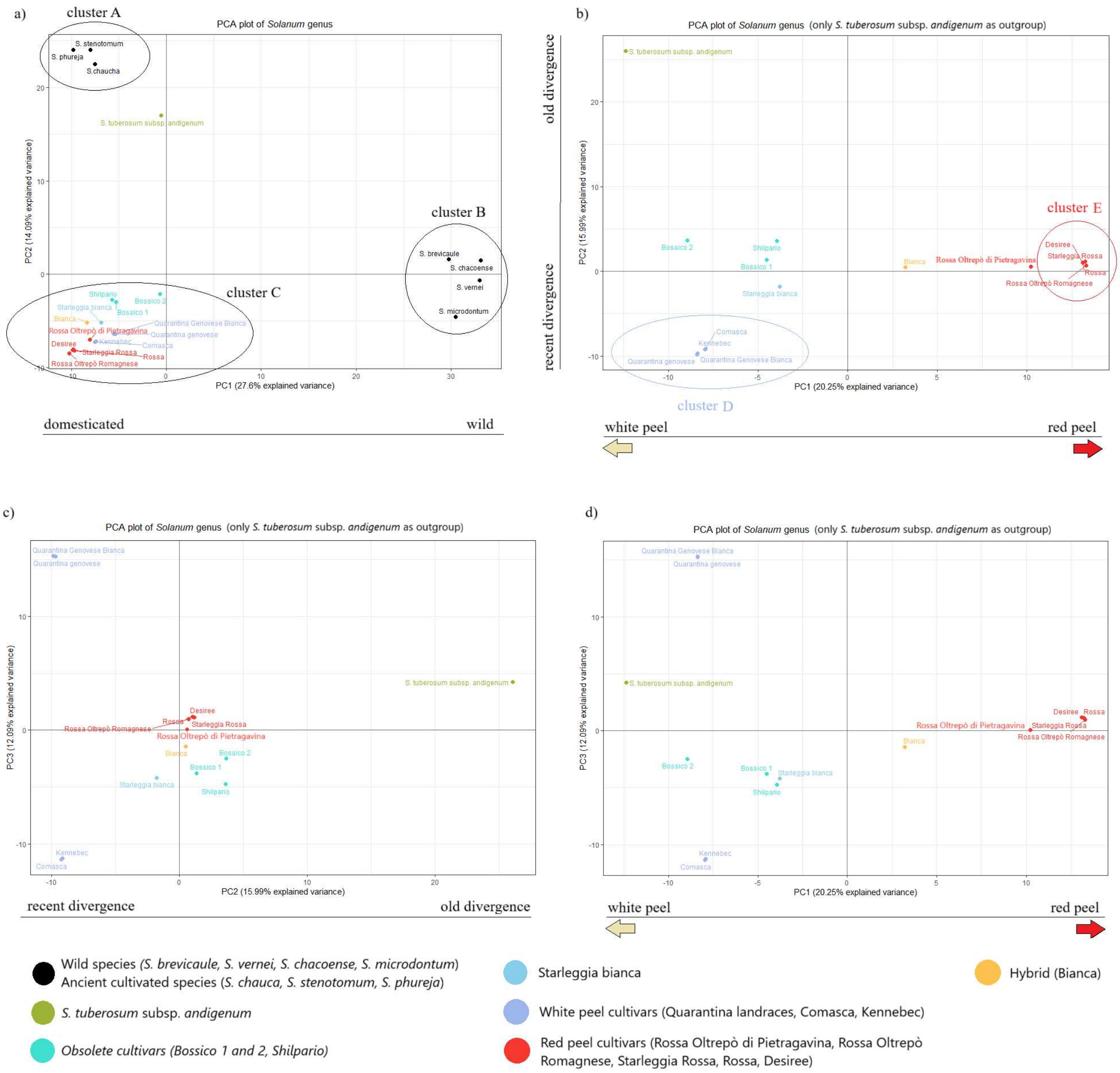


Figure 3.15: ***Solanum genus samples' PCA plots.*** *PCA plots of either (a) all Solanum genus samples in the study or only landraces and S. tuberosum subsp. andigenum as outgroup (b, c, d). We display three plots to show the relationship between the first three PCs, as from the fourth one the proportion of variance explained is* $< 10\%$. *Please refer to the legend to distinguish the samples: (i) black: wild/ancient species, (ii) green: S. tuberosum subsp. andigenum, (iii) cyan: obsolete landraces (Bossico and Shilpario), (iv) light blue: Starleggia Bianca, (v) blue: white peel landraces close to commercial Kennebec, (vi) red: red peel landraces close to commercial Desiree, (vii) yellow: hybrid landrace Bianca.*

To better appreciate the 15.99% of variance caught by PC2 (Fig. 3.15), we can integrate the information coming from the maximum likelihood phylogenetic tree (Figure 3.16), it is again possible to see an evident division between wild species and cultivated species/landraces, found at opposite ends of the tree: the former are located on the left branches, while the latter on the right[3]. It also shows that wild species, in green, grouped with high support ($bootstrap = 100$) and a big genetic distance, reflected by the length of the branch, from the rest of the samples. This guided our choice of rooting the tree on this clade (Fig. 3.17), considering them as an outgroup.

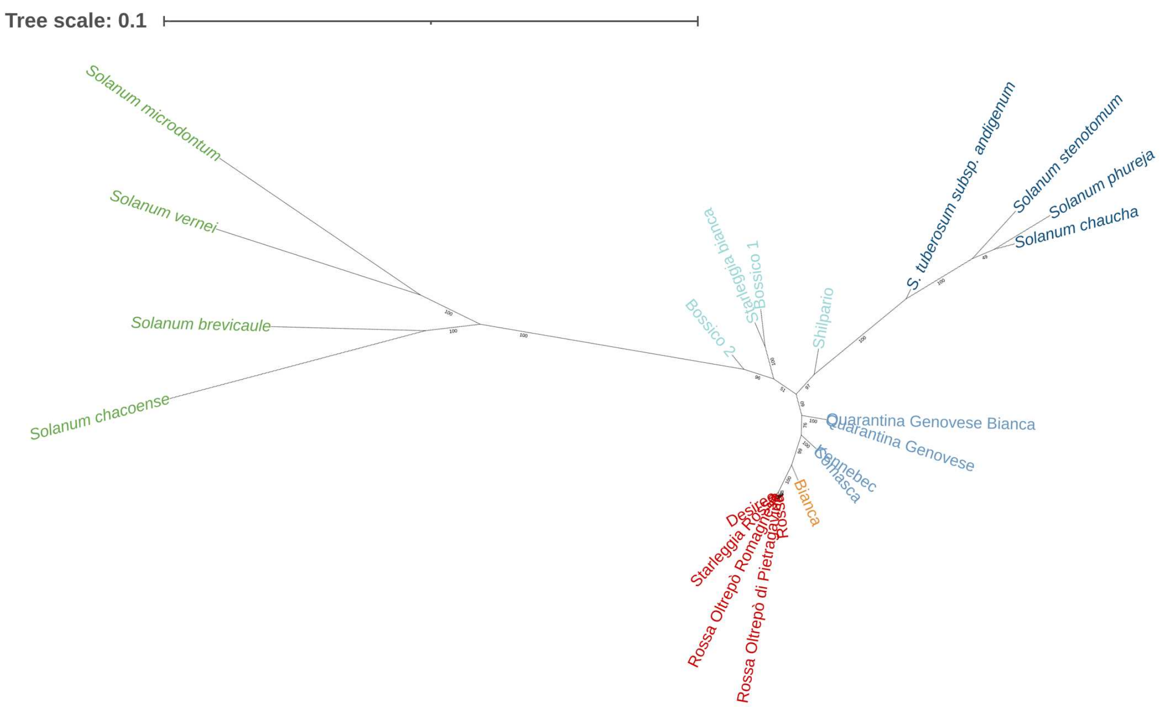


Figure 3.16: ***Unrooted maximum likelihood tree of Solanum genus and landraces.*** *Phylogenetic maximum likelihood (consensus) tree.*

The rooted tree in Figure 3.17 shows the same results as the unrooted tree (Fig. 3.16), with the addition of a putative root of the tree set on the wild species clade. The tree shows, with high bootstrap support, that Bossico 2 ($bootstrap = 96$), Bossico 1 and the cultivar Starleggia Bianca (together $bootstrap = 100$), are the closest to the wild species (i.e. *S. vernei*, etc...), and not, as we would expect, the Andean cultivated ones (i.e. *S. stenotomum*, etc...). Bossico 1, Bossico 2 and Starleggia Bianca are therefore phylogenetically closer to the Andean cultivated species. With an equally high support ($bootstrap = 97$), the obsolete landrace Shilpario from the Botanical Garden of Bergamo, separated from the cultivated Andean species long before the commercial variants Kennebec and Desiree. These relationships between cultivated Andean species and the obsolete landraces was caught by PC2 in the PCA plots (Fig. 3.15, b and c). All the red-peel cultivars consisted of very closely related genotypes with $bootstrap = 100$ and

[3] Left and right are relative to this specific tree which was plotted using the web tool iTOL, with custom rotation for visualization purposes.

very short branches length. Furthermore, the internal branches of this clade have low boostrap support; for example, Rossa Oltrepò Romagnese, Starleggia Rossa and Desiree separate from the other two red peel variants with a weak bootstrap support of 59, and the relationships are similar for the other landraces in this group. This suggested that red peel landraces are not old Italian landraces but clones or cultivars closely related to the commercial variant Desiree.

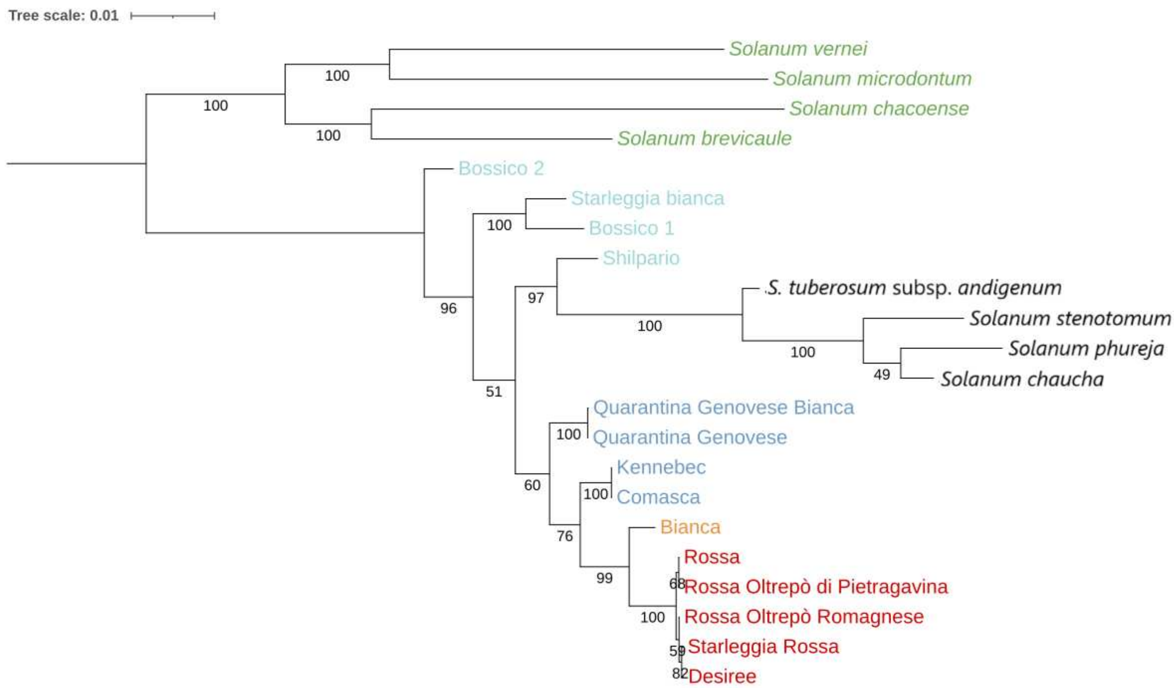


Figure 3.17: ***Rooted maximum likelihood tree of Solanum genus and landraces.*** *Phylogenetic maximum likelihood (consensus) tree rooted with the most distant clade of wild species. Tips colors correspond to those in the PCA plot (Fig. 3.15).*

The situation is more complicated when observing white peel variants: they do not cluster as close together as the red peel potatoes did in both PCA plots (Fig. 3.15) and in the phylogenetic trees (Fig. 3.16 and 3.17). Landrace Bianca is located halfway through white and red peel variants with a high support ($bootstrap = 99$) , suggesting being the result of hybridization between red and and white peel cultivars, or involved in the domestication of red cultivars. All the remaining landraces, even if all belong to the same white peel category, do not cluster in the same clade: we can confirm that Kennebec is more similar to Comasca and that the two Quarantina ($bootstrap = 100$) are close relatives of the commercial cultivar Kennebec as well, since their division in two different groups is not well supported ($bootstrap = 60$). As mentioned before, the remaining 4 landraces (Bossico 1 and 2, Starleggia Bianca and Shilpario) are not included in this cluster. Bossico 1, Bossico 2 and Starleggia Bianca are well separated from all the other cultivated species, both Andean and Italian white and red peel variants. Shilpario is, instead, more closely related to the Andean cultivated species ($bootstrap = 51$) than to the modern cultivated cultivars, and the relationship with the modern commercial cultivars is not even supported ($bootstrap = 51$), suggesting an older domestication origin of Shilpario than of Kennebec and Desiree.

3.3 *Anacamptis* spp.

The dataset of *Anacamptis* samples (Table 2.4) was explored using the same pipeline of *C. thyrsoides*, since the study setting was similar. After running `Stacks`: `denovo_map.pl`, we obtained a mean coverage of 17.93 ± 5.69 ($median = 17.24$) over 90 samples. In comparison to *C. thyrsoides* dataset that contained only one sample with less than 10x coverage, *Anacamptis* had six (B_GB1, B_GB2, P_BA1, P_FA3, P_MA2, P_VO2): however, their coverage fell just out of the threshold of 10 ($min = 8.77$) and moreover they were distributed among all populations. `gstacks` built consensus sequences for 5,462,291 loci, with an average contig length of 244.3 bp. Out of 565 million paired-end reads, 99.1% were successfully aligned on them, with a mean insert length of 266.1bp (±69.1). In total, 5,462,274 loci were genotyped with an effective mean coverage of 31.0x ± 14.0x (ranging from 9.4x to 77.7x).

3.3.1 Variant calling

After filtering, we identified 13,791 polymorphic non-rare SNPs: we retained more SNPs in comparison to *C. thyrsoides*, both in absolute values and in percentage (0.09% versus 0.25% of the genotyped loci). However, differently from *C. thyrsoides*, more samples showed an higher rate of missing data, with one sample, P_NI3, reaching 39.63% of missing data (Fig. 3.18).

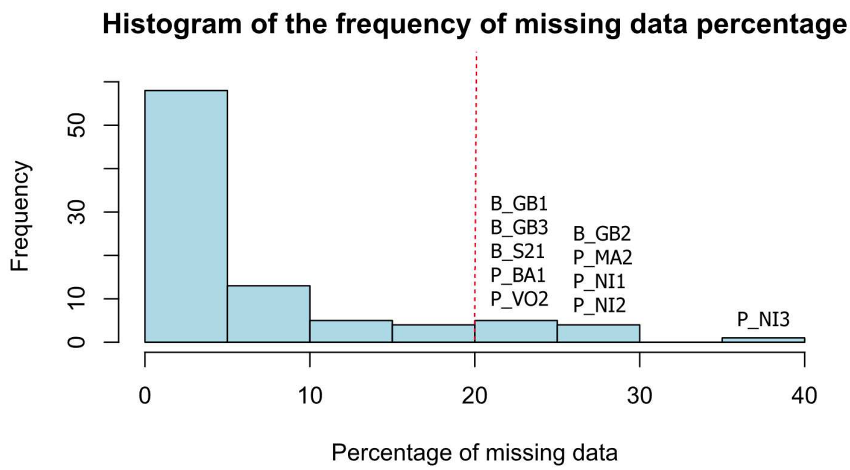


Figure 3.18: ***Histogram of missing data for Anacamptis samples.*** *This plot shows the distribution of missing data percentage: a red dashed line highlights a 20% threshold.*

We explored the distribution of missing data in higher detail, by plotting boxplots divided by population (Fig. 3.19, a) and by habitat type (Fig. 3.19, b). We have noticed that urban and natural populations of genus *Anacamptis* had a similar distribution of missing data: between the two, a total of 10 samples had over 20% of missing data, but they were all outliers so they are not specific to any type of habitat. When we compared the boxplots divided by populations we observed that those outliers corresponded to whole populations, namely B_GB, B_S an P_NI. This is not ideal since samples from these populations could cluster together based on their percentage of missing data and not on actual similarity between them. Since we are mostly interested in the comparison between urban and natural populations, we have decided to not discard these samples.

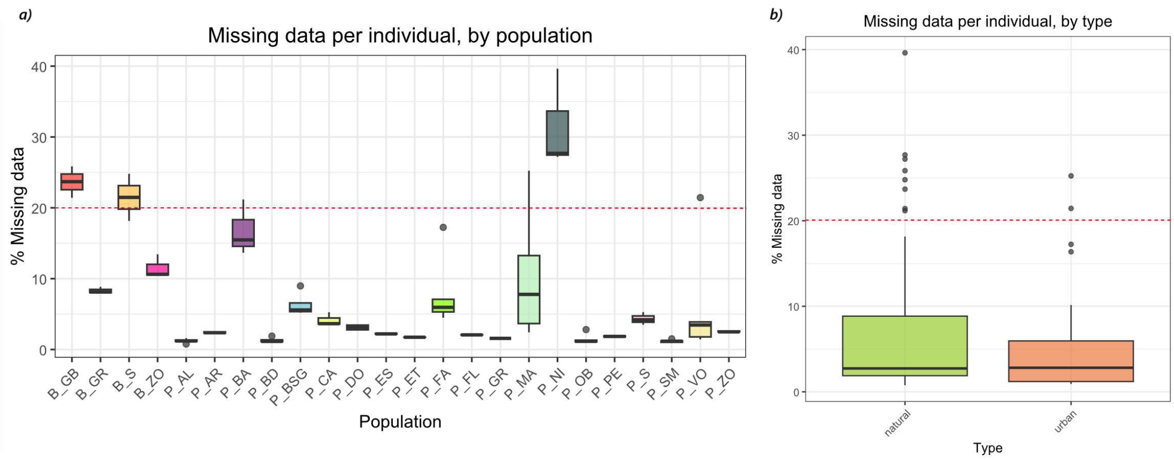
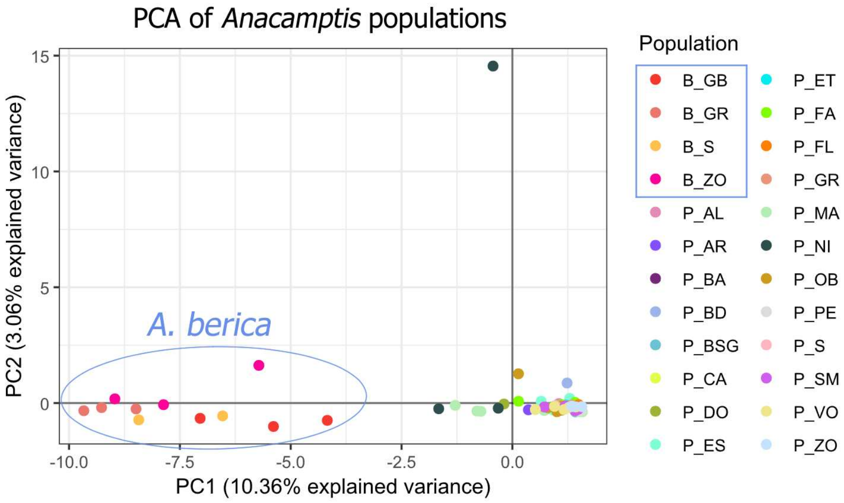


Figure 3.19: ***Boxplot of missing data for Anacamptis samples.*** *These two plots show missing data distribution in orchids samples, divided by population (a) and by type (b). A red dashed line highlights the 20% threshold.*

3.3.2 Spatial separation

The PCA plots in Figure 3.20 used the previously built matrix of 13791 SNPs. The percentage of explained variance drastically decreased to a 1.5%-2% plateau after PC2, indicating that showing more than these Principal Components would not be informative. The first component, explaining up to 10.36% of the total variance in the dataset, could mislead to divide natural, on negative values, and urban populations towards the origin (Fig. 3.20, bottom), but after coloring the samples by population (Fig. 3.20, top) it appeared obvious that it had caught the difference between *A pyramidalis* and *A. berica*: this separation by species has the highest proportion of variance explained. The samples of *A. pyramidalis* do not cluster in natural and urban populations even when *A. berica* was excluded from the analysis. PC2 is not able to separate the samples further: all the samples group around the x axis (y = 0), aside from one, P_NI3 that, as we have shown in Figure 3.18, is the sample with the largest percentage of missing data which is probably the reason of its distance from the others.



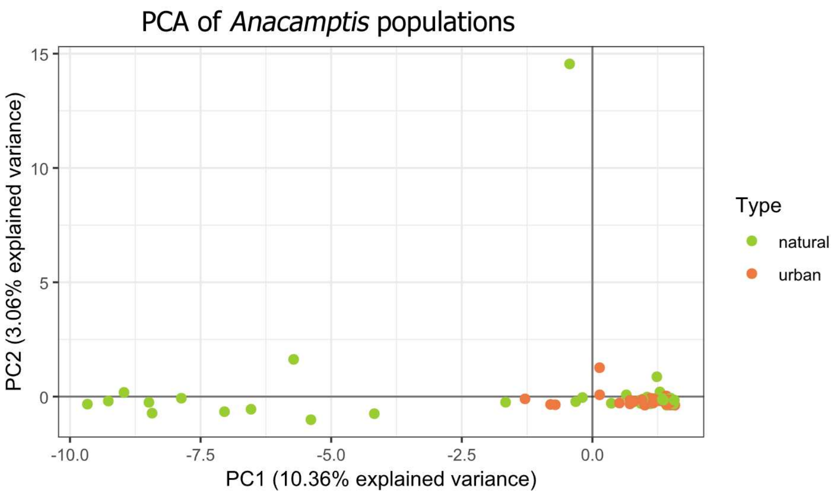


Figure 3.20: ***PCA plot for Anacamptis samples.*** *These plots are built on the same matrix of 13791 SNPs. On the top, samples are colored by population, on the bottom by type.*

It is possible to appreciate the same relationships when calculating a ML consensus tree (Fig. 3.21). *A. berica* did not form a clade as some samples (B_ZO3, B_GB1 and B_GB2) were more closely related to *A. pyramidalis* so it was impossible to root the tree on a putative *A. berica* clade. These samples separate from the main group of *A. berica* samples with high boostrap support: 93 for B_GB1, 90 for B_ZO3 and 94 for B_GB2. Nevertheless, these samples have longer branches in comparison to *A. pyramidalis* showing a bigger genetic distance.

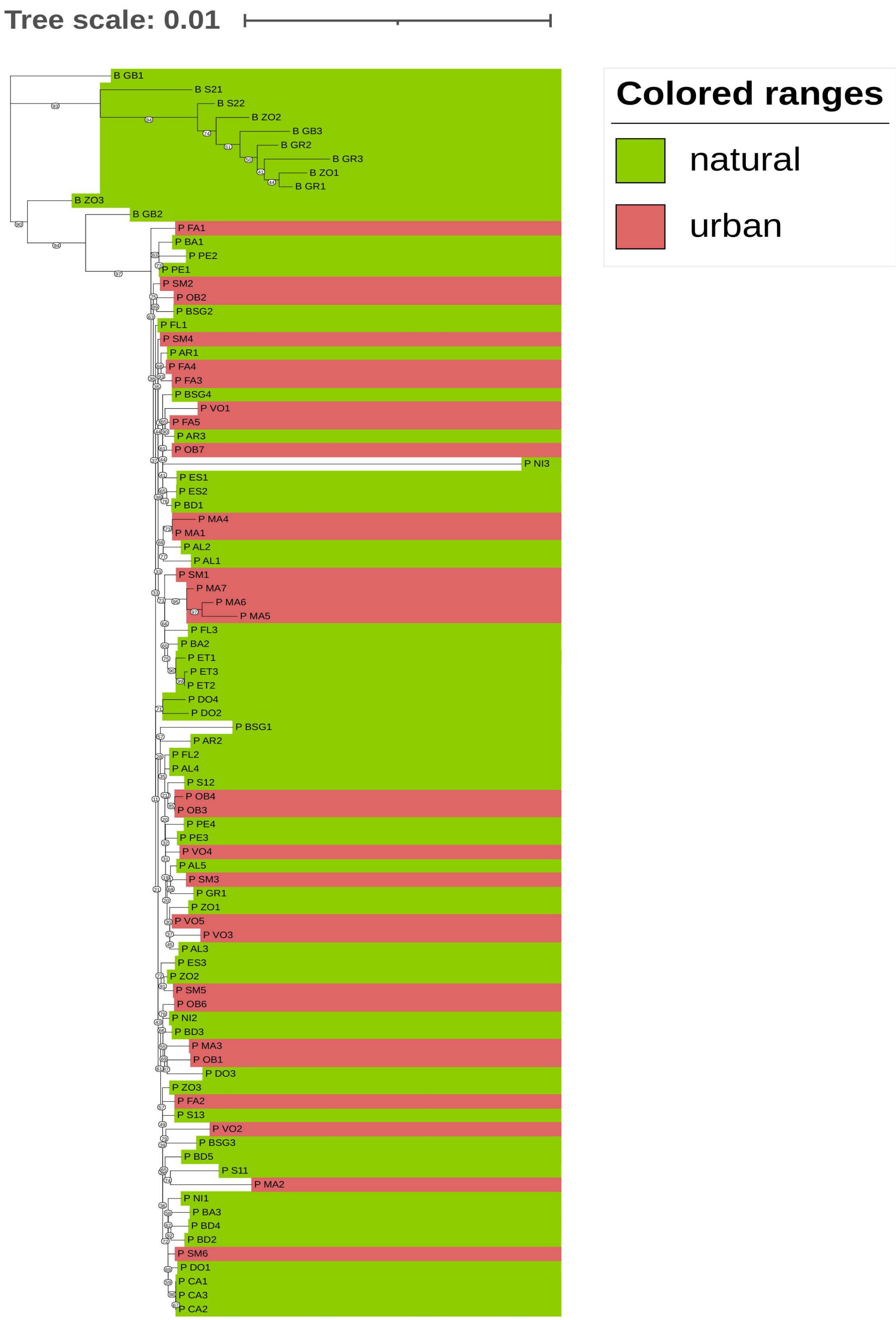


Figure 3.21: ***Rooted maximum likelihood tree of Anacamptis populations.*** *Unrooted phylogenetic maximum likelihood (consensus) tree built using* `iqtree` *(best-fit model = TVM+F+R3) and plotted using* `iTol`*. Colors ranges reflect the dichotomy of urban and natural populations, reporting Table 2.4 information.*

The longest branch length corresponds to P_NI3, but, as we discussed before, this might be simply an artefact of its large percentage of missing data. This sample should have probably been removed from the analysis. To reconstruct populations structure we followed a simpler approach than what was used for *C. thyrsoides*: we first explored a broad range of number of K clusters (reflecting the total number of samples in the dataset) using `Admixture`, and found the theoretical optimal value as the minimum 5-fold CV error (Fig. 3.22). We repeated the same analysis including and excluding outgroup *A. berica* and noticed that the CV error did not reach a clear minimum as in *C. thyrsoides* and instead fluctuated until it reached a plateau around $K = 50$ both with or without *A. berica*. The theoretical optimal values are $K = 80$ for the dataset including *A. berica* and $K = 71$ in the other case, but they are clearly one of the many low values of the plateaus reached by the software and are unlikely the correct number of ancestral groups.

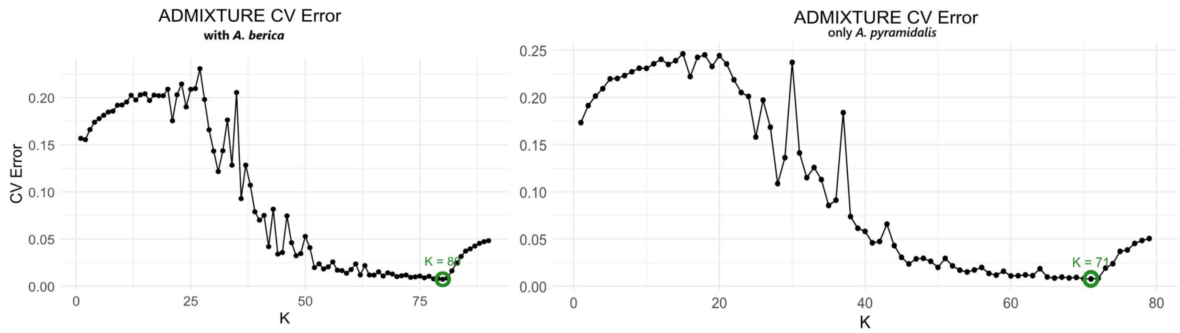


Figure 3.22: ***Cross validation error versus number of K clusters.*** *5-fold cross validation error as computed by* `Admixture` *when including populations of A. berica (left,* $K \in [1,90]$*) and when excluding them (right,* $K \in [1,79]$*). In green the minimum CV error value.*

Considering the complexity of the dataset and the uncertain result of the minimum CV values selected by `Admixture`, we explored our data using `STRUCTURE` testing K values chosen visually from Figures 3.22. We used the same *burnin* and *iterations* parameters as for *C. thyrsoides* but we raised the number of runs to 10. When including *A. berica* we explored $K = 2$ because it corresponded to a local minimum and we wanted to test if the population structure reflected the two species division like it had already happened to subspecies *thyrsoides* and *carniolica* and indicated by the PCA and ML tree. Then, we filtered the dataset to include only *A. pyramidalis* (we kept 15365 SNPs) and performed 10 runs with $K = 16$, which, again, corresponded to a local minimum. Results are shown in Figure 3.23. When $K = 2$, `STRUCTURE` catches some of the division between *A. berica* and *A. pyramidalis* (top plot), but the division between the two species is not as clear as it was between the *C. thyrsoides* subspecies: both samples of *A.*

berica and *A. pyramidalis* show admixture, including natural (P_BA) and urban (P_FA, P_MA) *A. pyramidalis* populations. When $K = 16$ some of the populations show little to no admixture: P_BA, P_BSG, P_CA, P_PE, P_NI show little to none while P_AR, P_DO, P_ES and P_S share only a little part of their ancestry with the rest of the samples.

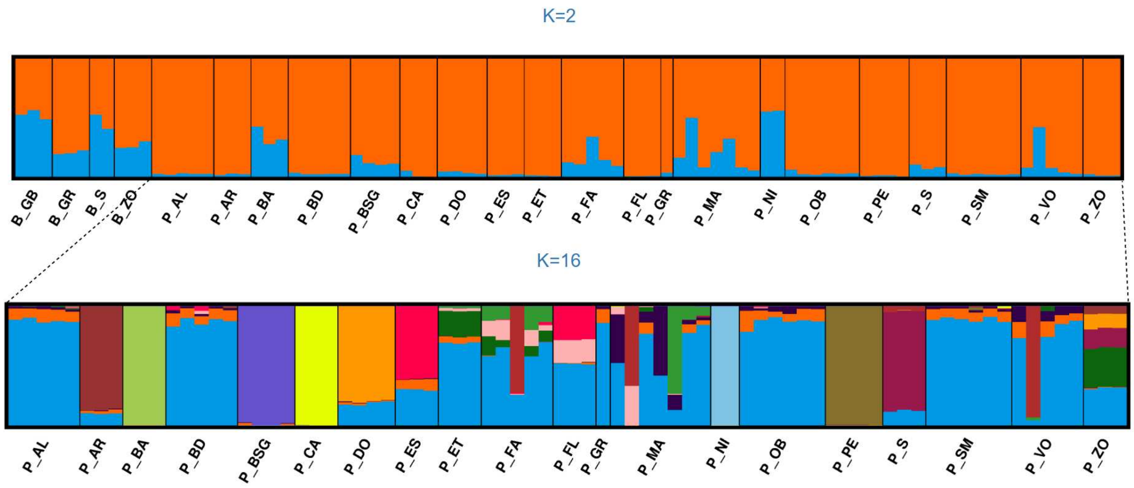


Figure 3.23: `STRUCTURE` ***results for Anacamptis populations structure.*** *Results of 10 runs of* `STRUCTURE`*, plotted using* `CLUMPAK`*. On top, full dataset and $K = 2$, on the bottom A. berica was excluded and $K = 16$.*

3.3.3 Inbreeding metrics

F_{IS} 95% indices were computed with R package `hierfstat` over 2000 iterations: the results are reported in Table 3.5.

Table 3.5: F_{IS} **95% confidence intervals for populations of *A. pyramidalis* and *A. berica*.**

Pop	$C.I._{95\%}$	Pop	$C.I._{95\%}$	Pop	$C.I._{95\%}$
B_GB	$[-0.2167, -0.1759]$	**P_ET**	$[-0.3041, -0.2376]$	**P_MA**	$[-0.0814, -0.0636]$
B_GR	$[-0.2754, -0.2340]$	**P_FA**	$[-0.0686, -0.0457]$	**P_NI**	$[-0.0723, -0.0423]$
B_S	$[-0.5238, -0.4292]$	**P_FL**	$[-0.1497, -0.1039]$	**P_OB**	$[-0.0603, -0.0418]$
B_ZO	$[-0.2778, -0.2332]$	**P_GR***	*not valid*	**P_PE**	$[-0.1668, -0.1228]$
P_AL	$[-0.0677, -0.0431]$	**P_BA**	$[-0.1050, -0.0683]$	**P_S**	$[-0.0882, -0.0574]$
P_AR	$[-0.0521, -0.0299]$	**P_BD**	$[-0.0525, -0.0322]$	**P_SM**	$[-0.1189, -0.0900]$
P_BSG	$[-0.0527, -0.0337]$	**P_CA**	$[-0.4046, -0.2873]$	**P_VO**	$[-0.0753, -0.0520]$
P_DO	$[-0.0664, -0.0450]$	**P_ES**	$[-0.0660, -0.0366]$	**P_ZO**	$[-0.0707, -0.0376]$

*None of the C.I. contain 0: with 95% confidence, all the populations have inbreeding factors weakly to strongly negative. *It was not possible to compute F_{IS} for P_GR as this population is represented by only one individual.*

No population shows inbreeding, but there is weak evidence that populations of *A. berica* have lower F_{IS} values than those of *A. pyramidalis* (Welch t-test p-value = 0.0213). Natural populations seem to be less inbred (p-value = 0.0123): this is not supported when *A. berica* is excluded from the analysis (p-value = 0.1124). After we tested F_{ST} estimation software on *Campanula* dataset, we used *Anacamptis* to compare the first software used (`populations`) against the one we chose as optimal for *C. thyrsoides* (`StAMPP`). `StAMPP` is able to compute p-values associated with each estimate of F_{ST} (in the bottom table 3.6 non significant values are highlighted in grey) so we still used this software to compute our results, reported in the top table 3.6. As for F_{IS}, population P_GR is too small (1 individual) for statistically significant computations. Out of 23 pairwise F_{ST} value, 10 have $p - value > 0.05$ indicating gene flow among the populations.

Table 3.6: **Pairwise F_{ST} values (above) and p-values (below).**

Pop	B_GB	B_GR	B_S	B_ZO	P_AL	P_AR	P_BA	P_BD	P_BSG	P_CA	P_DO	P_ES	P_ET	P_FA	P_FL	P_GR	P_MA	P_NI	P_OB	P_PE	P_S	P_SM	P_VO
B_GR	0.025																						
B_S	0.045	0.033																					
B_ZO	0.010	-0.001	0.047																				
P_AL	0.184	0.258	0.329	0.231																			
P_AR	0.127	0.200	0.260	0.176	0.021																		
P_BA	0.146	0.222	0.284	0.197	0.054	0.024																	
P_BD	0.190	0.259	0.330	0.233	0.017	0.012	0.036																
P_BSG	0.155	0.227	0.279	0.205	0.020	0.005	0.028	0.016															
P_CA	0.189	0.269	0.390	0.257	0.075	0.077	0.125	0.070	0.069														
P_DO	0.141	0.207	0.251	0.180	0.028	0.011	0.036	0.020	0.012	0.068													
P_ES	0.130	0.205	0.270	0.180	0.018	0.003	0.036	0.006	0.006	0.104	0.012												
P_ET	0.167	0.254	0.343	0.235	0.091	0.092	0.123	0.098	0.085	0.223	0.091	0.101											
P_FA	0.151	0.211	0.258	0.186	0.022	-0.001	0.023	0.016	0.018	0.056	0.024	0.002	0.053										
P_FL	0.164	0.244	0.329	0.224	0.035	0.027	0.063	0.011	0.022	0.153	0.031	0.037	0.144	-0.001									
P_GR	-0.043	0.106	0.154	0.092	0.051	-0.011	0.060	0.052	-0.022	0.362	-0.014	0.027	0.226	-0.028	0.172								
P_MA	0.147	0.198	0.234	0.174	0.031	0.008	0.027	0.026	0.020	0.037	0.026	0.007	0.064	0.027	0.013	-0.043							
P_NI	0.051	0.096	0.117	0.067	0.075	0.014	0.030	0.068	0.049	0.077	0.044	0.011	0.074	0.045	0.047	-0.207	0.064						
P_OB	0.184	0.237	0.303	0.213	0.021	0.010	0.034	0.011	0.020	0.055	0.029	0.007	0.085	0.009	0.011	0.023	0.032	0.068					
P_PE	0.204	0.273	0.356	0.250	0.057	0.057	0.064	0.054	0.057	0.142	0.069	0.072	0.160	0.056	0.091	0.159	0.053	0.090	0.055				
P_S	0.139	0.210	0.275	0.189	0.038	0.028	0.054	0.039	0.020	0.122	0.029	0.029	0.116	0.024	0.056	0.034	0.017	0.034	0.034	0.090			
P_SM	0.213	0.274	0.344	0.249	0.050	0.035	0.056	0.036	0.051	0.083	0.057	0.047	0.120	0.044	0.051	0.090	0.053	0.102	0.038	0.074	0.073		
P_VO	0.151	0.218	0.261	0.193	0.018	0.009	0.030	0.022	0.017	0.059	0.030	0.007	0.057	0.013	0.021	-0.023	0.033	0.052	0.013	0.054	0.033	0.048	
P_ZO	0.148	0.227	0.306	0.202	0.008	0.006	0.043	0.004	0.002	0.114	0.012	0.011	0.115	-0.010	0.037	0.063	0.010	0.026	0.002	0.066	0.031	0.040	0.002

Pop	B_GB	B_GR	B_S	B_ZO	P_AL	P_AR	P_BA	P_BD	P_BSG	P_CA	P_DO	P_ES	P_ET	P_FA	P_FL	P_GR	P_MA	P_NI	P_OB	P_PE	P_S	P_SM	P_VO
B_GR	–																						
B_S	–	–																					
B_ZO	0.015	0.626	–																				
P_AL	–	–	–	–																			
P_AR	–	–	–	–	–																		
P_BA	–	–	–	–	–	–																	
P_BD	–	–	–	–	–	–	–																
P_BSG	–	–	–	–	–	0.065	–	–															
P_CA	–	–	–	–	–	–	–	–	–														
P_DO	–	–	–	–	–	–	–	–	–	–													
P_ES	–	–	–	–	–	0.169	–	0.062	0.069	–	–												
P_ET	–	–	–	–	–	–	–	–	–	–	–	–											
P_FA	–	–	–	–	–	0.618	–	–	–	–	–	0.227	–										
P_FL	–	–	–	–	–	–	–	0.016	–	–	–	–	–	0.582									
P_GR	1.000	–	–	–	0.006	0.704	0.004	0.005	0.867	–	0.786	0.101	–	0.965	–								
P_MA	–	–	–	–	–	0.001	–	–	–	–	–	0.012	–	–	–	0.999							
P_NI	–	–	–	–	–	–	–	–	–	–	–	0.002	–	–	–	1.000	–						
P_OB	–	–	–	–	–	0.002	–	–	–	–	–	0.026	–	–	0.011	0.106	–	–					
P_PE	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–				
P_S	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	0.046	–	–	–	–			
P_SM	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–		
P_VO	–	–	–	–	–	0.001	–	–	–	–	–	0.022	–	–	–	0.906	–	–	–	–	–	–	
P_ZO	–	–	–	–	0.028	0.069	–	0.138	0.261	–	0.001	–	–	0.996	–	–	0.001	–	0.307	–	–	–	0.248

Pairwise F_{ST} values (top) and corresponding p-values (bottom), where '-' is used for p-values $< 10^{-16}$. Cells in grey indicate non-significative comparisons ($p > 0.05$).

To assess whether weighted and unweighted estimates of F_{ST} differ for *Anacamptis* genus as they did for *C. thyrsoides*, we computed a Wilcoxon paired test with $H_0 : \text{median}(F_{ST}^{\text{populations}} -$

$F_{ST}^{\text{StAMPP}}) = 0$ and H_A : median$(F_{ST}^{\text{populations}} - F_{ST}^{\text{StAMPP}}) \neq 0$. There is a weak statistical proof that the two estimates differ (p-value for the two-sided test = 0.01957): differently from *C. thyrsoides*, for orchids of genus *Anacamptis* there is statistical significance that median F_{ST} values for `populations` (which computes unweighted estimates) are larger than those computed by `StAMPP`, which utilizes a weighted formula (p-value for the one-sided test = 0.009785). The significance of this test, even if acceptable and higher than the one obtained for the two-sided test, is lower than the one obtained when comparing software performance in *C. thyrsoides* ($0.009785 > 2.842 \cdot 10^{-14}$), so more testing should be performed preferably on different subsets. It will be able to assess if it is a repeatable and predictable behavior or if over/under-estimating F_{ST} estimates depends on the species under study. Interestingly, when excluding population P_GR from the test, there was no evidence that `populations` overestimated results in comparison to `StAMPP` ($p - value = 0.1704$) or even that the two software estimated F_{ST} differently ($p - value = 0.3408$). We then compared average values (computed only on significative values) between populations of species *A. pyramidalis*, *A. berica* and between the two species. We found no support (Mann-Whitney U test $p - value = 0.2749$) that the average F_{ST} within populations of *A. pyramidalis* (0.053 ± 0.046) is different than the average F_{ST} within populations of *A. berica* (0.032 ± 0.015) , so they posses similar within-species gene flow. To characterize the source of variation in the dataset, we performed an AMOVA (Analysis of MOlecular VAriance) test using R package `poppr` (v2.9.8; Kamvar *et al.*, 2015), by grouping samples depending on, in hierarchical order: their species, urban/natural habitat provenance and population. The test assigned a negative value to the proportion of variance explained by the urban/natural dichotomy, confirming that there is no difference between the two groups: for this reason we repeated the same test by excluding this hierarchical level, and reported our findings in Table 3.7.

Table 3.7: **AMOVA results: components of covariance and Φ-statistics.**

Source of variation	**% of total**	**Statistic**	**Value (Φ)**
Between species	32.253	$\Phi_{\text{samples–total}}$	0.367
Between populations within species	4.451	$\Phi_{\text{samples-species}}$	0.066
Within samples	63.296	$\Phi_{\text{species–total}}$	0.323

AMOVA summary showing variance components, percentage of total variation, and fixation indices (Φ-statistics) across hierarchical levels.

We found that almost two thirds (63.296%) of the genetic variance can be found between different individuals. Only a small fraction (4.451%) can be attributed to their belonging to specific populations (within species) while almost a third of the variance is found between the two different species (32.253%). Moreover, this test gives insights on the ratio of variation between groups. ϕ coefficients are, in fact, F-statistics (Nei, 1977) analogous (Excoffier *et al.*, 1992) and provide population differentiation statistics: we expect a higher ϕ value for an higher amount of differentiation. In Table 3.7 we can appreciate that $\Phi_{\text{samples-total}} = 0.367$ shows moderate genetic difference between all individuals, $\Phi_{\text{samples-species}} = 0.066$ suggests small difference between populations of the same species, and $\Phi_{\text{species-total}} = 0.323$ highlights differences between species. We computed a final F_{ST} value between *A. pyramidalis* and *A. berica*, confirming moderate genetic distance ($F_{ST} = 0.253$, p-value $< 10^{-16}$, 99.5% C.I. = $[0.231, 0.273]$) between the two species: integrating this with the results of AMOVA testing, we can confirm that this genetic distance is caused by the combined effect of variance between individuals and between species, and is not influenced by their division in populations.

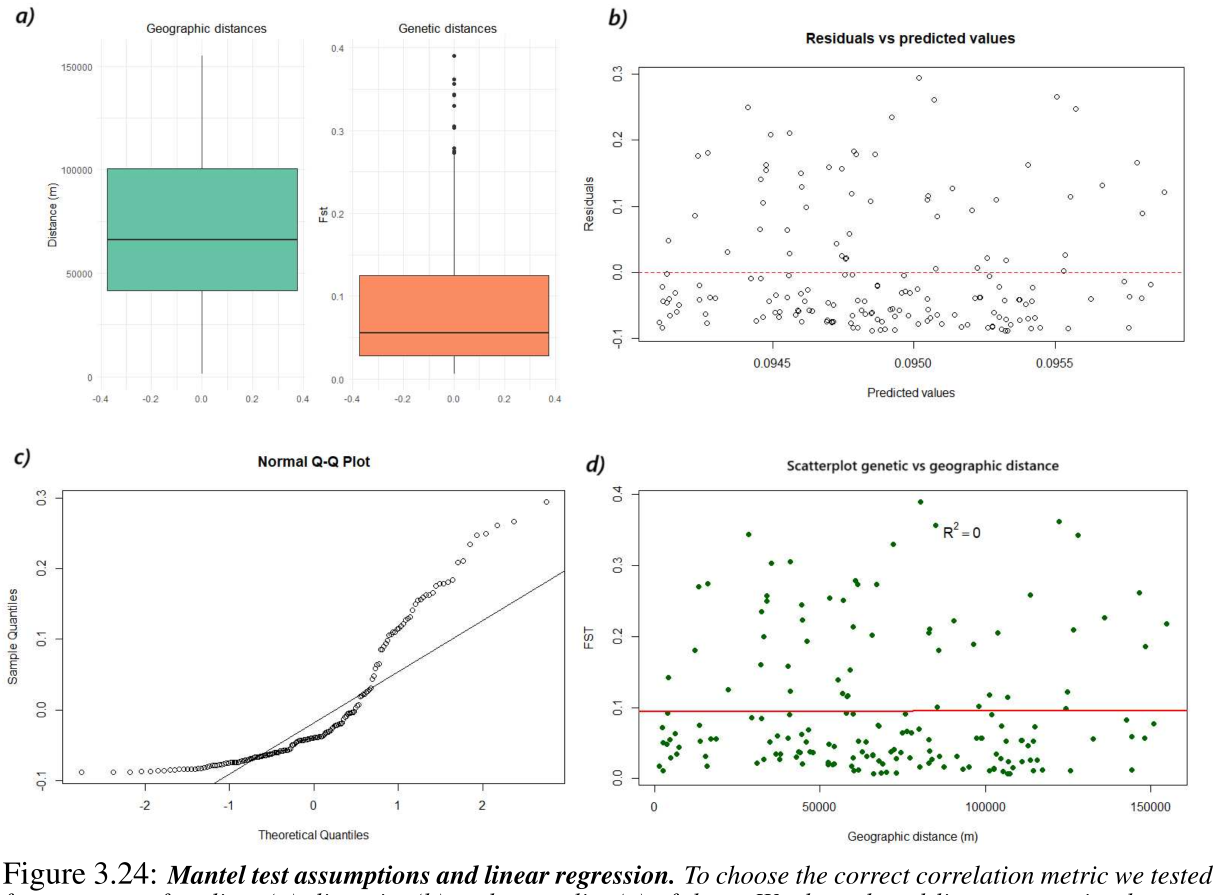


Figure 3.24: ***Mantel test assumptions and linear regression.*** *To choose the correct correlation metric we tested for presence of outliers (a), linearity (b) and normality (c) of data. We then plotted linear regression between genetic and geographic distance (d) as:* $9.408e-02+1.163e-08 \cdot distance\ (m)$, $p-value = 0.9513$.

We then used the fixation index results in Table 3.6 to perform Isolation by Distance testing on the populations, filtering out the whole species of *A. berica* and populations that had no signficant p-values both from the F_{ST} and geographic matrices. Before applying the Mantel test, as we did for *C. thyrsoides* we checked for correlation metrics assumptions, as reported in Figure 3.24. The dataset contained several outliers (Fig. 3.24, a), the data did not present evidence of heteroskedasticity (Breusch-Pagan Test p-value $= 0.5488$) and was approximately linearly distributed (plot b), but the distribution was not normal (plot c and Shapiro-Wilk test p-value $= 1.526e-13$). Following these results, we opted to use Spearman's correlation (ρ) over 9999 iterations: that confirmed no correlation ($\rho = -0.07398$, significance = 0.7647) between genetic and geographic distance in *Anacamptis* populations.

4. Discussion

4.1 *Campanula thyrsoides*

4.1.1 General observations

Choosing the appropriate restriction enzyme in GBS is important because of the absence of a size selection step (Sonah *et al.*, 2013). Our samples showed satisfactory quality after PCR amplification, but, considering that consistent fragment sizes are essential for consistency of data across runs and laboratories (K. R. Andrews *et al.*, 2016), we had to manually intervene even if GBS protocols usually do not require it. The fact that we improved the proportion of reads of the correct length using a size selection kit (refer to the Materials and Methods section) shows that a proper *C. thyrsoides* GBS protocol needs to be further investigated. In soybeans ApeKI has proven to be the most appropriate choice in comparison to enzymes like PstI, MseI and MspI (Sonah *et al.*, 2013), but the two species differ in number of chromosomes (2n = 2x with x = 12 for *C. thyrsoides* and 2n = 2x with x = 20 for soybean) and genome size (putative 4.83Gb for *C. thyroides* and 978.4 Mb for soybean[1]), which can influence the restriction enzyme efficacy. The choice of enzyme can largely benefit from the use of a reference genome (Sonah *et al.*, 2013), not available for *C. thyrsoides*. This is a limitation in our study and should be addressed in the future studies.

Species with short dispersal range usually have isolated populations: this is the case of *C. thyrsoides*, since most of its seeds (99.99%) are dispersed within 10 m from the mother plant

[1]Following v2 of the assembly, NCBI RefSeq assembly GCF_000004515.4.

(Ægisdóttir *et al.*, 2009). Isolation can lead to a strong genetic differentiation among populations (high F_{ST}) but also low genetic diversity within them (high F_{IS} and low π^2) due to small population sizes (Hughes *et al.*, 1999). Since our π estimates are based on SNPs, we cannot say if populations of subspecies *carniolica*, small and isolated at medium-high altitudes, have an overall low nucleotide diversity, but we can confirm that there is moderate differentiation (F_{ST}) between them. *C. thyrsoides* subspecies *carniolica* populations also showed extremely low inbreeding factors (F_{IS}), which confirms the presence of a good gene flow within populations that was already found in the subspecies *thyrsoides* despite its limited seed dispersal capacity (Ægisdóttir *et al.*, 2009). Subspecies *carniolica* might even have an advantage compared to subsp. *thyrsoides*, since a taller inflorescence may facilitate dispersal over larger distances (Scheepens *et al.*, 2011). Small populations size can be caused by both recent loss of habitat (Yin *et al.*, 2024) and monocarpic reproduction characteristic of the species, but monocarpy might be also the reason why *C. thyrsoides* subspecies *carniolica* populations show no inbreeding depression: genetic erosion is usually more pronounced after several generations (Aguilar *et al.*, 2008), so by reproducing once in their life, the risk of inbreeding is mitigated and distributed over time. This might be a strategy of *C. thyrsoides* to adapt to an habitat that forces fragmentation due to its geography (Ægisdóttir *et al.*, 2007). No population is, in fact, in need of immediate conservation actions as the metrics we observed can be explained by *C. thyrsoides* characteristic. Its ability to mate with its half-siblings (Ægisdóttir *et al.*, 2007) and across generations may explain high differentiation among populations (Buckler & Thornsberry, 2002). Being an open-pollinated, outcrossing and self-incompatible species, *C. thyrsoides* could have suffered from outbreeding[3] rather than inbreeding depression (Volis & Blecher, 2010) and this could be the reason for the negative F_{IS} values for all populations (Kanaka *et al.*, 2023). Due to the limited number of samples in the outgroup CTT we cannot say if the same conclusion can be made for the whole subspecies *thyrsoides*, but it appears relevant for the populations in our study and hybrids of subspecies *carniolica* and *thyrsoides*: hybrids are, though, currently not documented (Scheepens *et al.*, 2011).

The results of Mantel test showed that in populations of subspecies *carniolica* there is a strong

[2]Defined as the average sequence divergence between any two individuals for a given locus (Buckler & Thornsberry, 2002).

[3]Defined as reduction of fitness after hybridization events (Volis & Blecher, 2010).

correlation between genetic and geographic distance, differently from previous findings on subspecies *thyrsoides* (Frei *et al.*, 2012). Since post-glacial colonization can be tracked from the genetic structure of a species (Ilves *et al.*, 2016) and subspecies *thyrsoides* already showed proof of a post-glacial event (Kuss *et al.*, 2011), we suspect that subspecies *carniolica* might have gone through the same evolutionary process.

4.1.2 Populaiton-specific observations

Pian dei Resinelli

Historically speaking, the region of Lombardy (Italy) had only occurrence records of the subspecies *thyrsoides* but the PCA plots that include outgroup CTT (Fig. 3.6) clearly cluster the population of PDR (Pian dei Resinelli, Lombardy) with the other populations of subspecies *carniolica*, with PC3 (Fig. 3.6, b) clearly catching this separation. When the outgroup CTT of subspecies *thyrsoides* is included in the analysis, PDR clearly separates from CTT, but when this population is excluded (Fig. 3.6, d, e and f), PDR separates from the other populations of subsp. *carniolica* too, showing its own distinct gene pool. This shows an expansion in its historical habitat and highlights the need of updating the checklist, made in 2005 (Conti *et al.*, 2005).

Slovenian populations

Slovenian populations (RT and STE) group together very well: it was previously observed that in the lower mountains of Slovenia, subspecies *carniolica* showed a later (in comparison to *thyrsoides*) flowering duration (Scheepens *et al.*, 2011). A longer flowering period of those populations fits well with the phenology of alpine species in warmer climate, which already demonstrated, in other species, the same phenological behavior when subject to shift in snowmelt dates (Vorkauf *et al.*, 2021). Following the hypothesis that Slovenian populations might be more adapted to the climate and drought (Scheepens *et al.*, 2011), we integrated their low sampling altitude (245m - 420m) with their clear clustering in the PCA and concluded that our analysis might have caught some of the genetic information linked to the adjustment to their habitat. The reduced sequencing approach, though, can not confirm if these populations are generally more adapted to their habitat: higher adaptation is usually accompanied by lower genetic variety (Habel & Schmitt, 2012), but RT and STE π levels are neither lower than the rest of *carniolica* populations, nor their F_{IS} values are largely higher. Confirmation of this adaptation hypothesis requires a more detailed study, combining environmental data and ideally an annotated reference genome, to connect genomic regions to traits and enviromental selection, which can lead to conservation plans for populations carrying these genes (Supple & Shapiro, 2018).

Alpago (ALP)

It is good practice to choose a genetically close population when planning a translocation intervention in order to minimize outbreeding depression risk (Yin *et al.*, 2024). ALP is the closest population to SP using weighted F_{ST} estimates, but it shows admixture with populations that were not sampled in the study (Fig. 3.11). With its admixed structure, ALP has the potential of acting as a source of new genetic variation (Supple & Shapiro, 2018), which could be either positive or negative. Potentially deleterious alleles should be identified before trans-locating its individuals to another population and, since the target population of Monte Serva does not currently show breeding depression (its $F_{IS} < 0$), the risks outweigh the benefits.

Outgroup *thyrsoides* (CTT)

F_{IS} value seems to be slightly lower for population CTT (subspecies *thyrsoides*, Table 3.3). We could hypothesize that subspecies *thyrsoides*, which is more common, might have an higher level of gene flow between populations since it was previously observed (Ægisdóttir *et al.*, 2009): no study has, though, compared F_{IS} values between the two subspecies. Up until now, comparative studies between subspecies *thyrsoides* and *carniolica* have only focused on morphological differences (Scheepens *et al.*, 2011) and flowering behavior (Kuss *et al.*, 2011), while genetic studies on population structure has been carried only on subspecies *thyrsoides*

(Ægisdóttir *et al.*, 2009; Frei *et al.*, 2012). Because of the low number of samples in the outgroup CTT, we can not use this study to statistically test whether the lower F_{IS} values depend on CTT's belonging to subspecies *thyrsoides* or if it is the specific gene pool of this population. A comparative analysis between the two subspecies structure, increasing the number of populations and individuals sampled for both, could shed light on their evolutionary relationship.

4.2 *Solanum tuberosum* landraces

In our study GBS was able to capture many interesting evolutionary relationships in the set of potatoes of genus *Solanum* we selected: we discuss our findings in the context of local landraces conservation.

4.2.1 Wild and ancient species

Wild species (*S. vernei*, *S. microdontum*, *S. chacoense*, *S. brevicaule*) were correctly assessed as the most genetically distant samples, both in the PCA and phylogenetic trees: they cluster together as they are all wild, but in the tree (Fig. 3.17) they present very long branches, highlighting the large genetic difference even between them. This was expected since they are not currently used as a food source, so they are rarely cross-bred. On the other hand, all the Andean cultivated species easily cluster together with branches of shorter length, probably because when using different taxonomies *S. chaucha*, *S. phureja* and *S. stenotomum* are actually considered cultivars of *S. tuberosum* (Machida-Hirano, 2015). In some studies they are considered even hybrids: in Huamán and Spooner (2002), *S. chaucha* is reported to be likely an hybrid of *S. tuberosum* subsp. *andigenum* with either *S. stenotomum* or *S. phureja*, with the latter deriving from a later divergence event than the former. This is probably why these three species are separated from *S. tuberosum* subsp. *andigenum* with high bootstrap support (100), but their relationships are not resolved in the tree ($boostrap = 49$). Nevertheless, this further proves how interesting Bossico 1, Bossico 2 and Starleggia bianca separate so clearly from this group, highlighting the importance of their conservation as landraces that are different from modern cultivars of *S. tuberosum*.

4.2.2 Italian landraces

Quarantina genovese is one of the landraces that is currently conserved *ex-situ*, specifically by the "Consortium for the protection of the Genoese white Quarantina and the traditional potatoes of the Genoese mountain" (http:// www.quarantina.it/category/il-consorzio/), recognised by the International Treaty on Plant Genetic Resources for Food and Agricolture (ITPGRFA). Our phylogenetic analysis, though, collocated this type of potato genetically close to the commercial variant Kennebec. The clustering of Quarantina landraces with each other is evident in both the PCA plots (Fig. 3.15) and in the phylogenetic tree (Fig. 3.17) where their clade showed a perfect bootstrap support (100): from these results, we could suspect a strong genetic identity of this landrace. Recent studies on Italian local landraces of *S. tuberosum* have shown significant similarity of Quarantina Genovese Bianca with commercial variant Kennebec both morphologically (Giupponi *et al.*, 2020) and genetically (Martina *et al.*, 2025) like in our study. In the study by Martina *et al.* (2025), Quarantina also belonged to a sub-cluster containing a French white peel commercial cultivar (Institute de Beauvais). Although this commercial cultivar was shown to be genetically identical to Quarantina Genovese Bianca using microsatellites markers (Mandolino *et al.*, 2015), GBS analysis on the same species, with its enhanced resolution, caught a certain degree of differentiation (Martina *et al.*, 2025). We concluded that, although our two Quarantina landraces formed a Quarantina clade, they are too similar to commercial variants, so we could not consider them a genetically distinct landrace. This might be different for other Quarantina landraces not included in our study (i.e. Quarantina Prugnona and Ruba Spes, a red peel mutation of Quarantina Bianca), that already showed differences from Kennebec in morphology (Giupponi *et al.*, 2020), so we would not exclude the Quarantina landrance in its enirety from conservation.

The case of Comasca (CO) overlapping with Kennebec instead, seems more straightforward to understand. In the area of Como, many old cultivars have been replaced by the dutch commercial variety Kennebec: for example, the name "Patata di Martinengo" is currently applied to local production of Kennebec (Rossi, 2019). The same thing was happening until a couple of years ago to other cultivars: both names ”Patata bianca di Oreno” and ”Patata bianca di Rovetto”, were used to actually refer to the aforementioned commercial variant. Years ago, the name ”Patata bianca di Rovetto” was also used to refer to ”Bianca di Como” (Rossi, 2019), Italian for ”White peel variety of Como”, which is nowadays used as a synonym of ”Comasca”. It is therefore likely that the seeds we were given as ”Comasca” were mistakenly believed to be the local landrace ”Bianca di Como” that in 2007 has been promoted as local potato cultivation in the Brianza, by the municipality of Vimercate. The old ”Bianca di Como” was likely a ”Patata bianca di Rovetto”, which was simply another name for a Kennebec cultivar in Como.

Both Starleggia Bianca and Rossa were reported as local landaraces of Campodolcino (SO) (Rossi, 2019) and currently cultivated as such (Vegini *et al.*, 2022). Our study, instead, highlighted the genetic differentiation of only Starleggia Bianca, the white peel variant, while Starleggia Rossa (red peel) vastly overlaps with all the other red peel varieties. The red-peel varieties and the modern cultivar Desiree are genetically indistinguishable from each other in our tree (Fig. 3.17) and it is probably due to the fact that potatoes are cultivated by tuber (and not seed). The uniformity of these samples should raise concerns due to the inevitable genetic erosion and vulnerability that these cultivars might suffer from (Salgotra & Chauhan, 2023). The genetic difference between Starleggia Rossa and Starleggia Bianca was already suspected due to the already observed morphological differences (Rossi, 2019): our study suggests, too, that Starleggia Rossa is a clone of a commercial variant. This relationship however remains to be confirmed with a larger sample set: repeating this type of genetic study on a larger set of samples could reject this conclusion, and thus either promote or discourage conservation of this variety in seed banks.

Our analysis confirmed that ”Rossa Oltrepò di Pietragavina” is a clone of the Dutch commercial cultivar Desiree. This was alrealdy suspected due to many similarities in their morphological traits (Rossi, 2019).

Lastly, the intermediate position of landrace Bianca between white and red peel variants in both the PCA and phylogenetic tree, shows that it might be (i) an hybrid, probably the result of crossbreeding between white and red commercial cultivars or (ii) a variety that was used for red-peel potato domestication. This study highlights the power of GBS in catching genetic diversity and phylogenetic relationships also among very closely related cultivars.

4.3 *Anacamptis* spp.

Through our analysis we have shown that samples of *A. pyramidalis* and *A. berica* do not cluster together based on their population of provenance neither on their urban or natural origin. Through the PCA plots and phylogenetic tree, we can confidently say that the only clear division is between the two species, confirming the distinction already observed in the study of Doro (2020). Furthermore, it shows that the pipeline we built is robust to detect similarities without forcing subdivisions if they aren't supported. We can broadly confirm that population P_NI belonged to species *A. pyramidalis*, even if there was some uncertainty during sampling: if we exclude P_NI3 due to the high percentage of missing data, in the PCA the rest of the samples are closer to the *A. pyramidalis* group than the *A. berica* one (Fig: 3.20). *A. pyramidalis* can hybridize with other species of the same genus (Fay *et al.*, 2024) so the intermediate positioning of P_NI1 could have shown hybridization with *A. berica*. This scenario seems highly unlikely since it was documented that diploid individuals *A. pyramidalis* hardly hybridize with *A. berica* tetra-ploids ones (Pegoraro *et al.*, 2016). Not enough hybrids were sampled to either support or refute this hypothesis, so we leave the investigation to future studies.

When analysing the population structure we showed that for $K = 16$, P_BA, P_BSG, P_CA, P_PE, P_NI showed little to no admixture while P_AR, P_DO, P_ES and P_S shared only a little part of their ancestry with the rest of the samples (Figure 3.23). This does not reflect the urban versus natural division of the populations, rather it seems that the explanation could be the geographical location of the samples. The samples at the extremes of our sampled area, in Lombardy (P_PE and P_NI) and in the Adriatic Coast (P_DO and P_BSG) are the most distinct, suggesting a longitudinal gradient. The geography of the gene flow among these population remains however unclear as the value selected for K does not correspond exactly to the minimum CV error value, so we cannot confidently accept the hypothesis that this value of K describes populations' structure appropriately. To analyze the values of K that were found by `Admixture` to be statistically optimal (Fig. 3.22), it would be necessary a more in depth analysis of the plateau values in Figure 3.22, running STRUCTURE through a considerable range of Ks, performing several runs and then choosing the optimal value of K using either Evanno or Puechmaille method, or a combination of both. It would require more computational resources than what we had available, not to mention the bigger resulting dataset in comparison to the size of the sampled populations, that would most likely ensure enough statistical strength to the analysis.

All of the populations show low to moderate negative levels of F_{IS} (Table 3.5), pointing to the conclusion that our samples do not suffer from inbreeding depression and do not need an intervention for their conservation. At the same time F_{ST} values (Table 3.6) are generally low, pointing at frequent gene flow among populations. While we know that vegetative propagation in *A. pyramidalis* is possible but not common (Alec M. Pridgeon *et al.*, 2001), there are no information about self-incompatibility (SI) (Fay *et al.*, 2024), which would explain the high level of within population diversity. Interestingly this species is nectarless, which might discourage pollinators to visit nearby flowers and promote movement between populations (Ilves *et al.*, 2016): this, combined with *A. pyramidalis* high range of seed dispersal (Alec M. Pridgeon *et al.*, 2001) might be one of the reasons why this species did not develop a SI system. The low levels of F_{ST} we found are, additionally, in accordance to what was observed in a previous study by Ilves *et al.* (2016). When we performed IBD testing, we found no correlation between genetic and geographic distance among the population samples. In the same aforementioned study of Ilves *et al.* (2016), instead, an IBD pattern was obtained when comparing populations from distant countries in Europe, while no IBD was observed when comparing samples within the same country. By sampling only individuals from a restricted region of Italy, the range of our study was probably too limited to appreciate a geographic pattern, suggesting a large dispersal range for *Anacamptis pyramidalis* seeds that are dispersed by wind.

5. Conclusion and future perspectives

Monitoring of biodiversity vulnerability should be a top priority in the protection of plant species (Salgotra & Chauhan, 2023) as also established by the European Biodiversity Strategy for 2030 and the new Restoration Regulation. The GBS approach used in this study shows robustness in different non-model plants, performing exceptionally well both in the presence and absence of a reference genome and with the use of different restriction enzymes, and thus it can be used for biodiversity research. These are good news for laboratories looking to contain costs but of course interested in delivering high quality results and insight into wild and genomically unknown plants. The same steps could be applied both to animals and fungi, while we would not recommend a RADSeq approach on either Archea or Bacteria, due to the smaller genome they possess.

5.1 *C. thyrsoides* subspecies

Populations of subspecies *carniolica* are not endangered, but the small number of individuals (probably caused by monocarpy; Ægisdóttir *et al.*, 2009) requires conservation of their habitat. It is also unknown if the two subspecies of *C. thyrsoides* are interfertile and produce viable offsprings (Scheepens *et al.*, 2011): now that subspecies *carniolica* seems to be moving towards the habitat of subspecies *thyrsoides*, it is crucial that their differences are preserved. The National Park of Dolomiti Bellunesi is already correctly applying *in situ* translocation of genotypes within the genetic reserve conservation area (Salgotra & Chauhan, 2023), thereby protecting Monte Serva population (SP) in its territory. In doing so, the Park is preventing further fragmentation of the habitat due to summer cattle pasture (Frei *et al.*, 2012). The Park is also committed to increase Monte Serva population's size by either breeding *ex-situ* individuals of its population to be reintroduced in their habitat, or by genetic rescue, meaning trans-locating individuals from the closest populations (Supple & Shapiro, 2018) which our study shows to be Alpago (ALP) and Passo del Pura (PPURA). We would prioritize PPURA as one of the closest populations to SP currently not showing high level of admixture, unlike ALP. Our genetic study also ensured that the Park's intervention will not disrupt Monte Serva's ancestry, and we recommend other parks and institutions to follow a similar approach. An analogous genetic analysis should be repeated in about 10 years after to correctly quantify the efficacy of the intervention, the genetical analysis and the conservation efforts. Furthermore, despite parks and other associations' best intentions, because of climate change the potential of *in situ* conservation of several species in the wild is declining (IUCN/SSC, 2014), raising our concerns about subspecies *thyrsoides* too. Subspecies *carniolica*, being slightly more specialized for lower altitudes and higher temperatures, could be invading *thyrsoides* habitat due to the increase of temperature. Even if the latest genome-editing technologies might allow researches to improve species traits to adapt to the changing climate (Supple and Shapiro, 2018; Yin *et al.*, 2024), biodiversity protection is still co-dependent to its mitigation (Theissinger *et al.*, 2023). Instead of limiting our action on genome-engineering to try to reverse the negative effects of our impact on the Earth, we need to reduce human influence and preserve biodiversity, and genomics approaches can help us identify hotspots were to start focusing our efforts.

5.2 *S. tuberosum* local landraces

Project La Rava e la Fava addressed the conservation of *S. tuberosum* landraces correctly, asking for a genetic study to choose which landraces to focus their effort on. In our analysis on genus *Solanum*, we combined Italian and South American accessions, and could highlight a distinct phylogenetic identity for some of the local landraces in this study. To limit the loss of genetic diversity for *S. tuberosum* in our national territory, it might be useful to include currently obsolete cultivars (Bossico and Shilpario; Vegini *et al.*, 2022) to plant genetic resources (PGRs) collections, an *ex-situ* conservation method started in the mid-20th century, that aims at gathering all cultivated, wild relatives, traditional cultivars, landraces, and advanced breeding lines of plants (Salgotra & Chauhan, 2023). Other studies highlighted the need to prioritize local germplasm, underutilized crop species and traditionally local landraces (Salgotra & Chauhan, 2023). Considering the benefits (i.e. easier farming due to adaptation to local climate and creation of a genes reserve for future crops improvement) that can result from conserving local landraces , we stress the importance of focusing the efforts on Bossico and Shilpario. PGRs collections could benefit also from the addition of Starleggia Bianca, although it is already currently cultivated for conservation purposes by different societies like Mato Grosso Association and Legambiente Valchiavenna (Vegini *et al.*, 2022) so a conservation intervention is less urgent. In 2004, the Italian Ministry of Agriculture, Food Sovereignty and Forests (MiPAAF), following the International Treaty on Plant Genetic Resources for Food and Agriculture (ITPGRFA), financed the project RGV/FAO for the realization of the objectives in the Treaty. To do this, in 2013 it included around 17,000 accessions across 22 genera of crop plants in the Multilateral System of Access and Benefit-sharing (Ammassari, 2013), a system set in the context of the International Treaty that works as a global gene pool and facilitates access to plant genetic material. The Italian germplasm databanks are available at PlantaRes database (Network Nazionale delle Risorse Genetiche Vegetali per l'Alimentazione e l'Agricoltura) and at the Mediterranean Germplasm Database (MGD). Currently (September 2025), the MGD does not include any accession of *S. tuberosum*, and only provides germplasm exclusively as a support to research and education projects, and not for crop improvement[1]. PlantaRes database, instead, contains information about two landraces of *S. tuberosum* (Accession names Montecopiolo and Patata Blu), whose germplasm is currently stored at the Horticulture and Floriculture Center of Monsampolo del Tronto (AP). There is another center associated with CREA, at Bergamo (Cereal and Industrial Crop Research Center - Bergamo branch), so storing Bossico 1, Bossico 2, Starleggia Bianca and Shilpario seeds there can be a proper additional *ex situ* approach for the conservation of these local landraces, in line with both La Rava e la Fava and ITPGRFA projects.

[1]The MGD Policy can be found at web address: https://www.ibbr.cnr.it/mgd/?action=page&id=policy.

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8. Additional material

Here are listed and reported all the materials named and used for the preparation and analysis of data for *Campanula thyrsoides*: for the sake of being concise, we avoided going in detail in chapter 2. Materials and Methods. All information is reported here and, in the case of pipelines and additional files, is freely downloadable from the Github repository `flavialeotta/Master-thesis`.

8.1 Protocols

In this section are reported, in detail, the protocols used in the wet lab before receiving the data: they can give some insight on how the samples were prepared and treated, which can influence the quality of data, as previously discussed.

8.1.1 SILEX: CTAB-Silica extraction protocol

SILEX is a highly-throughput DNA extraction protocol, based on the standard CTAB method with a DNA silica matrix recovery. Here we report the steps taken to extract the DNA samples needed to produce the NGS data used in the previous analyses. (Vilanova *et al.*, 2020)

1. Grind the tissue with Qiagen TissueLyser;
2. Add 1 mL of CTAB extraction buffer and 14 mL of β-mercaptoethanol for each sample and vortex, until complete homogenization;
3. Incubate in a thermoblock at 65°C for 45 minutes;
4. Let the samples rest on ice for 5 minutes;

5. Add 700 μL of the protein precipitation buffer, which is a solution of Chloroform:Isoamyl Alcohol (24:1), and turn the tubes upside down to mix the solution;
6. Centrifuge at 11000 rpm for 5 minutes;
7. Transfer 800 μL of supernatant phase in a new 2 mL tube;
8. Add 2,5 μL of RNAse and incubate at 35°C for 20 minutes;
9. Add 480 μL of binding buffer (25M NaCl + 20% PEG 8000) and gently invert the tube by hand to mix the solution;
10. Add 700 μL of EtOH 100% and turn the tubes upside down;
11. Add 20 μL of silica matrix buffer[1] and shake for 5 minutes;
12. Centrifuge for a few seconds and discard the supernatant;
13. Add 700 μL of EtOH 70%;
14. Suspend silica by turning the tubes upside down and spin again for a few seconds;
15. Discard the supernatant and turn the tubes upside down on paper for 10 minutes to let ethanol fully evaporate;
16. Add 100 μL elution buffer (10 mM Tris HCl pH 8.0 and 1 mM EDTA pH 8.0) and shake gently by hand until the pellet is re-suspended;
17. Incubate at 65°C for 3-5 minutes;
18. Centrifuge at 14000 rpm for 10 minutes;
19. Pipette 70 μL of supernatant and store it in a 1.5 mL Eppendorf.

8.1.2 DNA digestion

DNA digestion protocol was used to digest the same quantity of DNA for each sample. Since the goal was to digest 50 ng of DNA and each sample had a different concentration, we calculated the volume of DNA to digest for each sample i as:

$$V_{DNA_i} = \frac{m\ (ng)}{[DNA]_i\ (ng/\mu L)} = \frac{50\ ng}{[DNA]_i\ (ng/\mu L)} \tag{8.1}$$

[1]The Silica matrix buffer must be prepared beforehand following the protocol by Vilanova *et al.* (Vilanova *et al.*, 2020).

by using the concentration of DNA as obtained using Qubit. The samples to digest ($50\mu L$ each), to be prepared on ice, had to contain:

1. first distilled and de-ionized water (ddH_2O) calculated as $44\mu L - V_{DNA}$ and reported in Table 8.1;
2. then $4\mu L$ of NEBuffer™ r3.1 10x; [2]
3. extracted DNA as calculated with formula 8.1 and reported in Table 8.1;
4. $2\mu L$ of ApeKI ($\frac{5U}{\mu L}$). [3]

The samples went through four different protocols for digestion (Table 8.1): the difference between them are minimal. Those that got digested for 4 hours and then incubated in a thermocycler (80°C lid, hold 20°C) for an additional 8 hours with more ApeKI and buffer, show better smears in digestion but only slightly better bands in amplification.

8.1.3 Adapters preparation

Adapters are divided in top and bottom strands as shown in Figure 2.5 in the Materials and Methods section: an adapter stock needs to be prepared as follows.

1. Resuspend lyphilized oligos in TE[4] buffer up to $200\mu M$;
2. In a PCR tube mix add $25\mu L$ of top strand DNA oligo, $25\mu L$ of bottom strand DNA oligo and $50\mu L$ of TE;
3. Perform annealing reaction in a thermocycler with following program:
   - 2' at 95°C;
   - Ramp from 95°C to 25 °C lowering the temperature by 0.1°C per second;
   - 30' at 25°C;
   - hold at 20°C.
4. Perform a first dilution. For barcoded adapters (top strand):

[2] 1X NEBuffer™ r3.1 contains: 100 mM NaCl, 50 mM Tris-HCl, 10 mM MgCl2, 100 µg/ml Recombinant Albumin (pH 7.9 @ 25°C).

[3] U stands for enzyme unit, and is related to the katal by the following: $1\ U = 16.67\ nkat$. (Nomenclature Committee of the International Union of Biochemistry (NC-IUB), 1979). We know that the catalytic activity of an enzyme is the increase in the rate of chemical reaction that the enzyme produces. The katal (symbol: kat) is that catalytic activity, that will raise the rate of reaction' by one mole per second.

[4] Eurofins' TE buffer contains: 10 mM Tris, 1 mM EDTA (pH 8 @ 25°C).

Table 8.1: *Digestion information*

Sample ID	$ng/\mu L$ qubit	μL sample	$\mu L\ H_2O$	Sample ID	$ng/\mu L$ qubit	μL sample	$\mu L\ H_2O$	Sample ID	$ng/\mu L$ qubit	μL sample	$\mu L\ H_2O$
ALP1	22.8	21.93	22.07	MAU8*	30.6	16.34	27.66	SP1I1†			
ALP2	23.0	21.74	22.26	MAU9*	15.5	32.26	11.74	SP2I2†			
ALP3	41.4	12.08	31.92	MAU10*	15.5	31.85	12.15	SP4I4†			
ALP5	56.0	8.93	35.07	MAU11*	107	4.67	39.33	SP4AI5†			
ALP6	21.2	23.58	20.42	PDR1*	37.4	13.37	30.63	SP4BI6†			
ALP7	36.8	13.59	30.41	PDR2*	54.0	9.26	34.74	SP5I7**	15.4	32.47	11.53
ALP8	26.2	19.08	24.92	PDR3*	30.2	16.56	27.44	SP6I8**	24.8	20.16	23.84
ALP9	31.6	15.82	28.18	PDR4*	34.2	14.62	29.38	SP7I9**	76.8	6.51	37.49
ALP10	44.6	11.21	32.79	PDR5*	60.4	8.28	35.72	SP7I10**	47.8	10.46	33.54
ALP11	13.6	36.76	7.24	PDR6†				SP7I11**	83.0	6.02	37.98
ALP12*	24.0	20.83	23.17	PDR7†				SP10I12**	60.6	8.25	35.75
ALP13†				PDR8*	31.2	16.03	27.97	SP11I13**	49.0	10.20	33.80
ALP14*	22.8	21.93	22.07	PDR9*	32.2	15.53	28.47	SP3I3†			
ALP15*	22.2	22.52	21.48	PDR10*	36.2	13.81	30.19	SP11I14*	50.4	9.92	34.08
CAN1†				PDR11*	53.2	9.40	34.60	SP12I15**	74.4	6.72	37.28
CAN3†				PDR12*	30.2	16.56	27.44	SP13I16**	73.0	6.85	37.15
CAN4†				PDR13*	70.4	7.10	36.90	SP14I17**	23.2	21.55	22.45
CAN5†				PDR14*	35.4	14.12	29.88	SP17I18**	61.1	8.18	35.82
CAN6†				PDR15*	27.6	18.12	25.88	SP17I19*	57.8	8.65	35.35
CAN7†				PPURA1*	45.4	11.01	32.99	SNEV1**	191.0	2.62	41.38
CAN9†				PPURA2*	49.6	10.08	33.92	SNEV2*	45.8	10.92	33.08
CAN10†				PPURA3*	52.0	9.62	34.38	SNEV3*	25.2	19.84	24.16
CAN11†				PPURA4**	50.0	10.00	34.00	SNEV4†			
CAN12†				PPURA5**	47.2	10.59	33.41	SNEV5**	72.4	6.91	37.09
CAN13†				PPURA6†				SNEV6**	55.8	8.96	35.04
CTT1*	71.4	7.00	37.00	PPURA7**	31.8	15.72	28.28	SNEV7	92.7	5.39	38.61
CTT2*	57.4	8.71	35.29	PPURA8**	25.2	19.84	24.16	SNEV8	51.6	9.69	34.31
CTT3*	49.4	10.12	33.88	PPURA9†				SNEV9**	28.8	17.36	26.64
LPASS1*	25.4	19.69	24.31	PPURA10**	36.4	13.74	30.26	SNEV10**	39.0	12.82	31.18
LPASS2*	50.8	9.84	34.16	PPURA11**	83.8	5.97	38.03	SNEV11**	71.4	7.00	37.00
LPASS3†				PPURA12†				SNEV12†			
LPASS4*	29.2	17.12	26.88	PPURA13**	24.2	20.66	23.34	SNEV13**	53.2	9.40	34.60
LPASS5*	192	2.60	41.40	PPURA14†				SNEV14**	28.8	17.36	26.64
LPASS6*	58.4	8.56	35.44	PPURA15**	16.0	31.25	12.75	SNEV15†			
LPASS7*	40.0	12.50	31.50	RT1†				STE1**	76.8	6.51	37.49
LPASS8*	29.2	17.12	26.88	RT2†				STE3**	73.2	6.83	37.17
LPASS9*	35.2	14.20	29.80	RT3**	18.5	27.03	16.97	STE4**	110.0	4.55	39.45
LPASS10*	41.8	11.96	32.04	RT4†				STE5*	80.0	6.25	37.75
LPASS11*	25.6	19.53	24.47	RT5†				STE6**	38.4	13.02	30.98
LPASS12*	32.6	15.34	28.66	RT6**	16.8	29.76	14.24	STE7**	126.0	3.97	40.03
LPASS13*	23.4	21.37	22.63	RT7 †				STE8$\star$			
LPASS14*	30.0	16.67	27.33	RT8†				STE9$\star$	31.1	16.08	27.92
LPASS15*	22.4	22.32	21.68	RT9**	74.0	6.76	37.24	STE10$\star$	36.6	13.66	30.34
MAU1*	34.0	14.71	29.29	RT10†				STE11$\star$	30.8	16.23	27.77
MAU2†				RT11**	31.4	15.92	28.08	STE13$\star$	25.8	19.38	24.62
MAU3†				RT12†				STE14$\star$	49.4	10.12	33.88
MAU4†				RT13†				STE15**	51.6	9.69	34.31
MAU5*	25.0	20.00	24.00	RT14**	44.6	11.21	32.79				
MAU6†				RT15**	112.0	4.46	39.54				
MAU7*	50.8	9.84	34.16	RT16**	50.8	9.84	34.16				

The samples went through different digestion protocols. Legend: () = standard protocol of 2 hours digestion, (*) = overnight digestion (8 hours), (**) = 4 hours digestion and additionally overnight (8 hours) after adding again the solution of ApeKI and NEBuffer r3.1 10x, ($\star$) = 4 hours digestion, (†) = they were not digested due to absence of DNA on gel, so their row is crossed out.

- Bring together in a 1.5 mL tube 5.6μL of annealed barcoded adapter and 995μL of TE nuffer;
- Vortex and spin down.

For the common adapters:

- Bring together in a 1.5 mL tube 100μL of annealed common adapter and 900μL of

TE buffer;
   - Vortex and spin down.

5. Quantify adapters using Qubit;
6. Mix each barcoded adapter (300 ng) with common adapter (300 ng) in a $1,5mL$ tube with 200 μL of TE buffer at 3 $ng/\mu L$ concentration;
7. Dilute the stock with water to obtain a concentration of 0.6 $ng/\mu L$ and store at -20 °C.

8.1.4 Adapters ligation

After the previous step, we have, for each sample, its digested DNA and working adapter stock containing its own barcode (complete list in Table 8.2). The ligation protocol is comprised of the following steps to be performed on ice:

1. Transfer 20 μL of each digested DNA into PCR tubes, along with 6 μL of adapter work solution (different for each sample) and 24 μL of MasterMix solution[5] (common for all samples);
2. Incubate in thermocycler with the following program:
   - 2h at 22°C (non heated lid);
   - 30' at 65°C:
   - hold at 4°C.

8.1.5 Pooling and cleanup

We used Monarch PCR and DNA cleanup kit.

1. Combine 5 μL of each sample into a single 1.5 mL tube. Since the protocol states that it is better to not purify together more than 48 sample, we mixed all of them together and then split them in half in two 1.5 mL tubes;
2. Add 5 volumes of binding buffer to each pool sample and mix by pipetting;
3. Wash column following kit is instructions;

[5]The MasterMix solution for 1 reaction contains: 17 μL of water, 5 μL of T4 DNA Ligase Buffer 10x and 1.6 μL of T4 DNA ligase.

4. Elute each column with NEB's elution buffer (EB), 1 μL every 2 samples present in the pool. Since we used the maximum amount of samples (48), we used 24 μL of EB;
5. Combine all colums in a single tube.

8.1.6 PCR amplification and final cleanup

We used Monarch PCR and DNA cleanup kit:

1. Add PCR MasterMix[6] for 8 amplifications and incubate in the thermocycler with the following program:
   - 2' at 95°C;
   - 18 cycles of:
     - 30" at 95°C;
     - 1' at 62°C;
     - 20" at 68°C;
   - 5' at 68°C;
   - hold at 8°C.
2. For the cleanup combine the 8 PCR into two 2 mL tubes;
3. Add 5 volumes of binding buffer to each pool sample and mix by pipetting (using 1 column per each 2 mL tube);
4. Wash column following kit is instructions;
5. Elute each column with 20 μL of NEB's elution buffer (EB);
6. Combine the two elutions in one single 1.5 mL tube (final volume = 40 μL).

8.1.7 Size selection

We used Sera-Mag Select kit from cytiva, with Dual-side size selection protocol with size distribution of 150-500 bp and average size 250bp.

Round 1 1. Add Sera-Mag Select reagent at an appropriate sample volume (V1) to capture DNA above a defined threshold. Following the kit instruction, a 150 to 500 bp size

[6] The PCR MasterMix for 1 reaction contains: 34 μL of H_2O, 10 μL of NEB 5x Taq Master Mix, 1 μL Forward primer (12.5 uM), 1 μL Reverse primer (12.5 uM) and 2 μL of clean pooled ligated DNA.

distribution requires a right side size limit of 0.65x. Thereby the reagent volume is calculated as:

$$V1 = sample\ input \cdot right\ side\ limit = 200\ \mu L \cdot 0.65x = 130\ \mu L \qquad (8.2)$$

2. Vortex the sample for 30 to 60s, briefly spin and incubate at room temperature for 5-10 minutes;
3. Place the tube on a magnet rack for 5 minutes or until the beads have fully settled;
4. Aspirate and retain the supernatant, discard the beads.

Round 2 1. Add an additional volume (V2) of Sera-Mag Select reagent to the supernatant from round 1 to allow for binding of DNA fragments above the left threshold. The volume is calculated as:

$$V2 = VL - V1 = 200\ \mu L - 130\ \mu L = 70\ \mu L \qquad (8.3)$$

2. Vortex the sample for 30 to 60 seconds, briefly spin and insubate at room temperature for 5-10 minutes;
3. Place the tube on a magnet rack for 5 minutes or until the beads have fully settled to the magnet;
4. Aspirate and discard the supernatant.

Washing beads 1. With the tubes on the magnet, wash twice with 180-200 μL 85% ethanol;
2. Take care to aspirate as much wash solution as possible after the second wash;
3. Dry the beads for 5-10 minutes at room temperature to remove any residual ethanol.

Elution of bound DNA 1. Add 50 μL of TE buffer, vortex for 30 to 60 seconda to resuspend the beads and briefly spin;
2. Incubate for 5-10 minutes at room temperature to elute DNA;
3. Return to the magnet for 5 minutes or until the beads have fully settled;
4. Carefully aspirate the supernatant and transfer to a fresh tube.

8.2 Pipeline

Both main bash scripts files and supporting ones mentioned in the Material and Methods chapter, are available at Github repository `flavialeotta/Master-thesis`.

Additional file: Barcodes

The files *barcodes1.tsv* and *barcodes2.tsv* contain the individual-specific barcodes, respectively for the two sequencing batches, as reported in Table 8.2.

Table 8.2: *Individual-specific barcodes*

Individual	Barcode	Invidual	Barcode	Invidual	Barcode
ALP1	CTCC	MAU10	CATCT	SP7I9	GAACTTC
ALP2	TGCA	MAU11	CCTAC	SP7I10	GGACCTA
ALP3	ACTA	PDR1	GAGGA	SP7I11	GTCGATT
ALP5	CAGA	PDR2	GGAAC	SP10I12	AACGCCT
ALP6	AACT	PDR3	GTCAA	SP11I13	AATATGC
ALP7	GCGT	PDR4	TAATA	SP11I14	ACGTGTT
ALP8	CGAT	PDR5	TACAT	SP12I15	ATTAATT
ALP9	GTAA	PDR8	TCGTT	SP13I16	ATTGGAT
ALP10	AGGC	PDR9	GGTTGT	SP14I17	CATAAGT
ALP11*	GATC	PDR10	CCAGCT	SP17I18	CGCTGAT
ALP12	CTCC	PDR11	TTCAGA	SP17I19	CGGTAGA
ALP14	TGCA	PDR12	TAGGAA	SNEV1	CTACGGA
ALP15	ACTA	PDR13	GCTCTA	SNEV2	GCGGAAT
CTT1	CAGA	PDR14	CCACAA	SNEV3	TAGCGGA
CTT2	AACT	PDR15	CTTCCA	SNEV5	TCGAAGA
CTT3	GCGT	PPURA1	GAGATA	SNEV6	TCTGTGA
LPASS1	CGAT	PPURA2	ATGCCT	SNEV7	TGCTGGA
LPASS2	GTAA	PPURA3	AGTGGA	SNEV8	ACGACTAC
LPASS4	AGGC	PPURA4	ACCTAA	SNEV9	TAGCATGC
LPASS5*	GATC	PPURA5	ATATGT	SNEV10	TAGGCCAT
LPASS6*	TCAC	PPURA7	ATCGTA	SNEV11	TGCAAGGA
LPASS7	TGCGA	PPURA8	CATCGT	SNEV13	TGGTACGT
LPASS8	CGCTT	PPURA10	CGCGGT	SNEV14	TCTCAGTC
LPASS9	TCACC	PPURA11	CTATTA	STE1	CCGGATAT
LPASS10	CTAGC	PPURA13	GCCAGT	STE3	CGCCTTAT
LPASS11	ACAAA	PPURA15	GGAAGA	STE4	AACCGAGA
LPASS12	TTCTC	RT3	GTACTT	STE5	ACAGGGAA
LPASS13	AGCCC	RT6	GTTGAA	STE6	ACGTGGTA
LPASS14	GTATT	RT9	TAACGA	STE7	CCATGGGT
LPASS15	CTGTA	RT11	TGGCTA	STE8	CGCGGAGA
MAU1	ACCGT	RT14	TATTTTT	STE9	CGTGTGGT
MAU5	GCTTA	RT15	CTTGCTT	STE10	GCTGTGGA
MAU7	GGTGT	RT16	ATGAAAC	STE11	GGATTGGT
MAU8	AGGAT	SP5I7	AAAAGTT	STE13	GTGAGGGT
MAU9	ATTGA	SP6I8	GAATTCA	STE14	TATCGGGA
				STE15	TTCCTGGA

Samples with an asterisk (*) are not part of the final results because of intermediate quality checks. An horizontal line after ALP11 separates the two sequencing batches.

They are divided in two different files because the multiplexing steps were performed separately at the reception of the sequenced data, as to not assign the wrong ID to the individuals sharing the same barcode (for example, ALP1 and ALP12).

Additional step: Renaming

The pipeline `stacks: denovo_map.pl` expects read files to be named in a specific way, namely `[Sample name].1.fq.gz` and `[Sample name].2.fq.gz` for each pair of forward and reverse reads. We noticed that in the previous filtering step, the function `stacks: process_radtags` automatic output naming did not comply to this expectation. We implemented a script that recursively renames the files in the correct way: this is not a step in the analysis, but it is necessary to prepare the files for `stacks: denovo_map.pl`.

Hyperparameters tuning for function `denovo_map.pl`

Here are reported the results from the tested combinations of parameters m, M and n for function `Stacks: denovo_map.pl` . The final metric is computed using equation 2.2, and it depends on the combinations explored: this means that even if the combinations explored give the same results, if any combination performance is added or removed from this dataset, the final metric score would change.

Table 8.3: **Results of different denovo_map.pl parameters combinations tested on a subset.**

m	M	n	μ	σ	SNPs	unique	Score	m	M	n	μ	σ	SNPs	unique	Score
3	4	1	8.16467	1.09779	8027	4907	5.9801	4	0	1	10.6793	1.34164	3058	2636	0.9258
3	2	2	8.24544	1.09503	7184	5037	5.8236	5	0	0	12.9834	1.41215	643	566	6.3205
4	1	2	11.0961	1.36403	4710	3462	0.7983	4	1	1	11.0128	1.33904	3838	3140	0.7953
5	3	0	13.8098	1.47637	3629	2329	6.2209	4	4	1	11.0606	1.28325	5877	3498	0.9170
3	3	2	8.19333	1.09503	7786	5022	5.9038	4	4	2	11.0797	1.27756	6115	3599	0.8954
4	3	2	11.1314	1.27684	5585	3566	0.8240	4	0	2	10.7920	1.37755	4119	3049	0.9396
3	2	0	8.17156	1.06492	5690	4200	5.9211	3	1	0	8.03122	1.05295	4043	3574	5.9150
4	4	4	11.0598	1.29820	6319	3653	0.9144	4	3	1	11.0991	1.27240	5357	3477	0.8327
5	1	3	13.8772	1.63090	3759	2589	1.0417	5	4	1	13.8071	1.44523	4346	2525	1.1127
5	4	0	13.7800	1.46564	4054	2349	6.3097	4	3	4	11.1448	1.29970	6124	3680	0.8640
4	1	4	11.1814	1.38593	5785	3712	0.8520	3	3	1	8.17867	1.09000	7313	4844	5.9110

Results after applying function `denovo_map.pl` with combinations of parameters (m, M and n) on a subset of 9 samples (ALP1 to ALP3, and ALP5 to ALP10) including mean coverage (μ), standard deviation (σ) across samples' coverage, total SNP counts (SNPs), number of unique loci (unique) and the final score (score) across different parameter settings. The best combination is highlighted in grey.

8.3 Software and libraries

For the analysis to be reproducible, in this section we specify all the software and libraries used, as well as their versions.

Server asimina: Computations for the pre-processing and analysis of reads were performed on a Supermicro server with 160 threads, 3 Tb of RAM and 11 Tb of disk space configured in RAID 6, with Ubuntu server OS 4.15.0-213-generic. Server `asimina` is available at IP address 159.149.160.43 but not freely accessible.

Software used (in alphabetical order): Admixture 1.3.0, bamaddrg 9baba65f88228e55639689a3cea38dd150e6284f, bedtools 2.29.0, bwa mem 0.7.19, conda 25.1.1, fastq-mcf 1.05, fastp 0.24.0, FastQC 0.11.5, freebayes 1.3.8, iqtree 2.4.0, MultiThreadFreeBayes 0.1, multiqc 1.27.1, samtools 1.21, stacks 2.68, STRUCTURE 2.3.4, vcf2phylip 2.0, vcftools 0.1.16.

Python: The optimization algorithm was implemented on Python version: 3.12.9.

Libraries used: Numpy 1.26.4, Pandas 2.2.2, Scikit-learn 1.5.0, Scikit-optimize 0.10.2.

R: Graphs generation and populations' genetic analysis were performed using R version 4.4.2, on RStudio version 2025.05.0+496.

Packages used:, adegenet 2.1.11, ape 5.8.1, BEDASSLE 1.6.1, caret 7.0.1, dplyr 1.1.4, geosphere 1.5.20, ggpubr 0.6.1, ggmap 4.0.1, ggplot2 3.5.1, ggrepel 0.9.6, gridExtra 2.3, hierfstat 0.5.11, lme4 1.1.37, lmtest 0.9.40, MuMIn 1.48.11, plotly 4.10.4, poppr 2.9.8, RColorBrewer 1.1.3, rnaturalearth 1.0.1, rnaturalearthdata 1.0.0, patchwork 1.3.1, plotly 4.11.0, poppr 2.9.7, readxl 1.4.5, reshape2 1.4.4, sf 1.0.19, StAMPP 1.6.3, tidyr 1.3.1, tidyverse 2.0.0, vcfR 1.15.0, vegan 2.7.1.

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